

Master Thesis

(φ, Γ) -Modules and Galois Cohomology

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Abstract

In this Master thesis we introduce the theory of (φ, Γ) -modules which establishes an equivalence between Galois representations of the absolute Galois group of a local field K and a class of modules equipped with certain semi-linear actions. We then follow [Her98] to give an explicit construction of a complex calculating the Galois cohomology groups of $\mathbb{Z}_p$ -representation using (φ, Γ) -modules. To emphasize the explicitness of the new description of Galois cohomology we give proofs of the local Euler characteristic formula and local Tate duality following [Her98] and [Her01].

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Chapter 1

Introduction

Let K be a finite extension of $\mathbb{Q}_p$ and fix an algebraic closure $\bar{\mathbb{Q}}_p = \bar{K}$. We are naturally interested in the structure of the absolute Galois group $G_K = \text{Gal}(\bar{K}/K)$. An approach to studying the absolute Galois group of a local field is through its representations, i.e., $\mathbb{Z}_p$ -modules with continuous linear G_K -actions, and their Galois cohomology.

If we let K_∞ denote the maximal p -th cyclotomic extension. We obtain the following tower of extensions.

$$\begin{array}{c}
 \bar{K} \\
 \left| \begin{array}{c} \text{ } \end{array} \right. \Bigg)_{H_K} \\
 K_\infty \\
 \left| \begin{array}{c} \text{ } \end{array} \right. \Bigg)_{\Gamma_K} \\
 K \\
 \left| \begin{array}{c} \text{ } \end{array} \right. \\
 \mathbb{Q}_p
 \end{array}$$

The cyclotomic character $\chi : G_K \longrightarrow \mathbb{Z}_p^*$ identifies Γ_K with an open subgroup of $\mathbb{Z}_p^*$.

Fontaine-Wintenberger's theory from [FW79] of the field of norms allows us to construct an isomorphism between H_K and the absolute Galois group of a characteristic p field denoted by $\mathbf{E}_K$. For a $\mathbb{Z}_p$ -representation of H_K , we can encode the H_K -action into the information of a module over a Cohen ring $\mathbf{A}_K$ of $\mathbf{E}_K$ with a semi-linear Frobenius action which we call a φ -module. In [Fon07], Fontaine introduced the theory of (φ, Γ_K) -modules which allows us to pass from the theory of $\mathbb{Z}_p$ -representations of G_K to certain modules over $\mathbf{A}_K$ with semi-linear Frobenius action and residual Γ_K -action. The more complicated module structure allows for rather explicit and

simple descriptions of the Frobenius and Γ_K -action.

In chapter 2 we briefly review the basics of Galois representations and group cohomology, in particular its description using a cochain complex and as a derived functor. Then we turn to the study of the aforementioned (φ, Γ_K) -modules in Chapter 3. We introduce the field of norms $\mathbf{E}_K$, Fontaine's period rings $\mathbf{A}_K$ and $\mathbf{B}_K$ and the Frobenius and G_K -action on them. For the case $K = \mathbb{Q}_p$, the ring $\mathbf{A}_{\mathbb{Q}_p}$ takes the simple form

$$\mathbf{A}_{\mathbb{Q}_p} = \left\{ \sum_{n \in \mathbb{Z}} a_n \pi^n : a_n \in \mathbb{Z}_p, a_n \rightarrow 0 \text{ for } n \rightarrow -\infty \right\}$$

where we can think of π as a formal variable. We can then define the category of (φ, Γ_K) -modules as finitely generated $\mathbf{A}_K$ -modules with semi-linear Frobenius and Γ_K -action. We will call a (φ, Γ_K) -module étale if the image of the Frobenius generates the module over $\mathbf{A}_K$.

Theorem 1.0.1 ([Fon07], 3.4.3). There is an equivalence of categories

$$\{\mathbb{Z}_p\text{-representations of } G_K\} \longleftrightarrow \{\text{étale } (\varphi, \Gamma_K)\text{-modules over } \mathbf{A}_K\}.$$

via the functor $V \mapsto \mathbf{D}(V) = (\mathbf{A} \otimes_{\mathbb{Z}_p} V)^{H_K}$.

Given the equivalence of categories it is a natural question to ask how Galois cohomology translates into the category of (φ, Γ_K) -modules. In [Her98], Herr introduced a very explicit complex of (φ, Γ_K) -modules, whose cohomology is precisely the Galois cohomology of the corresponding Galois representation. In chapter 4 we prove the following theorem using Grothendieck's theorem, stating that every effeacable δ -functor is universal.

Theorem 1.0.2 ([Her00], Theorem 4). Let V be a p -torsion representation of G_K and $M = \mathbf{D}(V)$ the corresponding (φ, Γ_K) -module. Let $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ denote the follow complex

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M \longrightarrow 0 \\ & & x & \longmapsto & ((\varphi - 1)x, (\gamma - 1)x) & & \\ & & & & (x, y) & \longmapsto & (\gamma - 1)x - (\varphi - 1)y \end{array}$$

Then the cohomology groups $H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(M))$ of this complex coincide with the Galois cohomology groups $H^i(G_K, V)$.

This description of Galois cohomology allows us to regard Galois cohomology purely on the side of (φ, Γ_K) -modules. The complex $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ gives us very explicit descriptions of the cohomology groups using which we prove the following two classical results in Galois cohomology only using the theory of (φ, Γ_K) -modules. The two proofs follow [Her98] and [Her01].

Theorem 1.0.3 ([Her98], Theorem B). Let V be a p -torsion representation of G_K . Then $H^i(G_K, V)$ are finite and

$$\chi(V) := \frac{\#H^0(G_K, V) \cdot \#H^2(G_K, V)}{\#H^1(G_K, V)} = p^{-[K:\mathbb{Q}_p]\ell(V)}$$

where $\ell(V)$ is the length of V (as a $\mathbb{Z}_p$ -module).

Theorem 1.0.4 ([Her98], Theorem C). Let V be a p -torsion representation of G_K and let $V^*(1) = \text{Hom}(V, \mu_{p^\infty})$. Then there is a canonical isomorphism

$$H^2(G_K, \mu_{p^\infty}) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p.$$

Further, for $0 \leq i \leq 2$, the cup-product

$$H^i(G_K, V) \times H^{2-i}(G_K, V^*(1)) \longrightarrow H^2(G_K, \mu_{p^\infty}) \cong \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing.

Chapter 2

Preliminaries

2.1 Group cohomology

In this section, we recall the definition of group cohomology which is an important tool to study Galois representations.

Definition 2.1.1. Let G be a group. A G -module M is an abelian group together with a map

$$\begin{aligned} G \times M &\longrightarrow M \\ (g, m) &\longmapsto gm \end{aligned}$$

such that for $g, g' \in G$, $m, m' \in M$

- (i) $g(m + m') = gm + gm'$
- (ii) $(gg')m = g(g'm)$ and $1_G m = m$.

We write Mod_G for the category of G -modules, the morphisms being G -equivariant group homomorphisms.

If G and M are topological groups, we are mostly concerned with so-called topological G -modules. In this thesis, we are mostly interested in the case where G and M are profinite groups.

Definition 2.1.2. Let G be a profinite group equipped with the profinite topology. A topological G -module M is a topological group M with a G -action, such that

$$\begin{aligned} G \times M &\longrightarrow M \\ (g, m) &\longmapsto gm \end{aligned}$$

is continuous. If M is finite, we equip M with the discrete topology.

We call a G -module M *discrete* if the G -action is continuous with respect to the discrete topology on M . We denote the category of discrete G -modules of G by $\text{Mod}_G^{\text{discrete}}$.

Definition 2.1.3. Let G be a finite or profinite group and M a topological G -module. For $n \geq 1$, let $C^n(G, M)$ denote the set of all continuous maps $G^n \rightarrow M$, and set $C^0(G, M) = M$. We define coboundary maps

$$\begin{aligned} d^n : C^n(G, M) &\longrightarrow C^{n+1}(M) \\ (d^n f)(g_1, \dots, g_{n+1}) &= g_1 f(g_2, \dots, g_{n+1}) + \sum_{i=1}^n (-1)^i f(g_1, \dots, g_i g_{i+1}, \dots, g_{n+1}) \\ &\quad + (-1)^{n+1} f(g_1, \dots, g_n) \end{aligned}$$

We define the r -th continuous cohomology group $H_{\text{cont}}^r(G, M)$ to be the r -th cohomology group of the cochain complex

$$C^0(G, M) \xrightarrow{d^0} C^1(G, M) \xrightarrow{d^1} C^2(G, M) \xrightarrow{d^2} \dots$$

In other words for

$$\begin{aligned} Z^r(G, M) &= \ker(d^r) && \text{r-cocycles} \\ B^r(G, M) &= \text{im}(d^{r-1}) && \text{r-coboundaries} \end{aligned}$$

we define

$$H_{\text{cont}}^r(G, M) := \frac{Z^r(G, M)}{B^r(G, M)}.$$

Remark 2.1.4. Note that if the action of G on M is trivial, then a 1-cocycle is just a group homomorphism $G \rightarrow M$ and all 1-coboundaries are trivial. We therefore have

$$H^1(G, M) = \text{Hom}_{\text{cont}}(G, M).$$

For a G -module M let $M^G = \{m \in M \mid gm = m \text{ for all } g \in G\}$ be the group of invariants under G . We then obtain the left exact functor

$$\begin{aligned} \text{Mod}_G &\longrightarrow \text{Ab} \\ M &\longmapsto M^G. \end{aligned}$$

If we are in a nice enough category, for example the category of abstract G -modules for G a finite group we can show that the cohomology groups we just defined are in fact the r -th right derived functors of the functor $M \mapsto M^G$. This structure of the cohomology functor will be very useful later. However, to ensure the existence of these derived functors, we need the category to have enough injectives. This is not the case for the category of topological G -modules for G a profinite group, but in the following we will show it is true for the category of discrete G -modules.

Lemma 2.1.5 ([NSW13], Proposition 1.1.8). Let G be a profinite group and let M be a G -module. Then the following conditions are equivalent

- (i) M is discrete.
- (ii) For every $m \in M$ the subgroup $Gm := \{g \in G \mid g(m) = m\}$ is open.
- (iii) $M = \bigcup M^U$, where U runs through the open subgroups of G .

Note that if G is finite this means that any G -module is discrete.

Proof. 1. (i) $\Rightarrow$ (ii) Let $m \in M$. By restriction we have a continuous function

$$G \times \{m\} \longrightarrow M$$

where the preimage of the open set $\{m\}$ is $Gm := \{g \in G \mid g(m) = m\}$. Therefore it is open by continuity.

- 2. (ii) $\Rightarrow$ (iii) Let $m \in M$ then Gm is open and thus $m \in M^{Gm} \subset \bigcup M^U$, where U runs through the open subgroups of G .

- 3. (iii) $\Rightarrow$ (i) Let $(g, m) \in G \times M$. To show continuity, we find a neighbourhood of (g, m) that maps to the set $\{g(m)\}$. Indeed, if $m \in M^U$ for an open subgroup U of G , then we have that the open set $gU \times \{m\}$ maps to $\{g(m)\}$. $\square$

Proposition 2.1.6 ([NSW13], Lemma 2.6.5). Let G be a profinite group. The category $\text{Mod}_G^{\text{discrete}}$ has enough injectives.

Proof. We use the fact that the category of abstract G -modules without taking the topology of G into account has enough injectives since it is equivalent to the category of $\mathbb{Z}[G]$ -modules, where $\mathbb{Z}[G]$ is the group ring. For an abstract G -module M let

$$M^{\text{discrete}} = \bigcup M^U$$

where U runs through all open subgroups of G . Using Lemma 2.1.5 we see that $M^{\text{discrete}} \in \text{Mod}_G^{\text{discrete}}$. Now let $M \in \text{Mod}_G^{\text{discrete}}$ and let I be an injective object in Mod_G with a G -monomorphism $f : M \hookrightarrow I$. Since f is G -equivariant we have that $f(M^U) \subset I^U$ for all open subgroups $U \subset G$. Therefore, if M is discrete, f factors through I^{discrete} . It remains to show that I^{discrete} is an injective object of $\text{Mod}_G^{\text{discrete}}$. For $M, N \in \text{Mod}_G^{\text{discrete}}$ and a monomorphism $M \hookrightarrow N$ we get since I is injective as an abstract G -module

$$\begin{array}{ccc} M & \hookrightarrow & N \\ \downarrow & \nearrow g & \\ I & & \end{array}$$

but once again, since N is discrete, g must factor through I^{discrete} . $\square$

Definition 2.1.7. Let G be a finite group or a profinite group. Let M be a discrete G -module. We define the i -th cohomology group of G with coefficients in M as the i -th right derived functor of $M \mapsto M^G$. In particular, for an injective resolution in the category $\text{Mod}_G^{\text{discrete}}$

$$0 \longrightarrow M \longrightarrow I^0 \xrightarrow{d^0} I^1 \longrightarrow \dots$$

we obtain the complex

$$0 \xrightarrow{d^{-1}} (I^0)^G \xrightarrow{d^0} (I^1)^G \longrightarrow \dots$$

and

$$H^i(G, M) = \frac{\ker(d^i)}{\text{im}(d^{i-1})}.$$

Proposition 2.1.8 ([Mil20], Proposition 1.17). The cohomology groups calculated by the continuous cochain complex and the derived functors coincide for discrete G -modules for G profinite or finite, i.e. there is a canonical isomorphism

$$H_{\text{cont}}^i(G, M) \cong H^i(G, M).$$

From now on, we will therefore simply write $H^i(G, M)$ for both continuous cohomology and the derived functors.

Remark 2.1.9. A property that follows from the cohomology groups being the i -th right derived functors of $M \mapsto M^G$ is that a short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

of G -modules induces a long exact sequence

$$0 \longrightarrow H^0(G, M') \longrightarrow H^0(G, M) \longrightarrow H^0(G, M'') \xrightarrow{\delta} H^1(G, M') \longrightarrow H^1(G, M) \longrightarrow \dots$$

We also obtain a long exact sequence of continuous cohomology using the cochain complex for topological G -modules which we can show directly using homological algebra.

As discussed, it is very useful to view the cohomology groups as derived functors. If M is a topological profinite G -module given as the inverse limit of finite, discrete G -modules M_n we can calculate the cohomology of M using the cohomology of M_n .

Definition 2.1.10. We say an inverse system (A_i, f_{ij}) satisfies the Mittag-Leffler condition if for every k there exists a $j \geq k$ such that for all $i \geq j$,

$$f_{kj}(A_j) = f_{ki}(A_i).$$

Remark 2.1.11. We can think of a system satisfying the Mittag-Leffler conditions as the transition maps being surjective. In fact, one can show that we can rewrite any system satisfying the Mittag-Leffler conditions as a system with surjective transition maps with same inverse limit. Further, we can show that any inverse system of finite abelian groups satisfies the Mittag-Leffler conditions.

Proposition 2.1.12 ([Lan05], Proposition 10.3). Assume that (A_n) is an inverse system satisfying the Mittag-Leffler conditions. Given an exact sequence

$$0 \longrightarrow (A_n) \longrightarrow (B_n) \longrightarrow (C_n) \longrightarrow 0$$

then

$$0 \longrightarrow \varprojlim A_n \longrightarrow \varprojlim B_n \longrightarrow \varprojlim C_n \longrightarrow 0$$

is exact.

Proposition 2.1.13 ([NSW13], Corollary 2.7.6). Let G be profinite and M a compact G -module such that

$$M = \varprojlim_{n \in \mathbb{N}} M_n,$$

i.e., M is the inverse limit of countably many finite, discrete G -modules. If $H^i(G, M_n)$ satisfy the Mittag-Leffler condition, we get an isomorphism

$$H^i(G, M) \xrightarrow{\sim} \varprojlim H^i(G, M_n).$$

Proof. Since we have

$$M = \varprojlim M_n$$

where M_n are finite G -modules equipped with the discrete topology. By ([NSW13], §7 2.7.5) we have a short exact sequence

$$0 \longrightarrow \varprojlim^1 H^{i-1}(G, M_n) \longrightarrow H^i(G, M) \longrightarrow \varprojlim H^i(G, M_n) \longrightarrow 0.$$

Since $H^i(G, M_n)$ with the induced transition maps on cohomology groups satisfy the Mittag-Leffler conditions for all $i \in \mathbb{N}$, by ([NSW13], §7 2.7.4) we have that $\varprojlim^1 H^i(G, M_n) = 0$ for all $i \in \mathbb{N}$ and thus we get an isomorphism

$$H^i(G, M) \xrightarrow{\sim} \varprojlim H^i(G, M_n). \quad \square$$

We end the section by introducing the pairing known as the *cup-product*.

Proposition 2.1.14 ([Mil20], Proposition 1.38). There exists one and only one family of bi-additive pairings

$$\begin{aligned} \cup : H^r(G, M) \times H^s(G, N) &\longrightarrow H^{r+s}(G, M \otimes_{\mathbb{Z}} N) \\ (m, n) &\longmapsto m \cup n \end{aligned}$$

defined for all G -modules M, N and all integers r, s satisfying the following conditions:

- (i) These maps become morphisms of functors when the two sides are regarded as covariant bifunctors on (M, N) .
- (ii) For $r = s = 0$, the pairing is

$$\begin{aligned} M^G \times N^G &\longrightarrow (M \otimes_{\mathbb{Z}} N)^G \\ (m, n) &\longmapsto m \otimes n \end{aligned}$$

- (iii) If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is an exact sequence of G -modules such that

$$0 \rightarrow M' \otimes_{\mathbb{Z}} N \rightarrow M \otimes_{\mathbb{Z}} N \rightarrow M'' \otimes_{\mathbb{Z}} N \rightarrow 0$$

is exact, then

$$(\delta m'') \cup n = \delta(m'' \cup n), \quad m'' \in H^r(G, M''), \quad n \in H^s(G, N)$$

where δ denotes the connecting homomorphism from the long exact sequence of cohomology.

- (iv) If $0 \rightarrow N' \rightarrow N \rightarrow N'' \rightarrow 0$ is an exact sequence of G -modules such that

$$0 \rightarrow M \otimes_{\mathbb{Z}} N' \rightarrow M \otimes_{\mathbb{Z}} N \rightarrow M \otimes_{\mathbb{Z}} N'' \rightarrow 0$$

is exact, then

$$m \cup \delta n = (-1)^r \delta(m \cup n''), \quad m \in H^r(G, M), \quad n'' \in H^s(G, N'').$$

For explicit cocycles ϕ and ψ for m and n , $m \cup n$ is represented by the cocycle

$$(g_1, \dots, g_{r+s}) \longmapsto \phi(g_1, \dots, g_r) \otimes g_1 \cdots g_r \psi(g_{r+1}, \dots, g_{r+s}).$$

Proposition 2.1.15 ([Mil20], Proposition 1.39). We have for $x \in H^r(G, M)$ and $y \in H^s(G, M)$

$$x \cup y = (-1)^{rs} (y \cup x).$$

if we identify $M \otimes_{\mathbb{Z}} N$ and $N \otimes_{\mathbb{Z}} M$.

2.2 Galois representations

Let G be a group. In our case, G will always be a profinite absolute Galois group of a local field K .

Definition 2.2.1. An $\mathbb{F}_p$ -representation V of G is a finite-dimensional $\mathbb{F}_p$ -vector space V equipped with a linear G -action, i.e. with a group homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V).$$

We sometimes call an $\mathbb{F}_p$ -representation of G a *mod p representation*.

Similarly, a $\mathbb{Z}_p$ -representation V of G is a finitely generated $\mathbb{Z}_p$ -module V equipped with a linear G -action. We call a $\mathbb{Z}_p$ -representation of G a *free* if V is a free $\mathbb{Z}_p$ -module.

Lastly, a $\mathbb{Q}_p$ -representation V of G is a finite-dimensional $\mathbb{Q}_p$ -vector space V equipped with a linear G -action. We also call a $\mathbb{Q}_p$ -representation of G a *p -adic representation*.

We denote the categories of $\mathbb{F}_p$, $\mathbb{Z}_p$ or $\mathbb{Q}_p$ -representations of G by $\text{Rep}_{\mathbb{F}_p}(G)$, $\text{Rep}_{\mathbb{Z}_p}(G)$ or $\text{Rep}_{\mathbb{Q}_p}(G)$.

If G is a topological group, we always require the G -action to be continuous, i.e. that the function

$$\begin{aligned} G \times V &\longrightarrow V \\ (g, v) &\longmapsto \rho(g)(v) \end{aligned}$$

is continuous. Instead of $\rho(g)(v)$ we typically write $g(v)$.

Remark 2.2.2. In this thesis, we require all representations to be finitely generated. However, we can of course also consider infinitely generated modules with a linear G -action, which are precisely $\mathbb{Z}_p[G]$ -modules, where $\mathbb{Z}_p[G]$ is the group ring.

Remark 2.2.3. Note that V carries the topology of a $\mathbb{F}_p$ or $\mathbb{Q}_p$ -vector space, or a finitely generated $\mathbb{Z}_p$ -module. For a given choice of generators $e_1, \dots, e_d$, a basis of open sets will be given by $U_1 e_1 + \dots + U_d e_d$, for U_i open sets in $\mathbb{F}_p$, $\mathbb{Q}_p$ or $\mathbb{Z}_p$.

Proposition 2.2.4. Let G be a profinite group. Every p -adic representation V of G contains a G stable $\mathbb{Z}_p$ -lattice (i.e., a finitely generated $\mathbb{Z}_p$ -module that generates V over $\mathbb{Q}_p$).

Proof. Choose any $\mathbb{Q}_p$ basis $e_1, \dots, e_d$ of V and let L be the $\mathbb{Z}_p$ -span of that basis. We claim that

$$A := \{g \in G \mid g(L) \subset L\}$$

is an open subgroup of G . Indeed, since the action is continuous and A is the preimage of the subgroup $H := \{f \in \text{GL}(V) \mid f(L) \subset L\}$ of $\text{GL}(V)$, it suffices to show H is open. If we identify $\text{GL}(V)$ with $\text{GL}(\mathbb{Q}_p)$ via the basis $e_1, \dots, e_d$, a system of neighbourhoods of the identity is given by $\text{id} + p^r \text{Mat}_n(\mathbb{Z}_p)$. Now for $f \in H$, any matrix in $g \in f + p^r \text{Mat}_n(\mathbb{Z}_p)$ will still satisfy $g(L) \subset L$ since L is a $\mathbb{Z}_p$ -lattice. Therefore, H is open and thus A is an open subgroup of G . Since G is a profinite group, any open subgroup is of finite index, and we can choose representatives $g_1, \dots, g_n$ of the cosets in G/H and define

$$T = \sum_{i=1}^n g_i(L).$$

Then T is still a $\mathbb{Z}_p$ -lattice. Further, T is stable under G since for any $h \in T$, $h(T) = \sum_{i=1}^n (hg_i)(L)$, but by definition of H there is a g_j such that $g_j^{-1}hg_i(L) \subset L$, i.e. $hg_i(L) \subset g_j(L)$. We deduce that $h(T) \subset T$. $\square$

Definition 2.2.5. We call a $\mathbb{Z}_p$ -representation of G a *p-torsion* representation if there exists an $n \in \mathbb{N}$ such that $p^n \cdot V = 0$. We denote the category of p -torsion representations of G by $\text{Rep}_{p\text{-tor}}(G)$. It is a subcategory of $\text{Rep}_{\mathbb{Z}_p}(G)$.

Example 2.2.6. 1. The simplest case of a Galois representation is the trivial representation given by a rank one $\mathbb{Z}_p$ -module (or $\mathbb{Q}_p$ -vector space) with trivial Galois action.

2. Let L be a finite Galois extension of $\mathbb{Q}_p$. Then L is a finite dimensional $\mathbb{Q}_p$ -vector space with $\mathbb{Q}_p$ -linear Galois action of $\text{Gal}(L/\mathbb{Q}_p)$, i.e. a p -adic representation of $\text{Gal}(L/\mathbb{Q}_p)$.

3. For G a profinite group, all free rank one $\mathbb{Z}_p$ representations are given by characters $\delta : G \rightarrow \mathbb{Z}_p^*$, i.e. a rank one $\mathbb{Z}_p$ -module with basis v , s.t. $g(v) = \delta(g)v$ for all $g \in G$. We investigate an important example of this for $G_K = \text{Gal}(\bar{K}/K)$ the absolute Galois group of some finite extension K of $\mathbb{Q}_p$. We define $\mathbb{Z}_p(1)$ to be the free rank one $\mathbb{Z}_p$ -representation given by the cyclotomic character $\chi : G_K \rightarrow \mathbb{Z}_p^*$ which satisfies $g(\zeta_{p^i}) = \zeta_{p^i}^{\chi(g)_i}$ for all $g \in G_K$ and $\zeta_{p^i} \in \mu_{p^\infty}$. In fact we have that

$$\mathbb{Z}_p(1) = \varprojlim \mu_{p^n}$$

where μ_{p^n} denotes the group of p^n -th roots of unity, equipped with the structure of a p -torsion $\mathbb{Z}_p$ -representation with $\alpha\zeta = \zeta^\alpha$ for $\alpha \in \mathbb{Z}_p$ and $\zeta \in \mu_{p^n}$ and the obvious G_K -action.

For many results it is very useful to work with the category of p -torsion representations instead of with $\mathbb{Z}_p$ -representations. However, since we want to regard Galois cohomology later, it is unfortunate that the category $\text{Rep}_{p\text{-tor}}(G)$ does not have enough injectives. We introduce a slightly larger category that has enough injectives.

Definition 2.2.7. We denote by $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G)$ the category of (not necessarily finitely generated) $\mathbb{Z}_p$ -modules with $\mathbb{Z}_p$ -linear G -action that are discrete G -modules.

Proposition 2.2.8. Let G be a profinite group. The category $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G)$ has enough injectives.

Proof. Completely analogous to Proposition 2.1.6 we can prove that the category of discrete G_K -modules with $\mathbb{Z}_p$ -linear action, denoted by $\text{Rep}_{\mathbb{Z}_p}^{\text{discrete}}(G)$, has enough injectives. Thus, for an element $V \in \text{Rep}_{p\text{-tor}}^{\text{discrete}}(G)$, let I be an injective element in $\text{Rep}_{\mathbb{Z}_p}^{\text{discrete}}(G)$ with monomorphism $V \hookrightarrow I$ and let $I^{p\text{-tor}}$ be the p -torsion part of I . Since the image of any p -torsion module under a homomorphism must be p -torsion we

find that the monomorphism $V \hookrightarrow I$ factors through $I^{p\text{-tor}}$ and $I^{p\text{-tor}}$ is an injective element in $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G)$. $\square$

Proposition 2.2.9. The category $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G_K)$ consists precisely of elements that can be written as direct limits of elements of $\text{Rep}_{p\text{-tor}}(G_K)$.

Proof. Recall that we defined $\text{Rep}_{p\text{-tor}}(G_K)$ to be the category of finitely generated p -torsion modules with $\mathbb{Z}_p$ -linear G_K -action. Let $V \in \text{Rep}_{p\text{-tor}}^{\text{discrete}}(G_K)$ and let $(f_i)_{i \in I}$ be a generating system of V . For $J \subset I$ let V_J be the $\mathbb{Z}_p[G_K]$ -submodule generated by $(e_i)_{i \in J}$. If J is finite, then $V_J \in \text{Rep}_{p\text{-tor}}(G_K)$. Via inclusion maps we obtain a directed system $(V_J)_{J \subset I \text{ finite}}$, where each V_J carries the discrete topology. Since V is discrete we can see that the direct limit is precisely V . On the other hand, any direct limit of finite representations is discrete. $\square$

2.3 Some results from Galois cohomology

For this part, we want to state some theorems of group cohomology for G a finite or profinite Galois group.

The first important theorem is usually known as Hilbert 90.

Theorem 2.3.1 ([Mil20], Proposition 1.22). Let $K \subset L$ be a finite Galois extension with Galois group G . Then $H^1(G, L^*) = 0$.

The next theorem is called the Euler-Characteristic formula.

Theorem 2.3.2 ([IS01], II.5, Theorem 5). Let K be a finite extension of $\mathbb{Q}_p$ and G_K its absolute Galois group. Let V be a p -torsion $\mathbb{Z}_p$ -representation of G_K . Then $H^i(G_K, V)$ are finite and

$$\chi(V) := \frac{\#H^0(G_K, V) \cdot \#H^2(G_K, V)}{\#H^1(G_K, V)} = p^{-[K:\mathbb{Q}_p]\ell(V)}$$

where $\ell(V)$ denotes the length of V as a $\mathbb{Z}_p$ -module.

Taking the inverse limit yields:

Corollary 2.3.3. Let K be a finite extension of $\mathbb{Q}_p$ and G_K its absolute Galois group. Let V be a free $\mathbb{Z}_p$ -representation of G_K . Then $H^i(G_K, V)$ are finitely generated and

$$\chi(V) := \text{rank } H^0(G_K, V) - \text{rank } H^1(G_K, V) + \text{rank } H^2(G_K, V) = -[K:\mathbb{Q}_p]\text{rank } V$$

where rank denotes the rank as a $\mathbb{Z}_p$ -module.

The following theorem resembles Poincaré-duality and is usually called Tate-duality.

Theorem 2.3.4 ([IS01], II.5, Theorem 2). Let K be a finite Galois extension of $\mathbb{Q}_p$ and G_K its absolute Galois group. Further, let μ_{p^∞} denote the group of all p -th power roots of unity, with the obvious G_K -action. Let V be a p -torsion representation of G_K and let $V^*(1) = \text{Hom}(V, \mu_{p^\infty})$. Then there is a canonical isomorphism $H^2(G_K, \mu_{p^\infty}) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p$ and for $0 \leq i \leq 2$, the cup-product

$$H^i(G_K, V) \times H^{2-i}(G_K, V^*(1)) \longrightarrow H^2(G_K, \mu_{p^\infty}) \cong \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing.

Remark 2.3.5. Note that for two representations V, W of G , the representation $\text{Hom}(V, W)$ is given by the $\mathbb{Z}_p$ -module $\text{Hom}_{\mathbb{Z}_p}(V, W)$ equipped with G -action given by $(g\phi)(v) = g(\phi(g^{-1}v))$ for $g \in G$, $\phi \in \text{Hom}_{\mathbb{Z}_p}(V, W)$ and $v \in V$.

Definition 2.3.6. The *p -cohomological dimension* of a profinite group G is the smallest non-negative integer n such that for any discrete G -module A , one has $H^q(G, A)(p) = 0$ for $q > n$.

Proposition 2.3.7 ([IS01], II Proposition 15). For K a finite extension of $\mathbb{Q}_p$ the group $G_K = \text{Gal}(\bar{K}/K)$ has p -cohomological dimension 2.

Example 2.3.8. We want to apply the previous results to calculate some cohomology groups. Let K be a finite Galois extension of $\mathbb{Q}_p$ and G_K its absolute Galois group. Let $V = \mathbb{Z}/p^n\mathbb{Z}$ be equipped with the trivial G_K -action and $W = \mu_{p^n}$ with the obvious G_K -action.

First recall the Kummer sequence

$$0 \longrightarrow \mu_{p^n} \longrightarrow \bar{K}^* \xrightarrow{x \mapsto x^{p^n}} \bar{K}^* \longrightarrow 0$$

which gives a long exact sequence of cohomology

$$\begin{aligned} 0 &\longrightarrow H^0(G_K, \mu_{p^n}) \longrightarrow H^0(G_K, \bar{K}^*) \longrightarrow H^0(G_K, \bar{K}^*) \\ &\longrightarrow H^1(G_K, \mu_{p^n}) \longrightarrow H^1(G_K, \bar{K}^*) \longrightarrow H^1(G_K, \bar{K}^*) \\ &\longrightarrow H^2(G_K, \mu_{p^n}) \longrightarrow H^2(G_K, \bar{K}^*) \longrightarrow H^2(G_K, \bar{K}^*) \end{aligned}$$

and using Hilbert 90 we obtain $H^1(G_K, \bar{K}^*) = 0$ and thus $H^1(G_K, \mu_{p^n}) = K^*/(K^*)^{p^n}$. Further, it is possible to show that $H^2(G_K, \bar{K}^*) \cong \mathbb{Q}_p/\mathbb{Z}_p$ which we will not show here. This then implies that $H^2(G_K, \mu_{p^n}) = \mathbb{Z}/p^n\mathbb{Z}$. It is clear that $H^0(G_K, \mu_{p^n}) = \mu_{p^n} \cap K$. We summarize

- (i) $H^0(G_K, \mu_{p^n}) = \mu_{p^n} \cap K$
- (ii) $H^1(G_K, \mu_{p^n}) = K^*/(K^*)^{p^n}$
- (iii) $H^2(G_K, \mu_{p^n}) = \mathbb{Z}/p^n\mathbb{Z}$.

We continue by showing that the Tate dual of $\mathbb{Z}/p^n\mathbb{Z}$ is precisely μ_{p^n} . Indeed, an element $\phi \in \text{Hom}_{\mathbb{Z}_p}(\mathbb{Z}/p^n\mathbb{Z}, \mu_{p^\infty})$ is uniquely determined $\phi(1)$, which has to be a p^n -th root of unity. Further, the G_K -action is given by $(g\phi)(v) = g(\phi(g^{-1}v)) = g(\phi(v))$ and thus $(g\phi)(1) = g(\phi(1))$ so it coincides with the usual Galois action on the p^n -th roots of unity.

We know that $H^0(G_K, \mathbb{Z}/p^n\mathbb{Z}) = \mathbb{Z}/p^n\mathbb{Z}$ and $H^2(G_K, \mu_{p^n}) = \mathbb{Z}/p^n\mathbb{Z}$. In this case, the Tate-duality pairing is just given by

$$\begin{aligned} \mathbb{Z}/p^n\mathbb{Z} \times \mathbb{Z}/p^n\mathbb{Z} &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, y) &\longmapsto \frac{x}{p^n} \cdot \frac{y}{p^n}. \end{aligned}$$

Further, we know that $H^0(G_K, \mu_{p^n}) = \mu_{p^n} \cap K$ will be cyclic of order p^k for $0 \leq k \leq n$. Since

$$H^2(G_K, \mathbb{Z}/p^n\mathbb{Z}) \times \mu_{p^n} \cap K \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing, $H^2(G_K, \mathbb{Z}/p^n\mathbb{Z}) \cong \mu_{p^n} \cap K$. In case $K = \mathbb{Q}_p$, we have that $H^2(G_K, \mathbb{Z}/p^n\mathbb{Z})$ is trivial.

Finally, using the Euler-Characteristic formula we can calculate that $H^1(G_K, \mathbb{Z}/p^n\mathbb{Z})$ has cardinality p^{2n} . Indeed, by Remark 2.1.4 since the action of G_K is trivial on $\mathbb{Z}/p^n\mathbb{Z}$ we have

$$H^1(G_K, \mathbb{Z}/p^n\mathbb{Z}) \cong \text{Hom}(G_K, \mathbb{Z}/p^n\mathbb{Z}).$$

For $K = \mathbb{Q}_p$ we can see that $\text{Hom}(G_K, \mathbb{Z}/p^n\mathbb{Z}) = \text{Hom}(\text{Gal}(\mathbb{Q}_p^{\text{ab}}/\mathbb{Q}_p), \mathbb{Z}/p^n\mathbb{Z})$ where $\mathbb{Q}_p^{\text{ab}}$ is the maximal abelian extension of $\mathbb{Q}_p$. Now $\text{Gal}(\mathbb{Q}_p^{\text{ab}}/\mathbb{Q}_p) \cong \hat{\mathbb{Z}} \times \mathbb{Z}_p^* = \prod_q \mathbb{Z}_q \times \mathbb{Z}_p^*$ and since there are no non-trivial homomorphisms $\mathbb{Z}_q \longrightarrow \mathbb{Z}/p\mathbb{Z}$ for $q \neq p$ we have

$$\text{Hom}(G_{\mathbb{Q}_p}, \mathbb{Z}/p^n\mathbb{Z}) \cong \text{Hom}(\mathbb{Z}_p \times \mathbb{Z}_p^*, \mathbb{Z}/p^n\mathbb{Z}) \cong \mathbb{Z}/p^n\mathbb{Z} \times \mathbb{Z}/p^n\mathbb{Z}$$

which indeed has order p^{2n} .

Definition 2.3.9. Let K be a finite extension of $\mathbb{Q}_p$ and let L/K be a finite extension. For a representation V of G_K we define the induced representation $\text{Ind}_L^K(V)$ of G_K to be

$$\text{Ind}_L^K(V) := \{f : G_K \longrightarrow V \mid f(h\sigma) = hf(\sigma) \text{ for all } h \in G_L, \sigma \in G_K\}$$

equipped with the G_K -action given by

$$(gf)(\sigma) = f(\sigma g).$$

In fact, the next proposition, known as Shapiro's lemma, tells us that we can relate the cohomology groups of a representation to the cohomology groups of its induced representation.

Proposition 2.3.10 ([Mil20], Proposition 1.11). Let K be a finite extension of $\mathbb{Q}_p$ and let L/K be a finite extension. Then for all $i \geq 0$

$$H^i(G_K, \text{Ind}_K^L(V)) \cong H^i(G_L, V).$$

It is often easier to work with p -torsion representations. The following results will help us reduce to the p -torsion case when proving results about Galois cohomology.

Corollary 2.3.11. For a $\mathbb{Z}_p$ -representation V of G_K the inverse system $H^i(G_K, V/p^n V)$ satisfies the Mittag-Leffler condition, and thus we get an isomorphism

$$H^i(G_K, V) \xrightarrow{\sim} \varprojlim H^i(G_K, V/p^n V).$$

Proof. By ([Tat10], Theorem 2.1) since $V/p^n V$ is p -torsion, the cohomology groups $H^i(G_K, V/p^n V)$ are finite. Since any inverse system of finite abelian groups satisfies the Mittag-Leffler conditions, by Proposition 2.1.13, we get the wanted isomorphism. $\square$

Example 2.3.12. For $K = \mathbb{Q}_p$ we can use Example 2.3.8 and Proposition 2.3.11 to calculate the cohomology of the trivial one-dimensional free $\mathbb{Z}_p$ -representation. We have

$$\begin{aligned} H^0(G_{\mathbb{Q}_p}, \mathbb{Z}_p) &= \varprojlim H^0(G_{\mathbb{Q}_p}, \mathbb{Z}/p^n \mathbb{Z}) = \varprojlim \mathbb{Z}/p^n \mathbb{Z} = \mathbb{Z}_p \\ H^2(G_{\mathbb{Q}_p}, \mathbb{Z}_p) &= \varprojlim H^2(G_{\mathbb{Q}_p}, \mathbb{Z}/p^n \mathbb{Z}) = 0. \end{aligned}$$

2.4 Iwasawa algebras

For finite groups G' we often consider the *group ring* $\mathbb{Z}_p[G']$ which is the ring of formal linear combinations of elements of G' with coefficients in $\mathbb{Z}_p$. Any G' -module that has a structure of a $\mathbb{Z}_p$ -module can also be seen as a $\mathbb{Z}_p[G']$ -module.

Definition 2.4.1. Let G be a profinite group. We define the *Iwasawa algebra* $\Lambda[G]$ of G to be

$$\Lambda[G] = \varprojlim \mathbb{Z}_p[G/H]$$

where H runs through all open normal subgroups and $\mathbb{Z}_p[G/H]$ are the group rings of G/H .

Note that in the profinite topology, the open normal subgroups of G are precisely the normal subgroups of finite index. The transition maps are induced by the quotient maps $G/H \twoheadrightarrow G/H'$ for $H' \subset H$. The Iwasawa algebra remembers both the topology of G and the topology of $\mathbb{Z}_p$. We equip it with the inverse limit topology, and each $\mathbb{Z}_p[G/H]$ carries the topology of a finitely generated $\mathbb{Z}_p$ -module. In this thesis, we will be interested in the Iwasawa algebra of an open subgroup Γ of $\mathbb{Z}_p^*$.

We regard the Iwasawa algebra of $\Gamma \cong \mathbb{Z}_p$. Let γ be a topological generator of Γ , i.e. $\Gamma = \langle \gamma \rangle$. The open normal subgroups of Γ are the subgroups $\Gamma_n := \langle \gamma^{p^n} \rangle$ for $n \in \mathbb{N}$, which are cyclic of order p^n generated by $\gamma + \Gamma_n$.

For T a formal variable, we have isomorphisms of $\mathbb{Z}_p$ -algebras

$$\begin{aligned} \mathbb{Z}_p[\Gamma/\Gamma_n] &\xrightarrow{\sim} \mathbb{Z}_p[T]/((1+T)^{p^n} - 1) \\ \gamma \bmod \Gamma_n &\longmapsto 1 + T \bmod (1+T)^{p^n} - 1 \end{aligned} \quad (2.1)$$

which are compatible with the transition maps of the inverse system.

Proposition 2.4.2. We have an isomorphism of $\mathbb{Z}_p$ -algebras, which is also a homeomorphism,

$$\begin{aligned} \mathbb{Z}_p[[T]] &\xrightarrow{\sim} \Lambda[\Gamma] \\ T &\longmapsto \gamma - 1 \end{aligned}$$

where $\Lambda[\Gamma]$ carries the inverse limit topology and $\mathbb{Z}_p[[T]]$ carries the (p, T) -topology.

Proof. Let $q_n = (1+T)^{p^n} - 1 \in (p, T)^n$. For each n , we have continuous maps

$$\mathbb{Z}_p[[T]] \longrightarrow \mathbb{Z}_p[[T]]/(q_n) \xrightarrow{\sim} \mathbb{Z}_p[T]/(q_n) \xrightarrow{\sim} \mathbb{Z}_p[\Gamma/\Gamma_n]. \quad (2.2)$$

where the first two maps are the obvious maps, sending $\sum a_n T^n$ to its class in $\mathbb{Z}_p[T]/(q_n)$ and the last arrow is (2.1). Thus there is a continuous homomorphism

$$\begin{aligned} \kappa : \mathbb{Z}_p[[T]] &\longrightarrow \Lambda[\Gamma] \\ T &\longmapsto \gamma - 1. \end{aligned}$$

We see that $\ker(\kappa) = \bigcap_{n \in \mathbb{N}} (q_n) \subset \bigcap_{n \in \mathbb{N}} (p, T)^n = \{0\}$. Further, $\mathbb{Z}_p[[T]]$ with the (p, T) -topology is compact, so the image under κ is closed and the image of κ must be dense since (2.2) is surjective for every $n \in \mathbb{N}$, thus κ is surjective. Since $\mathbb{Z}_p[[T]]$ is compact and $\Lambda[\Gamma]$ Hausdorff we can deduce that κ is indeed an isomorphism of topological $\mathbb{Z}_p$ -algebras. $\square$

In general, we are interested in the case where Γ is identified with an open subgroup of $\mathbb{Z}_p^* \cong C_{p-1} \times (1 + p\mathbb{Z}_p)$, which is always procyclic, since $\mathbb{Z}_p^*$ is. Therefore, Γ takes the form $C_n \times \mathbb{Z}_p$ where $n \mid p-1$. Then we can show that for γ a topological generator of Γ

$$\begin{aligned} \kappa' : \Lambda[\Gamma] &\xrightarrow{\sim} \mathbb{Z}_p[[T]][C_n] \\ \gamma &\longmapsto (1+T)\zeta. \end{aligned}$$

is an isomorphism, where $\mathbb{Z}_p[[T]][C_n]$ is the group ring of C_n with coefficients in $\mathbb{Z}_p[[T]]$ and ζ is a generator of C_n .

Remark 2.4.3. If $\Gamma \cong \mathbb{Z}_p$ and $\sigma = \gamma^a \in \Gamma$, $a \in \mathbb{Z}_p$, we have that $\frac{\sigma-1}{\gamma-1} = \frac{((\gamma-1)+1)^a-1}{\gamma-1} \in \Lambda[\Gamma] = \mathbb{Z}_p[[\gamma-1]]$ as a power series in $\gamma-1$.

Remark 2.4.4. Let X be a p -adically complete and complete topological group with a continuous Γ -action, such that $(\gamma-1)(U_i) \subset U_i$ for a basis $(U_i)_{i \in I}$ of neighbourhoods of 0 in X .

We show that we can extend the action on X to $\Lambda[\Gamma]$. To see that it satisfies to show that for all $x \in X$, $(\gamma-1)^r(x)$ tends to zero. In fact, for $x \in X$ we have that

$$(\gamma-1)^{p^n}(x) = \gamma^{p^n}(x) - \binom{p^n}{p^n-1} \gamma^{p^n-1}(x) + \cdots - x.$$

We see that all in between terms are divisible by p^n and $\gamma^{p^n}(x) - x$ tends to zero since $\langle \gamma^{p^n} \rangle$ are precisely the open normal subgroups of Γ , so γ^{p^n} tends to the identity of Γ . For any neighbourhood U of 0 in X , let n be such that $(\gamma-1)^{p^n}(x) \in U$, then, since $(\gamma-1)(U_i) \subset U_i$, we have $(\gamma-1)^r(x) \in U$ for all $r \geq p^n$.

Since X is complete, we see that all power series in $\gamma-1$ act on X and thus $\Lambda[\Gamma]$ acts on X continuously.

2.5 Witt vectors

We briefly review the construction of Witt vectors since this will be needed later. Let p be a prime number, and let $\underline{X} = (X_0, X_1, \dots)$ and $\underline{Y} = (Y_0, Y_1, \dots)$ be indeterminates.

Definition 2.5.1. The n -th *Witt polynomial* of $\underline{X}$ is

$$W_n(\underline{X}) = W_n(X_0, \dots, X_n) := \sum_{i=0}^n p^i X_i^{p^{n-i}} = X_0^{p^n} + pX_1^{p^{n-1}} + \cdots + p^n X_n.$$

Note that we can express $X_n \in \mathbb{Z}[\frac{1}{p}][W_0, \dots, W_n]$.

Lemma 2.5.2 ([GS13], II.6 Theorem 6). For every $\Phi \in \mathbb{Z}[X, Y]$ there exist unique $\Phi_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$ such that for all $n \in \mathbb{N}$

$$W_n(\Phi_0, \dots, \Phi_n) = \Phi(W_n(X_0, \dots, X_n), W_n(Y_0, \dots, Y_n)).$$

Definition 2.5.3. We define the polynomials $S_n, P_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$ as the polynomials associated to $X+Y \in \mathbb{Z}[X, Y]$ and $XY \in \mathbb{Z}[X, Y]$, i.e. inductively define via

$$\begin{aligned} W_n(S_0, \dots, S_n) &= W_n(X_0, \dots, X_n) + W_n(Y_0, \dots, Y_n) \\ W_n(P_0, \dots, P_n) &= W_n(X_0, \dots, X_n) \cdot W_n(Y_0, \dots, Y_n). \end{aligned}$$

For $\lambda \in \mathbb{Z}$ we further define $M(\lambda)_n \in \mathbb{Z}[X_0, \dots, X_n; Y_0, \dots, Y_n]$ to be the polynomials associated to $\lambda X \in \mathbb{Z}[X]$, i.e. inductively define via

$$W_n(M(\lambda)_0, \dots, M(\lambda)_n) = \lambda W_n(X_0, \dots, X_n)$$

Remark 2.5.4. We can derive explicit formulas for the case $n = 1$. By definition we have

$$S_0^p + pS_1 = X_0^p + pX_1 + Y_0^p + pY_1$$

and

$$P_0^p + pP_1 = (X_0^p + pX_1)(Y_0^p + pY_1).$$

We thus get

$$\begin{aligned} S_0 &= X_0 + Y_0 \\ S_1 &= X_1 + Y_1 + \frac{1}{p} \sum_{i=1}^{n-1} \binom{p}{i} X_0^i Y_0^{n-i} \\ P_0 &= X_0 Y_0 \\ P_1 &= pX_1 Y_1 + Y_0^p X_1 + Y_1 X_0^p. \end{aligned}$$

Proposition 2.5.5 ([FO22], Lemma 1.28). We assign the weight p^i to the i -th coordinate X_i . Then for $n \in \mathbb{N}$

1. $S_n = X_n + Y_n + \text{terms of degree } \geq 2$, of which all polynomials have weight p^n .
2. $P_n = p^n X_n Y_n + \text{terms of degree } \geq 3$, of which all polynomials have weight $2p^n$.
3. $P_n(X_0, 0, \dots, 0; Y_0, \dots, Y_n) = X_0^{p^n} Y_n$.
4. $M(p)_n \equiv X_{n-1}^p \pmod{p}$ for $n \geq 1$

Let A be an arbitrary commutative ring.

Definition 2.5.6. Let $n \in \mathbb{N}$. We define the ring of Witt vectors of length n , denoted by $W_n(A)$ with coefficients in A to be the set A^n with addition and multiplication, for $a = (a_0, \dots, a_{n-1}), b = (b_0, \dots, b_{n-1}) \in A^n$ given by

$$\begin{aligned} a + b &= (S_0(a, b), \dots, S_{n-1}(a, b)) \\ a \cdot b &= (P_0(a, b), \dots, P_{n-1}(a, b)). \end{aligned}$$

We further define restriction maps

$$\begin{aligned} \text{res} : W_{n+1}(A) &\longrightarrow W_n(A) \\ (a_0, \dots, a_n) &\longmapsto (a_0, \dots, a_{n-1}) \end{aligned}$$

which are surjective ringhomomorphisms. We then define

$$W(A) = \varprojlim W_n(A).$$

If A is a topological ring, we equip $W_n(A)$ with the product topology of A^n and $W(A)$ with the inverse limit topology.

Remark 2.5.7. Note that $W(A)$ has a $\mathbb{Z}$ -module structure via, for $a = (a_0, a_1, \dots) \in W(A)$ and $\lambda \in \mathbb{Z}$

$$\lambda(a_0, a_1, \dots) = (M(\lambda)_0(a), M(\lambda)_1(a), \dots)$$

which is compatible with the additive structure we defined.

Example 2.5.8 ([GS13], II.6). $W(\mathbb{F}_p) = \mathbb{Z}_p$ and $W_n(\mathbb{F}_p) = \mathbb{Z}/p^n\mathbb{Z}$.

Definition 2.5.9. (i) We define the *Verschiebung*

$$\begin{aligned} V : W(A) &\longrightarrow W(A) \\ (a_0, a_1, \dots) &\longmapsto (0, a_0, a_1, \dots). \end{aligned}$$

(ii) We define the *Teichmüller map*

$$\begin{aligned} [\cdot] : A &\longrightarrow W(A) \\ a &\longmapsto (a, 0, 0, \dots). \end{aligned}$$

(iii) If A is of characteristic p , we define the Witt vector *Frobenius map*

$$\begin{aligned} \sigma : W(A) &\longrightarrow W(A) \\ (a_0, a_1, \dots) &\longmapsto (a_0^p, a_1^p, \dots). \end{aligned}$$

Proposition 2.5.10. Let A be a commutative ring of characteristic p .

1. $(a_0, a_1, \dots) = \sum_{n=0}^{\infty} V^n([a_n])$
2. $V\sigma = \sigma V = p$

Proof. The first statement is clear. For the second, note that A has characteristic p so Proposition 2.5.5 implies that

$$p(a_0, a_1, \dots) = (0, a_0^p, a_1^p, \dots). \quad \square$$

Definition 2.5.11. We call a ring A of characteristic p *perfect* if the Frobenius map $x \mapsto x^p$ is an automorphism.

Corollary 2.5.12. If A is a perfect ring, then every element in $W(A)$ can be written uniquely in the form

$$(a_0, a_1, \dots) = \sum_{n=0}^{\infty} p^n [a_n^{p^{-n}}]$$

Further, the projection map

$$\begin{aligned} W(A) &\longrightarrow A \\ (a_0, a_1, \dots) &\longmapsto a_0 \end{aligned}$$

induces an isomorphism

$$W(A)/pW(A) \cong A.$$

Proof. By Proposition 2.5.10 we have $V^n(\sigma^n[a_n]) = V^n([a_n^{p^n}]) = p^n[a_n]$, and since the p -th power map is an automorphism we have $V^n([a_n]) = p^n[a_n^{p^{-n}}]$. It follows from Proposition 2.5.10 that

$$(a_0, a_1, \dots) = \sum_{n=0}^{\infty} V^n([a_n]) = \sum_{n=0}^{\infty} p^n [a_n^{p^{-n}}]$$

Using this description we see that the kernel of the projection map

$$\begin{aligned} W(A) &\longrightarrow A \\ (a_0, a_1, \dots) &\longmapsto a_0 \end{aligned}$$

is precisely $pW(A)$ and we obtain the remaining statement. $\square$

Chapter 3

p -adic Hodge theory and (φ, Γ) -modules

3.1 Preliminaries

We fix some notation for the rest of the thesis. We will always consider p an odd prime.

Let K be a local field with finite residue field k with characteristic p which implies that K is a finite extension of $\mathbb{Q}_p$. Let $K_0 = \text{Frac } W(k)$ be the field of fractions of the ring of Witt vectors of k . Then K_0 is the unique extension $\mathbb{Q}_p \subset K_0 \subset K$ such that $\mathbb{Q}_p \subset K_0$ is unramified and $K_0 \subset K$ is totally ramified. Also fix an algebraic closure $\bar{\mathbb{Q}}_p$ of $\mathbb{Q}_p$, which we will also denote by $\bar{K}$.

We choose and fix ζ_{p^n} compatible primitive p^n -th roots of unity, in the sense of $\zeta_{p^{n+1}}^p = \zeta_{p^n}$. We denote by μ_{p^n} the group of p^n -th roots of unity in $\bar{\mathbb{Q}}_p$ and set $\mu_{p^\infty} := \bigcup_{n \geq 1} \mu_{p^n}$. Let $K_n := K(\mu_{p^n})$, $K_{0n} := K_0(\mu_{p^n})$, and similarly, $K_\infty = K(\mu_{p^\infty})$, and $K_{0\infty} = K_0(\mu_{p^\infty})$.

$$\begin{array}{c}
 \bar{K} \\
 \left| \begin{array}{c} \left. \begin{array}{c} \bar{K} \\ K_\infty \end{array} \right\} H_K \\ K_\infty \\ \left. \begin{array}{c} K_\infty \\ K \end{array} \right\} \Gamma_K \\ K \\ K_0 \end{array} \right. \\
 K \\
 \left| \right. \\
 K_0
 \end{array}$$

We denote by $H_K := \text{Gal}(\bar{K}/K_\infty)$ and $\Gamma_K := \text{Gal}(K_\infty/K)$. Further, we denote

the cyclotomic character by $\chi : G_K \rightarrow \mathbb{Z}_p^*$, with $g(\zeta_{p^i}) = \zeta_{p^i}^{\chi(g)}$ for all $g \in G_K$ and $\zeta_{p^i} \in \mu_{p^\infty}$. We see that $\ker(\chi) = H_K$, so χ identifies Γ_K with an open subgroup of $\mathbb{Z}_p^*$. Also note that $\mathbb{Z}_p^* \cong (C_{p-1}) \times (1 + p\mathbb{Z}_p)$ and $\mathbb{Z}_p^* = \varprojlim (\mathbb{Z}/p\mathbb{Z})^*$ is procyclic.

Remark 3.1.1. Since $\mathbb{Z}_p^*$ is procyclic, we can find a topological generator, i.e. an element that generates a dense subgroup. Indeed, we can pick any lift of a generator of $(\mathbb{Z}/p^2\mathbb{Z})^*$. It will generate $(\mathbb{Z}/p^2\mathbb{Z})^*$ and also $(\mathbb{Z}/p^n\mathbb{Z})^*$ so it generates a dense subset.

3.2 Fields of Norms

Our first goal is to establish an isomorphism between H_K and the absolute Galois group $\text{Gal}(X_{K_\infty/K}^{\text{sep}}/X_{K_\infty/K})$ of a field of characteristic p , called the *field of norms* $X_{K_\infty/K}$. This was introduced by Fontaine and Wintenberger in [FW79]. Let

$$X_{K_\infty/K} = \varprojlim K_n$$

where $K_n = K(\mu_{p^n})$ and the transition maps are given by the norm maps $\mathcal{N}_{K_n/K_{n-1}}$.

If we consider elements of $X_{K_\infty/K}$ as sequences $x = (x^{(0)}, x^{(1)}, \dots)$ then the multiplication and addition are given by:

$$\begin{aligned} (xy)^{(n)} &= x^{(n)}y^{(n)} \\ (x+y)^{(n)} &= \lim_{m \rightarrow +\infty} \mathcal{N}_{K_{n+m}/K_n}(x^{(n+m)} + y^{(n+m)}) \end{aligned}$$

We can see that this field has characteristic p since

$$\lim_{m \rightarrow +\infty} \mathcal{N}_{K_{n+m}/K_n}(p) = \lim_{m \rightarrow +\infty} p^{p^{m-n}-1} = 0.$$

If now L/K is a finite Galois extension then $X_{L_\infty/L}/X_{K_\infty/K}$ is a finite Galois extension with $\text{Gal}(X_{L_\infty/L}/X_{K_\infty/K}) \cong \text{Gal}(L_\infty/K_\infty)$ where $L_\infty = L(\mu_{p^\infty})$. We deduce that $\text{Gal}(X_{K_\infty/K}^{\text{sep}}/X_{K_\infty/K}) \cong H_K$, where $X_{K_\infty/K}^{\text{sep}}$ denotes a separable closure of $X_{K_\infty/K}$.

For K an unramified extension of $\mathbb{Q}_p$, one can show that

$$X_{K_\infty/K} = k((\pi_K))$$

where π_K is some uniformizer of $X_{K_\infty/K}$. The local uniformizer we choose will generally be $\pi_K = (1, \zeta_p, \zeta_{p^2}, \dots) - (1, 1, \dots)$.

3.3 Characteristic p theory: φ -modules

We start by classifying representations of $G_{\mathbf{E}} = \text{Gal}(\mathbf{E}^{\text{sep}}/\mathbf{E})$ for some characteristic p field $\mathbf{E}$. So let $\mathbf{E}$ be a local field of characteristic p and let $\sigma : \mathbf{E}^{\text{sep}} \rightarrow \mathbf{E}^{\text{sep}}, x \mapsto x^p$

denote the absolute Frobenius. Note that the field of norms will in general not be a perfect field.

3.3.1 $\mathbb{F}_p$ -representations of $G_{\mathbf{E}}$

We will establish an equivalence between $\mathbb{F}_p$ -representations of $G_{\mathbf{E}}$ and so-called étale φ -modules over $\mathbf{E}$.

Definition 3.3.1. A φ -module over $\mathbf{E}$ is a finite-dimensional $\mathbf{E}$ -vector space M equipped with a σ -semilinear map

$$\varphi : M \longrightarrow M$$

where by σ -semilinear we mean that $\varphi(\lambda m) = \sigma(\lambda)\varphi(m) = \lambda^p \varphi(m)$, for all $\lambda \in \mathbf{E}$, $m \in M$. A φ -module M is called *étale* if $\varphi(M)$ generates M as an $\mathbf{E}$ -vector space. We denote the category of étale φ -modules over $\mathbf{E}$ by $\Phi M_{\mathbf{E}}^{\text{ét}}$.

Theorem 3.3.2 ([Fon07], 1.2.6). There is an equivalence of categories

$$\{\mathbb{F}_p\text{-representations of } G_{\mathbf{E}}\} \longleftrightarrow \{\text{étale } \varphi\text{-modules over } \mathbf{E}\}.$$

The functor is given by $V \mapsto \mathbf{D}(V)$ where $\mathbf{D}(V) = (\mathbf{E}^{\text{sep}} \otimes_{\mathbb{F}_p} V)^{G_{\mathbf{E}}}$ and the inverse functor is given by $\mathbf{D} \mapsto V(\mathbf{D}) = (\mathbf{E}^{\text{sep}} \otimes_{\mathbf{E}} \mathbf{D})^{\varphi=1}$.

Remark 3.3.3. We want to briefly explore why the functor is well-defined. Given a $\mathbb{F}_p$ -representation V we need to show that $\mathbf{D}(V)$ is a finite dimensional $\mathbf{E}$ -vector space and that the module is étale.

We start by regarding the structure of $\mathbf{D}(V)$. We have a diagonal action of $G_{\mathbf{E}}$ on $\mathbf{E}^{\text{sep}} \otimes_{\mathbb{F}_p} V$ and $\mathbf{E} \subset \mathbf{E}^{\text{sep}}$ is clearly fixed by $G_{\mathbf{E}}$. Thus we get a $\mathbf{E}$ -vector space structure. Furthermore, we have a σ -semilinear map given by $x \otimes v \mapsto x^p \otimes v$ on $\mathbf{E}^{\text{sep}} \otimes_{\mathbb{F}_p} V$ which commutes with the $G_{\mathbf{E}}$ -action so we get a Frobenius φ on $\mathbf{D}(V)$.

Using Hilbert 90 one can show that

$$\mathbf{E}^{\text{sep}} \otimes_{\mathbf{E}} \mathbf{D}(V) \xrightarrow{\sim} \mathbf{E}^{\text{sep}} \otimes_{\mathbb{F}_p} V \quad (3.1)$$

which shows $\dim_{\mathbf{E}} \mathbf{D}(V) < \infty$ since the RHS is clearly finite dimensional over $\mathbf{E}$.

To show that the $\mathbf{D}(V)$ is étale let $v_1, \dots, v_d$ be a basis of V over $\mathbb{F}_p$, write v_i for $1 \otimes v_i \in \mathbf{D}(V)$ and let $e_1, \dots, e_d$ a basis of $\mathbf{D}(V)$ over $\mathbf{E}$. We can write

$$e_i = \sum_{j=1}^d a_{ij} v_j$$

with $a_{ij} \in \mathbf{E}^{\text{sep}}$. By (3.1) we have that $e_1, \dots, e_d$ also generates $\mathbf{E}^{\text{sep}} \otimes V$ over $\mathbf{E}^{\text{sep}}$ thus $A = (a_{ij})_{i,j} \in \text{GL}_d(\mathbf{E}^{\text{sep}})$.

We have

$$\varphi(e_i) = \varphi\left(\sum_{j=1}^d a_{ij} v_j\right) = \sum_{j=1}^d \sigma(a_{ij}) \varphi(1 \otimes v_j) = \sum_{j=1}^d a_{ij}^p v_j = \sum_{j=1}^d b_{ij} e_j$$

where $B = (b_{ij})_{i,j}$ satisfies for all i

$$\sum_{j=1}^d a_{ij}^p v_j = \sum_{j=1}^d b_{ij} \sum_{k=1}^d a_{jk} v_k$$

so we deduce by uniqueness of coefficients that $BA = \varphi(A)$, where the φ -action on A is taken componentwise. Since we are in characteristic p , $\det(\varphi(A)) = \det(A)^p$ and thus $\det(B) = \det(A)^{p-1} \neq 0$. We conclude that $\varphi(\mathbf{D}(V))$ generates $\mathbf{D}(V)$ over $\mathbf{E}$, i.e. $\mathbf{D}(V)$ is étale.

Remark 3.3.4. We note that by (3.1) we also have

$$(\mathbf{E}^{\text{sep}} \otimes_{\mathbf{E}} \mathbf{D}(V))^{\varphi=1} \cong (\mathbf{E}^{\text{sep}} \otimes_{\mathbb{F}_p} V)^{\varphi=1} = V$$

since $\mathbf{E}^{\text{sep} \varphi=1} = \mathbb{F}_p$. We see that the functor $\mathbf{D} \mapsto V(\mathbf{D}) = (\mathbf{E}^{\text{sep}} \otimes_{\mathbf{E}} \mathbf{D})^{\varphi=1}$ indeed gives back the original representation.

3.3.2 $\mathbb{Z}_p$ -representations of $G_{\mathbf{E}}$

We lift the theory to the $\mathbb{Z}_p$ case. Again, let $\mathbf{E}$ be a not necessarily perfect local field of characteristic p . We start by picking a Cohen ring for $\mathbf{E}$, i.e. a DVR which we call $\mathbf{A}$ such that the residue field is given by $\mathbf{E}$. We denote its field of fractions by $\mathbf{B}$.

Remark 3.3.5. If $\mathbf{E}$ were perfect there is a unique choice of such a ring given by the Witt vectors $W(\mathbf{E})$. If $\mathbf{E}$ is not perfect there will be a choice involved, however we will later, for some $\mathbf{E}$, fix a convenient choice of a Cohen ring.

Now let $\hat{\mathbf{B}}$ be the p -adic completion of a maximal unramified extension of $\mathbf{B}$, with ring of integers $\hat{\mathbf{A}}$. The residue field of $\hat{\mathbf{A}}$ is $\mathbf{E}^{\text{sep}}$. Thus we have $\text{Gal}(\hat{\mathbf{B}}/\mathbf{B}) \cong \text{Gal}(\mathbf{E}^{\text{sep}}/\mathbf{E})$. We see that under this isomorphism we have a Galois action of $\text{Gal}(\mathbf{E}^{\text{sep}}/\mathbf{E})$ on $\hat{\mathbf{A}}$ and the invariants under $\text{Gal}(\mathbf{E}^{\text{sep}}/\mathbf{E})$ are $\mathbf{A}$.

Then there is a similar equivalence of categories:

Theorem 3.3.6 ([Fon07], 1.2.6). There is an equivalence of categories

$$\{\mathbb{Z}_p\text{-representations of } G_{\mathbf{E}}\} \longleftrightarrow \{\text{étale } \varphi\text{-modules over } \mathbf{A}\}.$$

The functor is given by $V \mapsto \mathbf{D}(V)$ where $\mathbf{D}(V) = (\hat{\mathbf{A}} \otimes_{\mathbb{Z}_p} V)^{G_{\mathbf{E}}}$.

Through lifting the absolute Frobenius $\sigma : \mathbf{E}^{\text{sep}} \rightarrow \mathbf{E}^{\text{sep}}, x \mapsto x^p$ on $\mathbf{E}^{\text{sep}}$ we obtain a Frobenius on $\hat{\mathbf{A}}$. On $\mathbf{A}$ it acts as the lift of the absolute Frobenius on $\mathbf{E}$. Then,

$\varphi : \hat{\mathbf{A}} \otimes_{\mathbb{Z}_p} V \rightarrow \hat{\mathbf{A}} \otimes_{\mathbb{Z}_p} V$, $a \otimes v \mapsto \sigma(a) \otimes v$ commutes with the $G_{\mathbf{E}}$ action and gives us a Frobenius on $\mathbf{D}(V)$.

Example 3.3.7. We have a quick look at the trivial representation $V = \mathbb{Z}_p$ of $G_{\mathbf{E}}$. Then $\mathbf{D}(V) = \hat{\mathbf{A}}^{G_{\mathbf{E}}} = \mathbf{A}$.

The inverse functor gives us $(\hat{\mathbf{A}} \otimes_{\mathbf{A}} \mathbf{A})^{\varphi=1} = \hat{\mathbf{A}}^{\varphi=1}$. Since the Frobenius on the residue field $\mathbf{E}^{\text{sep}}$ is just given as the usual absolute Frobenius σ and $\mathbf{E}^{\text{sep}\sigma=1} = \mathbb{F}_p$, we see that $\hat{\mathbf{A}}^{\varphi=1}/(p) = (\hat{\mathbf{A}}/(p))^{\varphi=1} \mathbb{F}_p$ and thus $\hat{\mathbf{A}}^{\varphi=1} = \mathbb{Z}_p = V$.

3.4 Rings of periods

Recall that K a finite extension of $\mathbb{Q}_p$ with residue field k and $K_0 = W(k)[1/p]$ is an unramified extension of $\mathbb{Q}_p$.

Let $\mathbf{C}$ be the p -adic completion of $\bar{K} = \bar{\mathbb{Q}}_p$. We define

$$\tilde{\mathbf{E}}^+ = \varprojlim_{x \mapsto x^p} \mathcal{O}_{\mathbf{C}}/(p)$$

where the transition maps are given by the p -th power map, i.e. $\tilde{\mathbf{E}}^+ = \{(x^{(0)}, x^{(1)}, \dots) : x^{(i+1)p} = x^{(i)}\}$ which is a perfect local ring of characteristic p . We let $\tilde{\mathbf{E}}$ be its field of fractions. Then $\tilde{\mathbf{E}}$ is an algebraically closed field of characteristic p , complete for the valuation given by $v_{\mathbf{E}}(x) = v_p(x^{(0)})$. Further, there is a componentwise G_{K_0} - or G_K -action on $\tilde{\mathbf{E}}$, since we can extend the action of G_{K_0} or G_K continuously on $\mathbf{C}$. If we consider elements of $\tilde{\mathbf{E}}$ as sequences $x = (x^{(0)}, x^{(1)}, \dots)$ then the multiplication and addition are given by:

$$\begin{aligned} (xy)^{(n)} &= x^{(n)}y^{(n)} \\ (x+y)^{(n)} &= \lim_{m \rightarrow +\infty} (x^{(n+m)} + y^{(n+m)})p^m \end{aligned}$$

Let $\varepsilon = (1, \zeta_p, \zeta_{p^2}, \dots) \in \tilde{\mathbf{E}}^+$. We then find the subfield $\mathbf{E}_{K_0} = k((\varepsilon - 1)) \subset \tilde{\mathbf{E}}$, where we note that all power series converge in $\tilde{\mathbf{E}}$ since $v_{\mathbf{E}}(\varepsilon - 1) = \frac{p}{p-1}$. $\mathbf{E}_{K_0}$ is then exactly the field of norms $X_{K_0\infty/K_0}$ of $K_0 \subset K_0\infty$.

Let $\mathbf{E}$ be a separable closure of $\mathbf{E}_{K_0}$ inside $\tilde{\mathbf{E}}$. We then have that $\text{Gal}(\mathbf{E}/\mathbf{E}_{K_0}) \cong H_{K_0}$ by the theory of the field of norms. In fact, the componentwise H_{K_0} -action coincides with the $\text{Gal}(\mathbf{E}/\mathbf{E}_{K_0})$ -action via this isomorphism.

Notation. We fix some general guidelines for notation to avoid confusion.

1. The subscript $_K$ will be used to denote invariants under H_K .
2. The superscript $^+$ will be used to denote taking only positive powers in power series. In some cases this results in being the ring of integers.

Now let

$$\begin{aligned}\mathbf{E}_K &= \mathbf{E}^{H_K} \\ \tilde{\mathbf{E}}_K &= \tilde{\mathbf{E}}^{H_K} \\ \mathbf{E}_K^+ &= \mathbf{E}^{+H_K} \\ \tilde{\mathbf{E}}_K^+ &= \tilde{\mathbf{E}}^{+H_K}.\end{aligned}$$

Note that $\mathbf{E}_K^+$ is precisely the ring of integers in $\mathbf{E}_K$ and $\tilde{\mathbf{E}}_K^+$ of $\tilde{\mathbf{E}}$. As we discussed in a previous section, since K is a finite extension of K_0 the field of norms $X_{K_\infty/K}$ is a finite Galois extension of $X_{K_{0\infty}/K_0} = \mathbf{E}_{K_0}$ with $\text{Gal}(X_{K_\infty/K}/\mathbf{E}_{K_0}) \cong \text{Gal}(K_\infty/K_{0\infty}) = H_{K_0}/H_K$. This in particular shows that

$$\mathbf{E}_K = \mathbf{E}^{H_K} = X_{K_\infty/K},$$

i.e. $\mathbf{E}_K$ is the field of norms associated to $K \subset K_\infty$ and $\text{Gal}(\mathbf{E}/\mathbf{E}_K) \cong H_K$.

Let $\tilde{\mathbf{A}} = W(\tilde{\mathbf{E}})$, and $\tilde{\mathbf{B}} = \tilde{\mathbf{A}}[1/p]$ its field of fractions. Then we can write any element $x \in \tilde{\mathbf{A}}$ in the form $\sum_{i=0}^{+\infty} p^i[x_i]$ and any element $x \in \tilde{\mathbf{B}}$ in the form $\sum_{i=-n}^{+\infty} p^i[x_i]$ for some $n \in \mathbb{N}$. For $k \in \mathbb{Z}$ we define

$$w_k(x) = \inf_{i \leq k} v_{\mathbf{E}}(x_i)$$

for $x = \sum_{i=-n}^{+\infty} p^i[x_i] \in \tilde{\mathbf{B}}$.

Proposition 3.4.1 ([CC98], I.3). Let $x = \sum_{i=0}^{+\infty} p^i[x_i] \in \tilde{\mathbf{A}}$ and $y = \sum_{i=0}^{+\infty} p^i[y_i] \in \tilde{\mathbf{A}}$.

- (i) $w_k(x) = +\infty \iff x \in p^{k+1}\tilde{\mathbf{A}}$
- (ii) $w_k(x+y) \geq \inf\{w_k(x), w_k(y)\}$ with equality of $w_k(x) \neq w_k(y)$
- (iii) $w_k(xy) \geq \inf_{i+j \leq k} w_i(x) + w_j(y)$
- (iv) $w_k(\varphi(x)) = pw_k(x)$

Proof. (i) If $x = \sum_{i=0}^{+\infty} p^i[x_i]$ and $w_k(x) = \inf_{i \leq k} v_{\mathbf{E}}(x_i) = +\infty$ then $[x_i] = 0$ for $i \leq k$ and thus $x \in p^{k+1}\tilde{\mathbf{A}}$.

(ii) Recall the definition of addition of Witt vectors. We have

$$x + y = \sum_{i=0}^{+\infty} p^i[z_i]$$

where $z_i = S_i(x_0, \dots, x_i^{p^i}; y_0, \dots, y_i^{p^i})^{p^{-i}}$ and S_i are the addition polynomials of the Witt vectors. From Lemma 2.5.4 we see that $S_i = x_i^{p^i} + y_i^{p^i} + Q_i$, where Q_i

is some polynomial in $x_0, \dots, x_i^{p^i}; y_0, \dots, y_i^{p^i}$ with only monomials of weight p^i in the sense of Proposition 2.5.4. Note that

$$v_{\mathbf{E}}(Q_i^{p^{-i}}) \geq \inf_{j \leq i} \inf \{v_{\mathbf{E}}(x_j), v_{\mathbf{E}}(y_j)\}$$

since Q_i consists of monomials of weight p^i . We can deduce that

$$\begin{aligned} v_{\mathbf{E}}(z_i) &= \inf \{v_{\mathbf{E}}(x_i), v_{\mathbf{E}}(y_i), v_{\mathbf{E}}(Q_i)\} \\ &\geq \inf_{j \leq i} \inf \{v_{\mathbf{E}}(x_j), v_{\mathbf{E}}(y_j)\} \end{aligned}$$

and we have $v_{\mathbf{E}}(z_i) = v_{\mathbf{E}}(x_i)$ or $v_{\mathbf{E}}(y_i)$ if $v_{\mathbf{E}}(x_i) = \inf_{j \leq i} \inf \{v_{\mathbf{E}}(x_j), v_{\mathbf{E}}(y_j)\}$ or $v_{\mathbf{E}}(y_i) = \inf_{j \leq i} \inf \{v_{\mathbf{E}}(x_j), v_{\mathbf{E}}(y_j)\}$. We deduce

$$w_k(x + y) = \inf_{i \leq k} v_{\mathbf{E}}(z_i) \geq \inf_{i \leq k} \inf \{v_{\mathbf{E}}(x_i), v_{\mathbf{E}}(y_i)\} = \inf \{w_k(x), w_k(y)\}.$$

Further we have that if $w_k(x) \neq w_k(y)$, say $w_k(x) < w_k(y)$, then there exists some i_0 such that x_{i_0} satisfies $v_{\mathbf{E}}(x_{i_0}) = w_k(x) = \inf_{i \leq k} \inf \{v_{\mathbf{E}}(x_i), v_{\mathbf{E}}(y_i)\}$. Then we see

$$w_k(x + y) = v_{\mathbf{E}}(z_{i_0}) = v_{\mathbf{E}}(x_{i_0}) = w_k(x).$$

(iii) For $x, y \in \tilde{\mathbf{A}}$ we have

$$\begin{aligned} w_k(xy) &= w_k\left(\sum_{i=0}^{+\infty} p^i \left(\sum_{l+m=i} [x_l][y_m]\right)\right) \\ &= w_k\left(\sum_{i=0}^k p^i \left(\sum_{l+m=i} [x_l][y_m]\right)\right) \\ &\geq \inf_{i \leq k, l+m=i} v_{\mathbf{E}}(x_l) + v_{\mathbf{E}}(y_m) \\ &\geq \inf_{i+j \leq k} w_i(x) + w_j(y). \end{aligned}$$

Note that in the third row we used the discussion from (ii) for the valuations of sums of elements in $\tilde{\mathbf{A}}$.

(iv)

$$w_k(\varphi(x)) = w_k\left(\sum_{i=-n}^{+\infty} p^i [x_i^p]\right) = \inf_{i \leq k} v_{\mathbf{E}}(x_i^p) = pw_k(x). \quad \square$$

We describe the topology of $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$. For $r \in \mathbb{R}$ and $k \in \mathbb{N}$ set $U_{r,k} := \{x \in \tilde{\mathbf{A}} \mid w_k(x) \geq r\}$. We then equip $\tilde{\mathbf{A}}$ with the topology given by $U_{r,k}$ as a basis of closed neighbourhood around 0. We equip $\tilde{\mathbf{B}} = \bigcup_{n \in \mathbb{N}} p^{-n} \tilde{\mathbf{A}}$ with the inductive limit topology. Then $\tilde{\mathbf{B}}$ is a complete local field with ring of integers $\tilde{\mathbf{A}}$ and residue field $\tilde{\mathbf{E}}$. Further, we the G_K -action on $\tilde{\mathbf{E}}$ induces a continuous G_K -action on $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ given by $g(\sum_{i=-n}^{+\infty} p^i [x_i]) = \sum_{i=-n}^{+\infty} p^i [g(x_i)]$. The action clearly still commutes with the lifted Frobenius.

Remark 3.4.2. The topology on $\tilde{\mathbf{A}}$ makes $\tilde{\mathbf{A}} \longrightarrow \tilde{\mathbf{E}}^{\mathbb{N}}$, $x \longmapsto (x_k)_{k \in \mathbb{N}}$ a homeomorphism.

Recall that $\varepsilon = (1, \zeta_p, \zeta_{p^2}, \dots) \in \tilde{\mathbf{E}}^+$ and let $\pi := [\varepsilon] - 1 \in W(\tilde{E})$.

By construction of Witt vectors and their Frobenius action we get that $\sigma([x]) = [x^p]$. Therefore $\sigma([\varepsilon]) = [\varepsilon^p]$ and

$$\sigma(\pi) = \sigma([\varepsilon]) - \sigma(1) = [\varepsilon^p] - 1 = (1 + \pi)^p - 1$$

Note that we used $[\varepsilon^p] = [\varepsilon]^p$ since the Teichmüllermap is multiplicative.

Let $g \in G_K$. Let $\chi : G_K \longrightarrow \mathbb{Z}_p^*$ be the cyclotomic character, i.e. $g(\zeta_{p^i}) = \zeta_{p^i}^{\chi(g)}$ for all $g \in G_K$ and $\zeta_{p^i} \in \mu_{p^\infty}$. Then we have

$$\begin{aligned} g(\pi) &= g([\varepsilon] - 1) = [g(\varepsilon)] - 1 \\ &= [(g(1), g(\zeta_p), g(\zeta_{p^2}), \dots)] - 1 \\ &= [(1^{\chi(g)}, \zeta_p^{\chi(g)}, \zeta_{p^2}^{\chi(g)}, \dots)] - 1 \\ &= [\varepsilon^{\chi(g)}] - 1 \\ &= (1 + \pi)^{\chi(g)} - 1. \end{aligned}$$

Let $\mathbf{A}_{K_0}$ denote the closure of $\mathcal{O}_{K_0}[\pi, \frac{1}{\pi}]$ in $\tilde{\mathbf{A}}$ which takes the form

$$\mathbf{A}_{K_0} = \left\{ \sum_{n \in \mathbb{Z}} a_n \pi^n : a_n \in \mathcal{O}_{K_0}, a_n \rightarrow 0 \text{ for } n \rightarrow -\infty \right\}$$

and is a discrete valuation p -ring with residue field $\mathbf{E}_{K_0} = k((\epsilon - 1))$. We have the Frobenius and G_K -action given by

$$\begin{aligned} \sigma(\pi) &= (1 + \pi)^p - 1, & \sigma(a) &= \text{Frob}_{K_0/\mathbb{Q}_p}(a) \text{ for } a \in \mathcal{O}_{K_0} \\ g(\pi) &= (1 + \pi)^{\chi(g)} - 1, & g(a) &= a \text{ for } a \in \mathcal{O}_{K_0}, g \in G_K \end{aligned}$$

where $\text{Frob}_{K_0/\mathbb{Q}_p} \in \text{Gal}(K_0/\mathbb{Q}_p)$ denotes the relative Frobenius of the unramified field extension $K_0/\mathbb{Q}_p$. In particular, we see that $\mathbf{A}_{K_0}$ and its field of fractions $\mathbf{B}_{K_0}[1/p]$ are closed under Frobenius and G_K -action. Note that the inherited topology on $\mathbf{A}_{K_0}$ is given by $p^n \mathbf{A}_{K_0} + T^n \mathbf{A}_{K_0}^+$ as open set around 0.

Now let $\mathbf{B}$ be the completion of the maximal unramified extension of $\mathbf{B}_{K_0}$ inside $\tilde{\mathbf{B}}$ and $\mathbf{A} = \mathbf{B} \cap \tilde{\mathbf{A}}$. Then $\mathbf{A}$ is a complete DVR with field of fractions $\mathbf{B} = \mathbf{A}[1/p]$ and residue field $\mathbf{E}$. Using that $\mathbf{E}$ is stable under Frobenius and G_K -action we can show that $\mathbf{A}$ and $\mathbf{B}$ are stable under Frobenius and G_K -action. Further, since $\mathbf{B}$ is the completion of the maximal unramified extension of $\mathbf{B}_{K_0}$, its Galois group is isomorphic to the one of the residue fields, i.e. $\text{Gal}(\mathbf{B}/\mathbf{B}_{K_0}) \cong \text{Gal}(\mathbf{E}/\mathbf{E}_{K_0}) \cong H_{K_0}$.

Similarly to before, let

$$\begin{aligned}\mathbf{A}_K &= \mathbf{A}^{H_K} \\ \tilde{\mathbf{A}}_K &= \tilde{\mathbf{A}}^{H_K} \\ \mathbf{B}_K &= \mathbf{B}^{H_K} \\ \tilde{\mathbf{B}}_K &= \tilde{\mathbf{B}}^{H_K}.\end{aligned}$$

The residue field of $\mathbf{B}_K$ is $\mathbf{E}_K$, the field of norms of $K \subset K_\infty$. Similarly, we have that $\text{Gal}(\mathbf{B}/\mathbf{B}_K) \cong \text{Gal}(\mathbf{E}^{\text{sep}}/\mathbf{E}_K) \cong H_K$.

For any finite extension L/K , [CC98] shows that $\mathbf{B}_L$ is an unramified extension of $\mathbf{B}_K$ of degree $[L : K]$ and if L/K is Galois then $\tilde{\mathbf{B}}_L/\tilde{\mathbf{B}}_K$ and $\mathbf{B}_L/\mathbf{B}_K$ are also Galois. Further, we have

$$\text{Gal}(\tilde{\mathbf{B}}_L/\tilde{\mathbf{B}}_K) = \text{Gal}(\mathbf{B}_L/\mathbf{B}_K) = \text{Gal}(\mathbf{E}_L/\mathbf{E}_K) = \text{Gal}(L_\infty/K_\infty) = H_L/H_K.$$

If $K = K_0$, i.e., if K is an unramified extension of $\mathbb{Q}_p$ then the definitions coincide.

Remark 3.4.3. If K is a ramified extension of $\mathbb{Q}_p$ the structure of $\mathbf{A}_K$ will be slightly more complicated. Let $L = K_\infty \cap K_0^{\text{ur}}$ (K_0^{ur} being the maximal unramified extension) and $\bar{\pi}_K$ be a uniformizer of $\mathbf{E}_K$ with $\bar{P}_{E_L}(X)$ its minimal polynomial over $\mathbf{E}_L$. Let $P_{E_L}(X)$ be a lift to $\mathcal{O}_L[[\pi]][X]$. Then, by Hensel's lemma there exists a unique element that satisfies $P_{E_K}(\tilde{\pi}_K) = 0$ and $\bar{\pi}_K \equiv \tilde{\pi}_K \pmod{p}$. It is now possible to show that

$$\mathbf{A}_K = \left\{ \sum_{n \in \mathbb{Z}} a_n \tilde{\pi}_K^n \mid a_n \in \mathcal{O}_L, a_n \rightarrow 0 \text{ for } n \rightarrow -\infty \right\}.$$

For details see [Ber02], III.1.1.

Remark 3.4.4. We also introduce

$$\tilde{\mathbf{A}}^+ = W(\tilde{\mathbf{E}}^+)$$

and further,

$$\mathbf{A}_{K_0}^+ = \mathbf{A}_{K_0} \cap \tilde{\mathbf{A}}^+ = \left\{ \sum_{n=0}^{+\infty} a_n \pi^n \mid a_n \in \mathcal{O}_{K_0} \right\} \subset \mathbf{A}_{K_0}.$$

3.5 Characteristic 0 theory: (φ, Γ) -modules

We now want to proceed to studying representations of G_K for K a finite extension of $\mathbb{Q}_p$. The results will be similar to our results about φ -modules in the characteristic $p > 0$ case.

3.5.1 $\mathbb{Z}_p$ -representations

Definition 3.5.1. For $\Gamma_K = \text{Gal}(\bar{K}/K_\infty)$, a (φ, Γ_K) -module over $\mathbf{A}_K$ is a finitely generated $\mathbf{A}_K$ -module with a σ -semilinear map φ and a semilinear action of Γ_K that commutes with φ .

A (φ, Γ_K) -module M is called *étale* if $\varphi(M)$ generates M as an $\mathbf{A}_K$ -module. We denote the category of étale (φ, Γ_K) -modules over $\mathbf{A}_K$ by $\Phi\Gamma\mathbf{M}_{\mathbf{A}_K}^{\text{ét}}$.

We obtain the following equivalence of categories.

Theorem 3.5.2 ([Fon07], 3.4.3). There is an equivalence of categories

$$\{\mathbb{Z}_p\text{-representations of } G_K\} \longleftrightarrow \{\text{étale } (\varphi, \Gamma_K)\text{-modules over } \mathbf{A}_K\}.$$

The functor is the same, $V \mapsto \mathbf{D}(V) = (\mathbf{A} \otimes_{\mathbb{Z}_p} V)^{H_K}$, with residual Γ_K -action (coming from the diagonal action of G_K) which commutes with the Frobenius (given by $\sigma \otimes id_V$).

Remark 3.5.3. Similar to (3.1) we can also show that

$$\mathbf{A} \otimes_{\mathbf{A}_K} \mathbf{D}(V) \xrightarrow{\sim} \mathbf{A} \otimes_{\mathbb{Z}_p} V \quad (3.2)$$

using Hilbert 90.

Remark 3.5.4. Using (3.2) we see that for V a free $\mathbb{Z}_p$ -representation of rank d , $\mathbf{A} \otimes_{\mathbf{A}_K} \mathbf{D}(V)$ is a free $\mathbf{A}$ -module of rank d . We deduce that $\mathbf{D}(V)$ is a free $\mathbf{A}_K$ -module of rank d and we have the following equivalence:

$$\{\text{free } \mathbb{Z}_p\text{-representations of } G_K \text{ of rank } d\} \longleftrightarrow \{\text{free étale } (\varphi, \Gamma_K)\text{-modules over } \mathbf{A}_K \text{ of rank } d\}.$$

3.5.2 p -adic representations

Definition 3.5.5. Let V be a finite-dimensional $\mathbf{B}_K$ -vector space. An $\mathbf{A}_K$ -lattice in V is a finitely generated sub- $\mathbf{A}_K$ -module L of V such that L generates V over $\mathbf{B}_K$, i.e. contains some $\mathbf{B}_K$ -basis of V .

Definition 3.5.6. A (φ, Γ_K) -module over $\mathbf{B}_K$ is a finite-dimensional $\mathbf{B}_K$ vector space equipped with a σ -semilinear φ -action and a semilinear Γ_K -action that commutes with the φ -action. We call a (φ, Γ_K) -module M over $\mathbf{B}_K$ *étale* if there exists a $\mathbf{A}_K$ -lattice $N \subset M$ such that N is stable under φ and $\varphi(N)$ generates N over $\mathbf{A}_K$.

Remark 3.5.7. Using exactly the same trick as in 2.2.4 we can choose an $\mathbf{A}_K$ -lattice $N \subset M$ such that N is at the same time stable under φ and G_K , and $\varphi(N)$ generates N over $\mathbf{A}_K$, i.e. N is an étale (φ, Γ_K) -module over $\mathbf{A}_K$.

Proposition 3.5.8 ([Fon07], 3.4.4). We have the following equivalence of categories

$$\{p\text{-adic representations of } G_K\} \longleftrightarrow \{\text{étale } (\varphi, \Gamma_K)\text{-modules over } \mathbf{B}_K\}.$$

via the functor $V \mapsto \mathbf{D}(V) = (\mathbf{B}_K \otimes_{\mathbb{Q}_p} V)^{H_K}$.

Remark 3.5.9. Given a p -adic representation V of G_K , let $T \subset V$ be a $\mathbb{Z}_p$ -lattice stable under G_K . Then $V = \mathbb{Q}_p \otimes_{\mathbb{Z}_p} T$ and $\mathbf{D}(V) = (\mathbf{B}_K \otimes_{\mathbb{Q}_p} V)^{H_K} = (\mathbf{A}_K \otimes_{\mathbb{Z}_p} T)^{H_K}[1/p] = \mathbf{D}(T)[1/p]$.

Similarly, given an étale (φ, Γ_K) -module M , let $N \subset M$ be a φ - and Γ_K -stable $\mathbf{A}_K$ -lattice. Then $N = \mathbf{D}(T)$ for some free $\mathbb{Z}_p$ -representation T of G_K . But then we see that $M = N[1/p] = \mathbf{D}(V)$ for $V = \mathbb{Q}_p \otimes_{\mathbb{Z}_p} T$.

The theory of (φ, Γ_K) -modules corresponding p -adic representations is therefore analogous to the one of free $\mathbb{Z}_p$ -representations.

3.5.3 p -torsion representations

A useful strategy when proving theorems about (φ, Γ) -modules is to look at modules obtained from p -torsion representations. Then we can use inverse limits to make statements about general $\mathbb{Z}_p$ -representations by Proposition 2.3.11.

Definition 3.5.10. The *length* of a module M is the length of the longest chain of submodules. If no longest chain of submodules exists, we say M has *infinite length*.

Definition 3.5.11. An étale p -torsion (φ, Γ) -module is an étale (φ, Γ) -module of finite length. We denote the category of p -torsion (φ, Γ) -modules over $\mathbf{A}_K$ by $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$.

Indeed, the p -torsion (φ, Γ) -modules correspond exactly to the p -torsion representations of G_K .

Theorem 3.5.12 ([Her98], Theorem 1.2). There is an equivalence of categories

$$\{p\text{-torsion representations of } G_K\} \longleftrightarrow \{\text{étale } p\text{-torsion } (\varphi, \Gamma_K)\text{-modules over } \mathbf{A}_K\}.$$

In particular, we see that an étale (φ, Γ) -module is p -torsion if and only if it is annihilated by a power of p .

Recall that we noted that the category $\text{Rep}_{p\text{-tor}}(G_K)$ or $\text{Rep}_{\mathbb{Z}_p}(G_K)$ does not actually have nice enough properties, i.e., does not have enough injectives. We therefore introduce the category of (φ, Γ_K) -modules equivalent to the category of discrete p -torsion representations of G_K .

Definition 3.5.13. We denote by $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$ the category consisting of inductive limits of étale p -torsion (φ, Γ) -modules.

Proposition 3.5.14. There is an equivalence of categories

$$\mathrm{Rep}_{p\text{-tor}}^{\mathrm{discrete}}(G_K) \longleftrightarrow \Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}.$$

In particular, the category $\Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$ has enough injectives.

Proof. Since the functor $\mathbf{D}$ is an equivalence of categories between p -torsion representations of G_K and étale p -torsion (φ, Γ_K) -modules, we can extend it to an equivalence of the categories consisting of all inductive limits of them. $\square$

3.6 Overconvergent rings

We very closely follow [CC98]. Note that although our ring $\mathbf{A}_{K_0}$ has a nice structure

$$\mathbf{A}_{K_0} = \left\{ \sum_{n \in \mathbb{Z}} a_n \pi^n \mid a_n \in \mathcal{O}_{K_0}, a_n \rightarrow 0 \text{ for } n \rightarrow -\infty \right\}.$$

the power series do not correspond to actual p -adic analytic functions $\sum_{n \in \mathbb{Z}} a_n T^n$. However, if we add some restrictions on the convergence of $a_n \rightarrow 0$ for $n \rightarrow -\infty$, we obtain analytic functions $\sum_{n \in \mathbb{Z}} a_n T^n$ that converge on some annulus $\rho < T < 1$.

We start by introducing a new index set. Let $\bar{\mathbb{R}} := \mathbb{R} \cup \{r^- \mid r \in \mathbb{R}\}$, with an order relation $<$ that satisfies $r_1 < r_2^- < r_2$ for all $r_1 < r_2$, $r_1, r_2 \in \mathbb{R}$. Let $\bar{\mathbb{R}}_+ := \mathbb{R}_+ \cup \{r^- \mid r \in \mathbb{R}_+\}$.

For $r \in \mathbb{R}_+$ define

$$\tilde{\mathbf{A}}_r^\dagger = \{x \in \tilde{\mathbf{A}} \mid w_k(x) + \frac{p}{p-1}rk \geq 0 \text{ for all } k \in \mathbb{N}\}$$

and

$$\tilde{\mathbf{A}}_{r^-}^\dagger = \{x \in \tilde{\mathbf{A}}_r^\dagger \mid \lim_{k \rightarrow \infty} w_k(x) + \frac{p}{p-1}rk = +\infty \text{ for all } k \in \mathbb{N}\} \subset \tilde{\mathbf{A}}_r^\dagger.$$

Since for $x = \sum_{i=0}^\infty p^i[x_i]$, $w_k(x) \geq 0$ for all $k \in \mathbb{N}$ is equivalent to $v_{\mathbf{E}}(x_k) \geq 0$, i.e. $x_k \in \tilde{\mathbf{E}}^+$, we can see that $\tilde{\mathbf{A}}_0^\dagger = W(\tilde{\mathbf{E}}^+) = \tilde{\mathbf{A}}^+$.

Further, for $r \in \mathbb{R}_+$ define

$$\tilde{\mathbf{B}}_r^\dagger = \tilde{\mathbf{A}}_r^\dagger \left[\frac{1}{p}\right] = \{x \in \tilde{\mathbf{B}} \mid w_k(x) + \frac{p}{p-1}rk \geq C \text{ for some } C \in \mathbb{R} \text{ and for all } k \in \mathbb{N}\}$$

and

$$\tilde{\mathbf{B}}_{r^-}^\dagger = \tilde{\mathbf{A}}_{r^-}^\dagger \left[\frac{1}{p}\right] = \{x \in \tilde{\mathbf{B}} \mid \lim_{k \rightarrow \infty} w_k(x) + \frac{p}{p-1}rk = +\infty \text{ for all } k \in \mathbb{N}\}.$$

Now let

$$\begin{aligned} \tilde{\mathbf{B}}^\dagger &= \bigcup_{r \in \mathbb{R}_+} \tilde{\mathbf{B}}_r^\dagger \\ \tilde{\mathbf{A}}_\infty^\dagger &= \bigcup_{r \in \mathbb{R}_+} \tilde{\mathbf{A}}_r^\dagger \end{aligned}$$

$$\begin{aligned}\tilde{\mathbf{A}}^\dagger &= \tilde{\mathbf{B}}^\dagger \cap \tilde{\mathbf{A}} \\ &= \{x \in \tilde{\mathbf{A}} \mid w_k(x) + \frac{p}{p-1}rk \geq C \text{ for some } C \in \mathbb{R}, r \in \mathbb{R}_+ \text{ and for all } k \in \mathbb{N}\}\end{aligned}$$

Remark 3.6.1. Note that using the new order relation on $\bar{\mathbb{R}}$, we have $\tilde{\mathbf{A}}_{r_1}^\dagger \subset \tilde{\mathbf{A}}_{r_2}^\dagger$ and $\tilde{\mathbf{B}}_{r_1}^\dagger \subset \tilde{\mathbf{B}}_{r_2}^\dagger$ whenever $r_1 < r_2$. Therefore, when defining $\tilde{\mathbf{B}}^\dagger$ it does not matter whether we sum over $r \in \mathbb{R}_+$ or $r \in \bar{\mathbb{R}}_+$.

Lemma 3.6.2 ([CC98], Lemma II.1.4). If $r \in \mathbb{R}_+$, an element $x = \sum_{k=0}^{+\infty} p^k[x_k]$ of $\tilde{\mathbf{A}}_r^\dagger$ is a unit in $\tilde{\mathbf{A}}_r^\dagger$ if and only if $v_{\mathbf{E}}(x_0) = 0$.

Proof. If $x = \sum_{k=0}^{+\infty} p^k[x_k]$ is a unit in $\tilde{\mathbf{A}}_r^\dagger$, then write $x^{-1} = \sum_{k=0}^{+\infty} p^k[y_k]$. Since we must have $x_0 y_0 = 1$ and also $v_{\mathbf{E}}(x_0) \geq 0$ and $v_{\mathbf{E}}(y_0) \geq 0$ since x, x^{-1} are both elements in $\tilde{\mathbf{A}}_r^\dagger$, we deduce that $v_{\mathbf{E}}(x_0) = 0$. Conversely, if $v_{\mathbf{E}}(x_0) = 0$ then, by dividing by $[x_0]$, we can assume that $x_0 = 1$ and $x = 1 - u$ where $u \in p\tilde{\mathbf{A}} \cap \tilde{\mathbf{A}}_r^\dagger$. Then $x^{-1} = \sum_{n \in \mathbb{N}} u^n$ is an element in $\tilde{\mathbf{A}}_r^\dagger$ since it is a closed subring of $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{A}}$ is p -adically complete. $\square$

As before, for $\alpha \in \tilde{\mathbf{A}}$ we write $\bar{\alpha}$ for its class in $\tilde{\mathbf{E}}$ and for $\beta \in \tilde{\mathbf{E}}$ we write $[\beta] \in \tilde{\mathbf{A}}$ for its Teichmüller lift.

Corollary 3.6.3 ([CC98], Lemma II.1.5). For $r \geq \frac{p-1}{p}$, $\frac{\pi}{[\bar{\pi}]}$ is a unit in $\tilde{\mathbf{A}}_r^\dagger$.

Proof. We can write $\pi = [\bar{\pi}] + p[\alpha_1] + p^2[\alpha_2] + \dots$ where α_i is a polynomial in $\epsilon^{p^{-i}} - 1$ with coefficients in $\mathbb{Z}$ and no constant term, using the addition polynomials of the Witt vectors. Thus $v_{\mathbf{E}}(\alpha_i) \geq v_{\mathbf{E}}(\epsilon^{p^{-i}} - 1) = \frac{1}{(p-1)p^{i-1}}$. Therefore, we can write $\pi = [\bar{\pi}](1 + p[a_1] + p^2[a_2] + \dots)$, with $v_{\mathbf{E}}(a_1) = v_{\mathbf{E}}(\alpha_1) - v_{\mathbf{E}}(\epsilon - 1) \geq -1$ and $v_{\mathbf{E}}(a_i) = v_{\mathbf{E}}(\alpha_i) - v_{\mathbf{E}}(\epsilon - 1) \geq -v_{\mathbf{E}}(\epsilon - 1) \geq -i$ for $i \geq 2$. Thus $\frac{\pi}{[\bar{\pi}]} \in \tilde{\mathbf{A}}_{\frac{p-1}{p}}^\dagger$ and by Lemma 3.6.2, $\frac{\pi}{[\bar{\pi}]}$ is a unit in $\tilde{\mathbf{A}}_{\frac{p-1}{p}}^\dagger$, and therefore a unit in $\tilde{\mathbf{A}}_r^\dagger$ for all $r \geq \frac{p-1}{p}$. $\square$

Proposition 3.6.4 ([CC98], Proposition II.1.2). (i) If $r \in \bar{\mathbb{R}}_+$, $\tilde{\mathbf{A}}_r^\dagger$ and $\tilde{\mathbf{B}}_r^\dagger$ are subrings of $\tilde{\mathbf{A}}$ and $\tilde{\mathbf{B}}$ stable under G_{K_0} and $\sigma(\tilde{\mathbf{A}}_r^\dagger) \subset \tilde{\mathbf{A}}_{pr}^\dagger$ and $\sigma(\tilde{\mathbf{B}}_r^\dagger) \subset \tilde{\mathbf{B}}_{pr}^\dagger$.

(ii) For $r \in \bar{\mathbb{R}}_+$ if α is an element of $\tilde{\mathbf{A}}^+$ such that for $\bar{\alpha}$ the reduction modulo p we have $v_{\mathbf{E}}(\bar{\alpha}) > 0$ then $\tilde{\mathbf{A}}_r^\dagger$ is complete and Hausdorff for the α -adic topology and $\tilde{\mathbf{A}}_r^\dagger$ is closed in $\tilde{\mathbf{A}}$.

(iii) $\tilde{\mathbf{B}}^\dagger$ is a subfield of $\tilde{\mathbf{B}}$ stable under G_{K_0} and σ .

Proof. (i) The fact that $\tilde{\mathbf{A}}_r^\dagger$ and $\tilde{\mathbf{B}}_r^\dagger$ are subrings follows directly from Proposition 3.4.1. Further, we have that for all $g \in G_{K_0}$ and $x \in \tilde{\mathbf{E}}$, $v_{\mathbf{E}}(g(x)) = v_{\mathbf{E}}(x)$ and thus for all $x \in \tilde{\mathbf{A}}$, $r \in \mathbb{N}$, $w_k(g(x)) = w_k(x)$. Using that for all $x \in \tilde{\mathbf{A}}$, $k \in \mathbb{N}$ we have $w_k(\sigma(x)) = pw_k(x)$ we get that $\sigma(\tilde{\mathbf{A}}_r^\dagger) \subset \tilde{\mathbf{A}}_{pr}^\dagger$ from Proposition 3.4.1.

(ii) We start by assuming $\alpha = [\bar{\alpha}]$ and let $s = v_{\mathbf{E}}(\bar{\alpha})$. To show Hausdorffness, it suffices to show that $x \in \alpha^i \tilde{\mathbf{A}}_r^\dagger$ for all $i \geq 0$ implies $x = 0$. Indeed, for $y \in \tilde{\mathbf{A}}_r^\dagger$ we have $w_k(\alpha^i y) \geq w_k(y) + is \geq -\frac{p}{p-1}kr + is$ for all $i \geq 0$ and thus $w_k(\alpha^i y) = +\infty$ for all $k \in \mathbb{N}$. But by Proposition 3.4.1 this implies $x \in p^{k+1} \tilde{\mathbf{A}}$ for all $k \in \mathbb{N}$ and thus $x = 0$. For completeness, let $(a_i)_{i \in \mathbb{N}}$ be a sequence of elements in $\tilde{\mathbf{A}}_r^\dagger$. We have by the previous discussion that for all $k \in \mathbb{N}$, $w_k(\alpha^i a_i) \rightarrow +\infty$ for $i \rightarrow +\infty$. Therefore, the sum $\sum_{i \in \mathbb{N}} \alpha^i a_i$ converges to some element $a \in \tilde{\mathbf{A}}$ with $w_k(a) \geq \inf_{i \in \mathbb{N}} w_k(a_i) + is$. We see now if $r \in \mathbb{R}_+$ that $a_i \in \tilde{\mathbf{A}}_r^\dagger$ implies $w_k(a) \geq \inf_{i \in \mathbb{N}} w_k(a_i) + is \geq \frac{p}{p-1}kr$ and $a \in \tilde{\mathbf{A}}_r^\dagger$, and if $r \in \bar{\mathbb{R}}_+ - \mathbb{R}_+$ then $a_i \in \tilde{\mathbf{A}}_r^\dagger$ implies $w_k(a) - \frac{p}{p-1}kr \geq \inf_{i \in \mathbb{N}} w_k(a_i) - \frac{p}{p-1}kr + is$ which tends to $+\infty$ since for all $i \in \mathbb{N}$, $w_k(a_i) - \frac{p}{p-1}kr \rightarrow +\infty$ for $k \rightarrow +\infty$ and again $a \in \tilde{\mathbf{A}}_r^\dagger$.

Now we drop the condition that $\alpha = [\bar{\alpha}]$. Let $\alpha \in \tilde{\mathbf{A}}_r^\dagger$ and $r > 0$. We can find some element $\beta \in \tilde{\mathbf{E}}^+$ satisfying $0 < v_{\mathbf{E}}(\beta) < \inf(r, v_{\mathbf{E}}(\bar{\alpha}))$. Then $\frac{\alpha}{[\beta]} \in \tilde{\mathbf{A}}_r^\dagger$ and since $\tilde{\mathbf{A}}_r^\dagger$ is complete and Hausdorff with respect to the $[\beta]$ -adic topology, it also is with respect to the α -adic topology. If $r = 0$ we can use that $\alpha = [\bar{\alpha}] + pu$ and $\tilde{\mathbf{A}}_0^\dagger = \tilde{\mathbf{A}}^+$ is complete with respect to both the $[\alpha]$ -adic and p -adic topology, so also for the α -adic topology.

Lastly, we have that for $r \in \mathbb{R}_+$, $\tilde{\mathbf{A}}_r^\dagger$ is closed as the intersection of the closed sets $U_{k,rk}$ in $\tilde{\mathbf{A}}$.

(iii) By the previous parts, $\tilde{\mathbf{B}}^\dagger = \bigcup_{r \in \mathbb{R}_+} \tilde{\mathbf{B}}_r^\dagger$ is a subring of $\tilde{\mathbf{B}}$ stable under G_{K_0} and σ . It remains to show that $\tilde{\mathbf{B}}^\dagger$ is a field, i.e., that every non-zero element is invertible. We can write any element $x \in \tilde{\mathbf{B}}_r^\dagger$ as $x = p^{-n} \sum_{k \in \mathbb{N}} p^k [x_k]$ where $\sum_{k \in \mathbb{N}} p^k [x_k] \in \tilde{\mathbf{A}}_r^\dagger$. Further, if we let k_0 the smallest integer such that $x_{k_0} \neq 0$ we have $x = p^{-n+k_0} [x_{k_0}] u$, where $u = 1 + \sum_{k \in \mathbb{N}} p^k [y_k]$, $y_k = \frac{x_{k+k_0}}{x_{k_0}}$. We see that $v_{\mathbf{E}}(y_k) \geq v_{\mathbf{E}}(x_{k+k_0}) - v_{\mathbf{E}}(x_{k_0}) \geq -\frac{p}{p-1}(k+k_0)r - v_{\mathbf{E}}(x_{k_0}) \geq -\frac{p}{p-1}ks$ for $s = r + \frac{k_0 r}{k} + \frac{p-1}{pk} \sup(0, v_{\mathbf{E}}(x_{k_0}))$. Thus u is a unit in $\tilde{\mathbf{A}}_s^\dagger$. Further, p is invertible in $\tilde{\mathbf{B}}^\dagger$ since $\frac{1}{p} \in \tilde{\mathbf{B}}_0^\dagger$ and $[x_{k_0}]$ is invertible in $\tilde{\mathbf{B}}^\dagger$ since for $t = \frac{p}{p-1}v_{\mathbf{E}}(x_{k_0})$ we have $[x_{k_0}^{-1}] \in \tilde{\mathbf{B}}_t^\dagger$. We conclude that $\tilde{\mathbf{B}}^\dagger$ is a field. $\square$

We now equip $\tilde{\mathbf{A}}_r^\dagger$ with the π -adic topology, with respect to which it is complete Proposition 3.6.4 since $v_{\mathbf{E}}(\bar{\pi}) = v_{\mathbf{E}}(\epsilon - 1) = \frac{p}{p-1} > 0$. Further, the G_{K_0} and σ -actions are continuous by Proposition 3.6.4. We then equip $\tilde{\mathbf{B}}_r^\dagger = \bigcup_{n \in \mathbb{N}} p^{-n} \tilde{\mathbf{A}}_r^\dagger$ with the inductive limit topology and also $\tilde{\mathbf{B}}^\dagger = \bigcup_{r > 0} \tilde{\mathbf{B}}_r^\dagger$ with the inductive limit topology.

Remark 3.6.5. The definition is motivated by us wanting the power series to converge. In fact, if the reader is familiar with the ring $\mathbf{B}_{\text{dR}}^+$ then we can show that for $x = \sum_{i=0}^{+\infty} p^i [x_i] \in \tilde{\mathbf{B}}$ the following are equivalent

- (i) $\sum_{i=0}^{+\infty} p^i [x_i]$ converges in $\mathbf{B}_{\text{dR}}^+$

- (ii) $\sum_{i=0}^{+\infty} p^i x_i^{(0)}$ converges in $\mathbf{C}$
- (iii) $\lim_{k \rightarrow +\infty} (k + v_{\mathbf{E}}(x_k)) = +\infty$
- (iv) $x \in \tilde{\mathbf{B}}_{\frac{p-1}{p}}^{\dagger}$

For details, see ([CC98] II.1.1).

We now set, for $r \in \bar{\mathbb{R}}_+ \cup \{\infty\}$

$$\mathbf{A}^{\dagger} = \mathbf{A} \cap \tilde{\mathbf{A}}^{\dagger}, \quad \mathbf{B}^{\dagger} = \mathbf{B} \cap \tilde{\mathbf{B}}^{\dagger}, \quad \mathbf{A}_r^{\dagger} = \mathbf{A} \cap \tilde{\mathbf{A}}_r^{\dagger}, \quad \mathbf{B}_r^{\dagger} = \mathbf{B} \cap \tilde{\mathbf{B}}_r^{\dagger}.$$

From the previous discussion we can directly deduce

- Proposition 3.6.6** ([CC98], Proposition II.1.6). (i) $\mathbf{B}^{\dagger}$ is a field stable under G_{K_0} and σ .
- (ii) If $r \in \bar{\mathbb{R}}_+$, the subrings $\mathbf{A}_r^{\dagger}$ and $\mathbf{B}_r^{\dagger}$ are stable under G_{K_0} and $x \in \mathbf{A}_r^{\dagger}$ if and only if $\sigma(x) \in \mathbf{A}_{pr}^{\dagger}$ (similarly, $x \in \mathbf{B}_r^{\dagger}$ if and only if $\sigma(x) \in \mathbf{B}_{pr}^{\dagger}$).
- (iii) If $r \in \bar{\mathbb{R}}_+$, $\mathbf{A}_r^{\dagger}$ is complete and Hausdorff with respect to the π -adic topology.

Following the notation from before, for $r \in \bar{\mathbb{R}}_+ \cup \{\infty\}$ we set

$$\mathbf{A}_K^{\dagger} = (\mathbf{A}^{\dagger})^{H_K}, \quad \mathbf{B}_K^{\dagger} = (\mathbf{B}^{\dagger})^{H_K}, \quad \mathbf{A}_{r,K}^{\dagger} = (\mathbf{A}_r^{\dagger})^{H_K}, \quad \mathbf{B}_{r,K}^{\dagger} = (\mathbf{B}_r^{\dagger})^{H_K}.$$

We now want to have a closer look at the rings $\mathbf{A}_{r,K}^{\dagger}$ for the case $K = K_0$, i.e., K and unramified extension of $\mathbb{Q}_p$. Recall that we have for $\pi = [\epsilon] - 1$

$$\mathbf{A}_{K_0} = \left\{ \sum_{n \in \mathbb{Z}} a_n \pi^n \mid a_n \in \mathcal{O}_{K_0}, a_n \rightarrow 0 \text{ for } n \rightarrow -\infty \right\}.$$

Definition 3.6.7. We define the following rings of Laurent series. Let

1. $\mathcal{A}_{K_0}^{[0,1]}$ be the ring of power series with coefficients in $\mathcal{O}_{K_0}$, i.e. the ring of analytic functions that converge on the open unit disk.
2. $\mathcal{A}_{K_0}^{[1,1]}$ be the ring of formal Laurent series of the form $\sum_{n \in \mathbb{Z}} a_n T^n$ with coefficients in $\mathcal{O}_{K_0}$ that satisfy $a_n \rightarrow 0$ for $n \rightarrow -\infty$.
3. $\mathcal{A}_{K_0}^{[\rho,1]}$ (or $\mathcal{A}_{K_0}^{(\rho,1)}$) be the ring of Laurent functions with coefficients in $\mathcal{O}_{K_0}$ that converge on the annulus $\rho \leq |T| < 1$ (or $\rho < |T| < 1$) and are bounded in norm by 1, i.e. the ring of analytic functions that converge on the annulus and are bounded in norm by 1.

Proposition 3.6.8. The map that sends a Laurent series $f(T) = \sum_{n \in \mathbb{Z}} a_n T^n$ to $f(\pi)$ induces isomorphisms

1. from $\mathcal{A}_{K_0}^{[0,1]}$ to $\mathbf{A}_{K_0}^+$,
2. from $\mathcal{A}_{K_0}^{[1,1]}$ to $\mathbf{A}_{K_0}$,
3. from $\mathcal{A}_{K_0}^{[\rho(r),1]}$ to $\mathbf{A}_{r^-,K_0}^\dagger$ and $\mathcal{A}_{K_0}^{(\rho(r),1)}$ to $\mathbf{A}_{r,K_0}^\dagger$ where $\rho(r) = p^{-\frac{1}{r}}$ and $r \in \mathbb{R}_+$, $r \geq \frac{p-1}{p}$.

Proof. The first two statements are obvious, and the last one is equivalent to the next lemma. $\square$

Remark 3.6.9. Most authors define the rings of overconvergent elements $\mathbf{A}_{r,K_0}^\dagger$ slightly different, for $r \in \mathbb{R}_+$ there is no distinction between r and r^- and the condition that $w_k(x) + \frac{p}{p-1}kr \geq 0$ is dropped. That will then result in the ring of overconvergent elements corresponding to the ring of analytic functions that converge on $\rho(r) \leq |T| < 1$ without the condition that they are bounded by 1. That way $\mathbf{A}_{K_0}^\dagger$ is just defined as the union of the $\mathbf{A}_{r,K_0}^\dagger$ and coincides with our definition of $\mathbf{A}_{K_0}^\dagger$.

Further note that we use $\frac{p}{p-1}$ in our definition of the overconvergent rings since $v_{\mathbf{E}}(\epsilon - 1) = \frac{p}{p-1}$, which will make some proofs slightly easier, but we might as well define the rings using $w_k(x) + kr \geq 0$.

Lemma 3.6.10. Let $r \in \bar{\mathbb{R}}_+$ with $r \geq \frac{p-1}{p}$, then for $x = \sum_{n \in \mathbb{Z}} a_n \pi^n \in \mathbf{A}_{K_0}$ the following are equivalent

- (i) $x \in \mathbf{A}_{r,K_0}^\dagger$
- (ii) $\begin{cases} rv_p(a_n) + n \geq 0 \text{ for all } n \in \mathbb{Z} & \text{if } r \in \mathbb{R}_+ \\ rv_p(a_n) + n \geq 0 \text{ for all } n \in \mathbb{Z} \text{ and } rv_p(a_n) + n \rightarrow +\infty \text{ for } n \rightarrow -\infty & \text{if } r \in \bar{\mathbb{R}}_+ - \mathbb{R}_+ \end{cases}$

Proof. First, note that

$$\mathbf{A}_{K_0}^+ = \left\{ \sum_{n=0}^{+\infty} a_n \pi^n \mid a_n \in \mathcal{O}_{K_0} \right\} = \mathbf{A}_{0,K_0}^\dagger$$

since we established that $\tilde{\mathbf{A}}^+ = \tilde{\mathbf{A}}_0^\dagger$. We can also deduce it by seeing that $\pi \in \mathbf{A}_{0,K_0}^\dagger$, $\mathcal{O}_{K_0} \subset \mathbf{A}_{0,K_0}^\dagger$ and that $\mathbf{A}_{0,K_0}^\dagger$ is complete with respect to the π -adic topology.

We thus have that an element $x = \sum_{n \in \mathbb{Z}} a_n \pi^n \in \mathbf{A}_{K_0}$ is in $\mathbf{A}_{r,K_0}^\dagger$ if and only if $x' = \sum_{n < 0} a_n \pi^n \in \mathbf{A}_{r,K_0}^\dagger$. If $k \in \mathbb{N}$, let $y_k = p^{-k} \sum_{n < 0, v_p(a_n) = k} a_n \pi^n$ and $n_k = \inf\{n < 0 \mid v_p(a_n) = k\}$. Note that $n_k > -\infty$ since by definition of $\mathbf{A}_{K_0}$, $v_p(a_n) \rightarrow 0$ for $n \rightarrow -\infty$. Then we have $y_k = [\bar{\pi}]^{n_k} u$ where

$$u = \frac{a_{n_k}}{p^k} \left(\frac{\pi}{[\bar{\pi}]} \right)^{n_k} \left(1 + \sum_{n > n_k, v_p(a_n) = v_p(a_{n_k})} \frac{a_n}{a_{n_k}} \pi^{n-n_k} \right).$$

We can see that u is a unit in $\tilde{\mathbf{A}}_{\frac{p-1}{p}}$ since $\frac{a_{n_k}}{p^k}$ and $1 + \sum_{n > n_k, v_p(a_n)=v_p(a_{n_k})} \frac{a_n}{a_{n_k}} \pi^{n-n_k}$ are units in $\mathbf{A}_{K_0}^+ = \mathbf{A}_{0, K_0}^\dagger$ and $\frac{\pi}{[\pi]}$ is a unit in $\tilde{\mathbf{A}}_{\frac{p-1}{p}}^\dagger$. Further, we know that by Lemma 3.4.1

$$w_0(y_k) = v_{\mathbf{E}}(\bar{y}_k) = w_0([\pi]^{n_k} u) = w_0([\pi]^{n_k}) + w_0(u) = n_k \frac{p}{p-1}$$

since $w_0(u) = 0$ and

$$w_i(y_k) = w_i([\pi]^{n_k} u) \geq w_0([\pi]^{n_k}) + w_i(u) \geq v_{\mathbf{E}}(\bar{y}_k) - i$$

for $i \geq 1$. Furthermore, since $x = \sum_{i=0}^{+\infty} p^i y_i$, we have

$$w_k(x) \geq \inf_{0 \leq i \leq k} w_k(p^i y_i) = \inf_{0 \leq i \leq k} w_{k-i}(y_i)$$

with equality if $w_0(y_k) < w_{k-i}(y_i)$ for all $0 \leq i \leq k-1$ by Lemma 3.4.1.

Lemma 3.6.11 ([CC98], Lemma II.2.3). Let $s \geq 0$ and $(u_k)_{k \in \mathbb{N}}, (v_k)_{k \in \mathbb{N}}$ two sequences in $\mathbb{R}$ satisfying the following conditions

1. $v_k \geq \inf_{0 \leq i \leq k} u_i + s(k-i)$
2. $v_k = u_k$ if $u_k < u_i + s(k-i)$ for all $0 \leq i \leq k-1$

then

1. $u_k \geq 0, \forall k \in \mathbb{N} \iff v_k \geq 0, \forall k \in \mathbb{N}$
2. if $s > 0, \lim_{k \rightarrow +\infty} u_k = +\infty \iff \lim_{k \rightarrow +\infty} v_k = +\infty$.

Corollary 3.6.12. For $r \geq \frac{p-1}{p}$ and $x \in \mathbf{A}_K$, we have

1. for $r \in \mathbb{R}_+, x \in \mathbf{A}_{r, K}^\dagger$ if and only if $v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr \geq 0$ for all $k \in \mathbb{N}$
2. for $r \in \bar{\mathbb{R}}_+ - \mathbb{R}_+, x \in \mathbf{A}_{r, K}^\dagger$ if and only if $v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr \geq 0$ for all $k \in \mathbb{N}$ and $\lim_{k \rightarrow +\infty} v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr = +\infty$.

Proof. This is an application of the previous lemma, for $s = \frac{pr}{p-1} - 1$ and $v_k = w_k(x') + \frac{p}{p-1}kr, u_k = v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr$. Note that $u_k < u_i + s(k-i)$ for all $0 \leq i \leq k-1$ implies

$$w_0(y_k) = v_{\mathbf{E}}(\bar{y}_k) < v_{\mathbf{E}}(\bar{y}_i) - (k-i) \leq w_{k-i}(y_i)$$

for all $0 \leq i \leq k-1$. Then we have $w_k(x') = w_0(y_k)$ and therefore $v_k = u_k$. $\square$

We finish the proof by noting that using $v_{\mathbf{E}}(\bar{y}_k) = \inf_{v_p(a_n)=k} n \frac{p}{p-1}$ (if $y_k = 0$ we set both sides to be $+\infty$) we see that $v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr \geq 0$ for all $k \in \mathbb{N}$ is equivalent to $rv_p(a_n) + n \geq 0$ for all $n < 0$. Similarly, $\lim_{k \rightarrow +\infty} v_{\mathbf{E}}(\bar{y}_k) + \frac{p}{p-1}kr = +\infty$ is equivalent to $\lim_{k \rightarrow +\infty} \inf_{v_p(a_n)=k} n + rv_p(a_n) = +\infty$ which in turn is equivalent to $\lim_{n \rightarrow -\infty} rv_p(a_n) + n = +\infty$ since $a_n \rightarrow 0$ for $n \rightarrow -\infty$. $\square$

Proposition 3.6.13 ([CC98], Proposition II.4.1). If L/K is a finite extension, then $\mathbf{B}_L^\dagger$ is an extension of $\mathbf{B}_K^\dagger$ of degree $[L_\infty : K_\infty]$ and if L/K is Galois then $\mathbf{B}_L^\dagger/\mathbf{B}_K^\dagger$ is also Galois and $\text{Gal}(\mathbf{B}_L^\dagger/\mathbf{B}_K^\dagger) = \text{Gal}(L_\infty/K_\infty)$.

Proof. See [CC98], Proposition II.4.1. □

3.7 Overconvergent representations

We very closely follow [CC98]. Using the new overconvergent rings, we can also define the equivalent (φ, Γ_K) -modules.

Definition 3.7.1. Let V be a $\mathbb{Z}_p$ -representation of G_K , we define

$$\begin{aligned} \mathbf{D}^\dagger(V) &= (\mathbf{A}^\dagger \otimes_{\mathbb{Z}_p} V)^{H_K} \\ \mathbf{D}_r^\dagger(V) &= (\mathbf{A}_r^\dagger \otimes_{\mathbb{Z}_p} V)^{H_K} \text{ for } r \in \mathbb{R}_+. \end{aligned}$$

We call a $\mathbb{Z}_p$ -representation V of G_K *overconvergent* if for some $r > 0$, $\mathbf{D}_r^\dagger(V)$ generates $\mathbf{D}(V)$ over $\mathbf{A}_K$.

Remark 3.7.2. Note that $\mathbf{D}_r^\dagger(V)$ is stable under Γ_K -action and the Frobenius φ sends $\mathbf{D}_r^\dagger(V)$ to $\mathbf{D}_{pr}^\dagger(V)$.

Remark 3.7.3. Equivalently, for p -adic representations V we have

$$\begin{aligned} \mathbf{D}^\dagger(V) &= (\mathbf{B}^\dagger \otimes_{\mathbb{Q}_p} V)^{H_K} \\ \mathbf{D}_r^\dagger(V) &= (\mathbf{B}_r^\dagger \otimes_{\mathbb{Q}_p} V)^{H_K} \text{ for } r \in \mathbb{R}_+. \end{aligned}$$

We call a p -adic representation V of G_K *overconvergent* if for some $r > 0$, $\mathbf{D}_r^\dagger(V)$ generates $\mathbf{D}(V)$ over $\mathbf{B}_K$ or equivalently $\dim_{\mathbf{B}_K^\dagger} \mathbf{D}^\dagger(V) = \dim_{\mathbb{Q}_p} V$.

Remark 3.7.4. In this [CC98], Cherbonnier and Colmez show that in fact all p -adic representations are overconvergent.

3.8 Induced (φ, Γ) -modules

Definition 3.8.1. Let L be a finite extension of K . Let M be an étale (φ, Γ_L) -module over $\mathbf{A}_L$. We define the *induced* (φ, Γ_K) -module to be

$$\text{Ind}_{\Gamma_L}^{\Gamma_K}(M) = \{f : \Gamma_K \longrightarrow M \mid f(hg) = h(f(g)) \text{ for all } h \in \Gamma_L\}$$

where we consider $\Gamma_L \cong \text{Gal}(K_\infty/L \cap K_\infty)$ as a subgroup of Γ_K . Note that then we have $[\Gamma_K : \Gamma_L] = [L \cap K_\infty : K]$. We equip $\text{Ind}_{\Gamma_L}^{\Gamma_K}(M)$ with the structure of a

(φ, Γ_K) -module over $\mathbf{A}_K$ in the following way:

$$\begin{aligned} (\lambda f)(g) &= g(\lambda)f(g) && \text{for all } \lambda \in \mathbf{A}_K, g \in \Gamma_K \\ \gamma(f)(g) &= f(\gamma g) = f(g\gamma) && \text{for all } \gamma, g \in \Gamma_K \\ \varphi(f)(g) &= \varphi(f(g)) && \text{for all } g \in \Gamma_K. \end{aligned}$$

We define $\text{Ind}_{\Gamma_L}^{\Gamma_K}(M)$ analogously for M an étale (φ, Γ_L) -module over $\mathbf{B}_L$ or M an overconvergent étale (φ, Γ_L) -module over $\mathbf{A}_L^\dagger$ or $\mathbf{B}_L^\dagger$.

Remark 3.8.2. If M is a finite-dimensional $\mathbf{B}_L$ vector space, then we see that

$$\begin{aligned} \dim_{\mathbf{B}_K}(\text{Ind}_{\Gamma_L}^{\Gamma_K}(M)) &= [\mathbf{B}_L : \mathbf{B}_K] \cdot [\Gamma_K : \Gamma_L] \cdot \dim_{\mathbf{B}_L}(M) \\ &= [L_\infty : K_\infty] \cdot [L \cap K_\infty : K] \cdot \dim_{\mathbf{B}_L}(M) \\ &= [L : L \cap K_\infty] \cdot [L \cap K_\infty : K] \cdot \dim_{\mathbf{B}_L}(M) \\ &= [L : K] \cdot \dim_{\mathbf{B}_L}(M) \end{aligned}$$

where in the third row we used $[L_\infty : K_\infty] = [L : L \cap K_\infty]$ since $LK_\infty = L_\infty$. The same holds for the rank if M is a free $\mathbf{A}_L$ module or $\mathbf{E}_L$ vector space.

Proposition 3.8.3. Let L be a finite extension of K . If V is a p -adic representation (or a free $\mathbb{Z}_p$ -adic representation) of G_L then

$$\text{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}(V)) = \mathbf{D}(\text{Ind}_L^K(V)).$$

Proof. We first show the statement for p -adic representations. Define a map

$$\begin{aligned} \kappa : \mathbf{D}(\text{Ind}_L^K(V)) &= (\mathbf{B} \otimes_{\mathbb{Q}_p} \text{Ind}_L^K(V))^{H_K} \longrightarrow \text{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}(V)) \\ \sum b_i \otimes f_i &\longmapsto (\bar{g} \mapsto \sum g(b_i) \otimes f_i(g)) \end{aligned}$$

where $g \in G_K$ is a lift of $\bar{g} \in \Gamma_K$. Note that the map κ is induced by a $\mathbb{Q}_p$ -linear map

$$\begin{aligned} \bar{\kappa} : \mathbf{B} \otimes_{\mathbb{Q}_p} \text{Ind}_L^K(V) &\longrightarrow \{f : G_K \longrightarrow \mathbf{B} \otimes_{\mathbb{Q}_p} V \mid f(hg) = h(f(g)) \text{ for all } h \in G_L\} \\ b \otimes f &\longmapsto (g \mapsto g(b) \otimes f(g)). \end{aligned}$$

For elements $\sum b_i \otimes f_i \in (\mathbf{B} \otimes_{\mathbb{Q}_p} \text{Ind}_L^K(V))^{H_K}$ we have that the right-hand side factors through Γ_K and maps into $\mathbf{D}(V)$.

κ is well-defined: First we show that the right-hand side of $\bar{\kappa}$ is the same for different elements $b \otimes f$ representing the same element in $\mathbf{B} \otimes_{\mathbb{Q}_p} \text{Ind}_L^K(V)$. This follows since the G_K -action on $\mathbf{B}$ is $\mathbb{Q}_p$ -linear, i.e.

$$g(\lambda b) \otimes (\lambda^{-1}f)(g) = g(b) \otimes f(g).$$

Let $g, g' \in G_K$ be lifts of the same element in Γ_K , then $g' = gh$ for some $h \in H_K$. We have

$$\sum g'(b_i) \otimes f_i(g') = \sum gh(b_i) \otimes f_i(gh) = \sum g(h(b_i)) \otimes (hf_i)(g) = \sum g(b_i) \otimes f_i(g)$$

where the last equality follows from the fact that $\sum h(b_i) \otimes hf_i = \sum b_i \otimes f_i$ since $\sum b_i \otimes f_i$ is H_K -invariant.

We show that $\sum g(b_i) \otimes f_i(g)$ indeed defines an element of $\mathbf{D}(V)$ for $g \in G_K$. For $h \in H_L \subset H_K$ we have

$$h(\sum g(b_i) \otimes f_i(g)) = \sum hg(b_i) \otimes h(f_i(g)) = \sum (hg)(b_i) \otimes f_i(hg) = \sum g(b_i) \otimes f_i(g)$$

since hg is also a lift of g . In the second equality we used that $h(f_i(g)) = f_i(hg)$ since $f_i \in \text{Ind}_L^K(V)$.

It remains to show that $\bar{g} \mapsto \sum g(b_i) \otimes f_i(g)$ defines an element in $\text{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}(V))$. Let $\bar{h} \in \Gamma_L$ and h a lift of $\bar{h}$ in G_L . Then we have

$$\sum hg(b_i) \otimes f_i(hg) = \sum hg(b_i) \otimes h(f_i(g)) = \bar{h}(\sum g(b_i) \otimes f_i(g)).$$

κ is injective: It suffices to show that $\bar{\kappa}$ has trivial kernel. But this is easy to prove since if for all $g \in G_K$, $g(b) \otimes f(g) = 0$ we have either $f = 0$ or $g(b) = 0$ for all $g \in G_K$ which implies $b = 0$. This implies that κ is also injective.

κ is a map of (φ, Γ_K) -modules: First, κ commutes with the φ -action since

$$\begin{aligned} \kappa(\varphi(\sum b_i \otimes f_i))(\bar{g}) &= \kappa(\sum \sigma(b_i) \otimes f_i)(\bar{g}) = \sum g(\sigma(b_i)) \otimes f_i(g) \\ &= \sum \sigma(g(b_i)) \otimes f_i(g) = \varphi(\kappa(\sum b_i \otimes f_i)(\bar{g})) \\ &= \varphi(\kappa(\sum b_i \otimes f_i))(\bar{g}). \end{aligned}$$

We show κ is $\mathbf{B}_K$ -linear and Γ_K -equivariant. For $\alpha \in \mathbf{B}_K$ we have

$$\begin{aligned} \kappa(\alpha(\sum b_i \otimes f_i))(\bar{g}) &= \kappa(\sum \alpha b_i \otimes f_i)(\bar{g}) = \sum g(\alpha b_i) \otimes f_i(g) \\ &= \sum g(\alpha)g(b_i) \otimes f_i(g) = g(\alpha)(\kappa(\sum b_i \otimes f_i)(\bar{g})) \\ &= (\alpha(\kappa(\sum b_i \otimes f_i)))(\bar{g}). \end{aligned}$$

For $\gamma \in \Gamma_K$ we have

$$\begin{aligned} \kappa(\gamma(\sum b_i \otimes f_i))(\bar{g}) &= \kappa(\sum \gamma(b_i) \otimes \gamma(f_i))(\bar{g}) = \sum g(\gamma(b_i)) \otimes f_i(g\gamma) \\ &= \kappa(\sum b_i \otimes f_i)(\bar{g}\gamma) = (\gamma(\kappa(\sum b_i \otimes f_i)))(\bar{g}). \end{aligned}$$

Using Remark 3.8.2 we can deduce that κ is in fact a $\mathbf{B}_K$ -linear isomorphism.

For a free $\mathbb{Z}_p$ -representation V , we analogously regard the map κ for the p -adic representation $V \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ induced by

$$\begin{aligned} \bar{\kappa} : \mathbf{B} \otimes_{\mathbb{Z}_p} \text{Ind}_L^K(V) &\longrightarrow \{f : G_K \longrightarrow \mathbf{B} \otimes_{\mathbb{Z}_p} V \mid f(hg) = h(f(g)) \text{ for all } h \in G_L\} \\ b \otimes f &\longmapsto (g \mapsto \sum g(b) \otimes f(g)). \end{aligned}$$

Note that $g(b) \otimes f(g)$ lies in $\mathbf{A} \otimes_{\mathbb{Z}_p} V$ for all $g \in G_K$ if and only if $b \otimes f \in \mathbf{A} \otimes_{\mathbb{Z}_p} \text{Ind}_L^K(V)$. Therefore, κ induces an isomorphism of (φ, Γ_K) -modules between $\text{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}(V))$ and $\mathbf{D}(\text{Ind}_L^K(V))$. $\square$

Corollary 3.8.4. Let L be a finite extension of K . If V is an overconvergent p -adic representation (or a free $\mathbb{Z}_p$ -adic representation) of G_L then

$$\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V)) = \mathbf{D}^\dagger(\mathrm{Ind}_L^K(V)).$$

Proof. We have

$$\begin{aligned} \dim_{\mathbf{B}_K^\dagger}(\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V))) &= [\mathbf{B}_L^\dagger : \mathbf{B}_K^\dagger] \cdot [\Gamma_L : \Gamma_K] \cdot \dim_{\mathbf{B}_L^\dagger}(\mathbf{D}^\dagger(V)) \\ &= [L_\infty : K_\infty] \cdot [L \cap K_\infty : K] \cdot \dim_{\mathbf{B}_L^\dagger}(\mathbf{D}^\dagger(V)) \\ &= [L : L \cap K_\infty] \cdot [L \cap K_\infty : K] \cdot \dim_{\mathbf{B}_L^\dagger}(\mathbf{D}^\dagger(V)) \\ &= [L : K] \cdot \dim_{\mathbf{B}_L^\dagger}(\mathbf{D}^\dagger(V)) \\ &= \dim_{\mathbf{B}_K^\dagger}(\mathbf{D}^\dagger(\mathrm{Ind}_L^K(V))). \end{aligned}$$

where we used $[\mathbf{B}_L^\dagger : \mathbf{B}_K^\dagger] = [L_\infty : K_\infty]$ from Proposition 3.6.13. We now see that $\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V))$ and $\mathbf{D}^\dagger(\mathrm{Ind}_L^K(V))$ are both $\mathbf{B}_K^\dagger$ -vector spaces of maximal dimension inside $\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}(V)) = \mathbf{D}(\mathrm{Ind}_L^K(V))$. Further, κ from Proposition 3.8.3 maps $\mathbf{D}^\dagger(\mathrm{Ind}_L^K(V))$ into $\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V))$ since $\bar{g} \mapsto \sum g(b_i) \otimes f_i(g)$ is in $\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V))$ if $b_i \in \mathbf{B}^\dagger$. We deduce that $\mathrm{Ind}_{\Gamma_L}^{\Gamma_K}(\mathbf{D}^\dagger(V)) = \mathbf{D}^\dagger(\mathrm{Ind}_L^K(V))$. $\square$

3.9 One-dimensional (φ, Γ) -modules

Let $K = \mathbb{Q}_p$. We want to study one-dimensional (φ, Γ) -modules over $\mathbf{E}_{\mathbb{Q}_p}$. It turns out that their structure is very simple to describe.

Let $\delta : G_{\mathbb{Q}_p} \rightarrow \mathbb{F}_p^*$ be a continuous character. Since $\mathbb{F}_p^*$ is abelian, δ factors through $G_{\mathbb{Q}_p}^{\mathrm{ab}}$, the Galois group of the maximal abelian extension of $\mathbb{Q}_p$, denoted by $\mathbb{Q}_p^{\mathrm{ab}}$. By local class field theory we know that

$$\mathbb{Q}_p^{\mathrm{ab}} = \mathbb{Q}_p^{\mathrm{ur}} \cdot \mathbb{Q}_{p^\infty}$$

where $\mathbb{Q}_p^{\mathrm{ur}}$ is the maximal unramified abelian extension of $\mathbb{Q}_p$. Since $\mathbb{Q}_p^{\mathrm{ur}} \cap \mathbb{Q}_{p^\infty} = \mathbb{Q}_p$ we deduce that $G_{\mathbb{Q}_p}^{\mathrm{ab}} \cong \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ur}}/\mathbb{Q}_p) \times \Gamma_{\mathbb{Q}_p}$. Note that $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ur}}/\mathbb{Q}_p) \cong \mathrm{Gal}(\mathbb{F}_p^{\mathrm{sep}}/\mathbb{F}_p) \cong \hat{\mathbb{Z}}$ is topologically cyclic with canonical generator $\mathrm{Frob}_{\mathbb{Q}_p}$.

Definition 3.9.1. Let $\delta : G_{\mathbb{Q}_p} \rightarrow \mathbb{F}_p^*$ be a continuous character. We define $M(\delta) = \mathbf{E}_{\mathbb{Q}_p} \cdot v$ to be a one-dimensional $\mathbf{E}_{\mathbb{Q}_p}$ -vector space, equipped with a σ -semilinear Frobenius φ and $\Gamma_{\mathbb{Q}_p}$ -action given by

$$\begin{aligned} \varphi(\lambda \cdot v) &= \sigma(\lambda) \delta(\mathrm{Frob}_{\mathbb{Q}_p}^{-1}) \cdot v \\ \gamma(\lambda \cdot v) &= \gamma(\lambda) \delta(\gamma) \cdot v \end{aligned}$$

Since the vector space is one-dimensional, this indeed defines an étale $(\varphi, \Gamma_{\mathbb{Q}_p})$ -module.

Definition 3.9.2. Let $\delta : G_{\mathbb{Q}_p} \longrightarrow \mathbb{F}_p^*$ be a continuous character. We define the mod p representation $\mathbb{F}_p(\delta)$, which is a one-dimensional $\mathbb{F}_p$ -vector space $\mathbb{F}_p \cdot e$ with linear $G_{\mathbb{Q}_p}$ -action given by $g \cdot e = \delta(g)e$.

Clearly, any one-dimensional $\mathbb{F}_p$ -representation is given by such a character since $\mathrm{GL}_1(\mathbb{F}_p) = \mathbb{F}_p^*$.

Proposition 3.9.3. Let $\delta : G_{\mathbb{Q}_p} \longrightarrow \mathbb{F}_p^*$ be a continuous character. Then

$$\mathbf{D}(\mathbb{F}_p(\delta)) = M(\delta).$$

Proof. We have $\mathbf{D}(\mathbb{F}_p(\delta)) = (\mathbf{E} \otimes_{\mathbb{F}_p} \mathbb{F}_p(\delta))^{H_{\mathbb{Q}_p}}$. We can write every element of $\mathbf{E} \otimes_{\mathbb{F}_p} \mathbb{F}_p(\delta)$ as $b \otimes e$ for some $b \in \mathbf{E}$ and e a basis of $\mathbb{F}_p(\delta)$. The $G_{\mathbb{Q}_p}$ -action is then given by $g(b \otimes e) = g(b) \otimes g(e) = \delta(g)g(b) \otimes e$.

If we look at invariants under the action of $H_{\mathbb{Q}_p}$, we find that $\mathbf{D}(\mathbb{F}_p(\delta))$ consists of elements $b \otimes e$ such that $b = \delta(h)h(b)$ for all $h \in H_{\mathbb{Q}_p}$. Since the $H_{\mathbb{Q}_p}$ -action factors through $\mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ab}}/\mathbb{Q}_{p\infty}) \cong \mathrm{Gal}(\mathbb{Q}_p^{\mathrm{ur}}/\mathbb{Q}_p)$ it suffices to check that $b = \delta(\mathrm{Frob}_{\mathbb{Q}_p}^{-1})\mathrm{Frob}_{\mathbb{Q}_p}^{-1}(b)$ where we view $\mathrm{Frob}_{\mathbb{Q}_p}^{-1} \in G_{\mathbb{Q}_p}^{\mathrm{ab}}$ such that $\mathrm{Frob}_{\mathbb{Q}_p}^{-1}$ fixes $\mathbb{Q}_{p\infty}$.

Let us first assume the following lemma and prove it afterwards.

Lemma 3.9.4. There is a basis $b \otimes e$ of $\mathbf{D}(\mathbb{F}_p(\delta))$, with $b \in \mathbf{E}$ such that

$$\begin{aligned} \mathrm{Frob}_{\mathbb{Q}_p}^{-1}(b) &= \delta(\mathrm{Frob}_{\mathbb{Q}_p})b \\ \gamma(b) &= b \quad \text{for all } \gamma \in \Gamma_{\mathbb{Q}_p}, \\ \sigma(b) &= \delta(\mathrm{Frob}_{\mathbb{Q}_p}^{-1})b \end{aligned}$$

Using this lemma we have that $b \otimes e$ is a basis for $\mathbf{D}(\mathbb{F}_p(\delta))$ with φ -action given by

$$\varphi(b \otimes e) = \sigma(b) \otimes e = \delta(\mathrm{Frob}_{\mathbb{Q}_p}^{-1})(b \otimes e)$$

and residual action of $\Gamma_{\mathbb{Q}_p}$ given by

$$\gamma(b \otimes e) = \gamma(b) \otimes \gamma(e) = \delta(\gamma)(b \otimes e)$$

Thus we have exactly the structure of $M(\delta)$. □

Lemma 3.9.5. $\mathbf{E}$ contains $\mathbb{F}_p$ and $\mathbb{F}_p^{\mathrm{sep}}$ as subfields. For $x \in \mathbb{F}_p^{\mathrm{sep}} \subset \mathbf{E}$ we have that the action of any lift $\mathrm{Frob}_{\mathbb{Q}_p} \in G_{\mathbb{Q}_p}$ of $\mathrm{Frob}_{\mathbb{Q}_p} \in G_{\mathbb{Q}_p}^{\mathrm{ab}}$ is given by $\mathrm{Frob}_{\mathbb{Q}_p}(x) = x^p = \sigma(x)$, i.e. as the usual Frobenius action.

Proof. Clearly, any characteristic p field contains $\mathbb{F}_p$ as a subfield. Further, we constructed $\mathbf{E}$ as the separable closure of $\mathbb{F}_p((\epsilon - 1))$ which therefore contains $\mathbb{F}_p^{\mathrm{sep}}$. Recall

that $\tilde{\mathbf{E}}^+ = \{(x^{(0)}, x^{(1)}, \dots) : x^{(i+1)p} = x^{(i)}, x^{(i)} \in \mathcal{O}_{\mathbf{C}}/(p)\}$ with componentwise $G_{\mathbb{Q}_p}$ action. But since identify $\mathbb{F}_p^{\text{sep}}$ with sequences $(x^{(0)}, x^{(1)}, \dots)$ with $x^{(i)} \in \mathbb{F}_p^{\text{sep}}$ and on $\mathbb{F}_p^{\text{sep}}$ we have that any lift of $\text{Frob}_{\mathbb{Q}_p}$ acts as the p -th power map, we deduce the Lemma. $\square$

Proof of Lemma 3.9.4. By Hilbert 90 we know that any $H^1(\text{Gal}(\mathbb{F}_p^{\text{sep}}/\mathbb{F}_p), \mathbb{F}_p^{\text{sep}*}) = 0$, and this further tells us that any element $a \in \mathbb{F}_p^*$ can be written as $a = \frac{\text{Frob}_{\mathbb{F}_p}(b)}{b} = \frac{b^p}{b}$ for some $b \in \mathbb{F}_p^{\text{sep}*}$. For $a = \delta(\text{Frob}_{\mathbb{Q}_p}^{-1})$ we obtain $b \in \mathbb{F}_p^{\text{sep}*} \subset \mathbf{E}$ satisfying

$$\begin{aligned} \text{Frob}_{\mathbb{Q}_p}^{-1}(b) &= \delta(\text{Frob}_{\mathbb{Q}_p})b \\ \gamma(b) &= b \quad \text{for all } \gamma \in \Gamma_{\mathbb{Q}_p}, \\ \sigma(b) &= \delta(\text{Frob}_{\mathbb{Q}_p}^{-1})b \end{aligned}$$

where the second line follows since the Galois action of $\Gamma_{\mathbb{Q}_p}$ is trivial on $\mathbb{F}_p^{\text{sep}} \subset \mathbf{E}$. Also note that we used that since δ is multiplicative we have $\delta(\text{Frob}_{\mathbb{Q}_p}^{-1}) = \frac{1}{\delta(\text{Frob}_{\mathbb{Q}_p})}$.

Using the previous Lemma we find that $b \otimes e \in \mathbf{D}(\mathbb{F}_p(\delta))$ is a basis of the wanted form, in particular, the first row implies it indeed defines an element of $\mathbf{D}(\mathbb{F}_p(\delta))$. $\square$

Corollary 3.9.6. All 1-dimensional étale $(\varphi, \Gamma_{\mathbb{Q}_p})$ -modules over $\mathbf{E}_{\mathbb{Q}_p}$ are given by $M(\delta)$ for some continuous character $\delta : G_{\mathbb{Q}_p}^* \rightarrow \mathbb{F}_p^*$.

Proof. This directly follows from the equivalence of categories between mod p -representations of dimension 1 and étale (φ, Γ) -modules over $\mathbf{E}_{\mathbb{Q}_p}$ of dimension 1. $\square$

Chapter 4

Cohomology of (φ, Γ) -modules

Given the equivalence of categories it is a natural question to ask how Galois cohomology translates into the category of (φ, Γ_K) -modules. We will see that a very explicit complex of (φ, Γ_K) -modules calculates precisely the Galois cohomology of the corresponding Galois representation. This explicit description, purely using the theory of (φ, Γ_K) -modules, allows us to give alternative proofs for the Euler-Characteristic formula and local Tate duality, following [Her98] and [Her01].

4.1 Cohomology of φ -modules

Recall that $\mathbf{E}_K$ is a field of characteristic p and $\mathbf{E}$ a separable closure of $\mathbf{E}_K$.

Proposition 4.1.1. For $i > 0$, $H^i(G_{\mathbf{E}_K}, \mathbf{E} \otimes_{\mathbb{F}_p} V) = 0$.

Proof. By (3.1) we have $H^1(G_{\mathbf{E}_K}, \mathbf{E} \otimes_{\mathbb{F}_p} V) \cong H^1(G_{\mathbf{E}_K}, \mathbf{E} \otimes_{\mathbf{E}_K} \mathbf{D}(V))$. Now the $G_{\mathbf{E}_K}$ -action on $\mathbf{D}(V)$ is trivial, therefore as a $G_{\mathbf{E}_K}$ -module $\mathbf{E} \otimes_{\mathbf{E}_K} \mathbf{D}(V)$ is isomorphic to $\bigoplus_{i=1}^d \mathbf{E} v_i$ for $v_1, \dots, v_d$ a basis of $\mathbf{D}(V)$ over $\mathbf{E}_K$. The statement thus follows from $H^i(G_{\mathbf{E}_K}, \mathbf{E}) = 0$ for $i > 0$. $\square$

We have the classic Artin-Schreier-sequence of $G_{\mathbf{E}_K}$ -modules

$$0 \longrightarrow \mathbb{F}_p \longrightarrow \mathbf{E} \xrightarrow{\varphi-1} \mathbf{E} \longrightarrow 0.$$

Let V be a $\mathbb{F}_p$ -representation of $G_{\mathbf{E}_K}$, then we can tensor the exact sequence with V to get another short exact sequence

$$0 \longrightarrow V \longrightarrow \mathbf{E} \otimes_{\mathbb{F}_p} V \xrightarrow{\varphi-1} \mathbf{E} \otimes_{\mathbb{F}_p} V \longrightarrow 0.$$

From the long exact sequence of cohomology, using that $H^0(G_{\mathbf{E}_K}, \mathbf{E} \otimes_{\mathbb{F}_p} V) = (\mathbf{E} \otimes_{\mathbb{F}_p} V)^{G_{\mathbf{E}_K}} = \mathbf{D}(V)$ we have

$$0 \longrightarrow H^0(G_{\mathbf{E}_K}, V) \longrightarrow \mathbf{D}(V) \xrightarrow{\varphi-1} \mathbf{D}(V) \longrightarrow H^1(G_{\mathbf{E}_K}, V) \longrightarrow 0$$

where we used that $H^1(G_{\mathbf{E}_K}, \mathbf{E} \otimes_{\mathbb{F}_p} V) = 0$. We directly deduce that

$$\begin{aligned} H^0(G_{\mathbf{E}_K}, V) &= V^{G_{\mathbf{E}_K}} = \mathbf{D}(V)^{\varphi=1} \\ H^1(G_{\mathbf{E}_K}, V) &= \frac{\mathbf{D}(V)}{(\varphi-1)(\mathbf{D}(V))}. \end{aligned}$$

If we use that all higher cohomology groups vanish, we have derived the following theorem, which is phrased in a more formal way:

Theorem 4.1.2 ([Her00], Theorem 2). Let $M \in \Phi M_{\mathbf{E}}^{\text{ét}}$ and let $\mathcal{C}^\bullet(M)_\varphi$ denote the following complex

$$\mathcal{C}^\bullet(M)_\varphi : 0 \longrightarrow M \xrightarrow{\varphi-1} M \longrightarrow 0$$

For $(H^i)_{i \in \mathbb{N}}$ the cohomological functor we obtain a cohomological functor $(\mathcal{H}^i := H^i \circ \mathcal{C}_\varphi^\bullet)_{i \in \mathbb{N}}$ from $\Phi M_{\mathbf{E}_K}^{\text{ét}}$ to the category of abelian groups.

Then the functor $(\mathcal{H}^i \circ \mathbf{D})_{i \in \mathbb{N}}$ from $\text{Rep}_{\mathbb{F}_p}(G_{\mathbf{E}_K})$ to the category of abelian groups coincides with the Galois cohomology functor $(H^i(G_{\mathbf{E}_K}, \cdot))_{i \in \mathbb{N}}$.

For the case of free $\mathbb{Z}_p$ -representations V we can show a very similar statement. Recall that $\mathbf{A}$ is a Cohen ring for $\mathbf{E}$. The Artin-Schreier sequence lifts to a short exact sequence of $G_{\mathbf{E}_K}$ -modules

$$0 \longrightarrow \mathbb{Z}_p \longrightarrow \mathbf{A} \xrightarrow{\varphi-1} \mathbf{A} \longrightarrow 0.$$

Proceeding just as above we obtain

$$0 \longrightarrow V \longrightarrow \mathbf{A} \otimes_{\mathbb{Z}_p} V \xrightarrow{\varphi-1} \mathbf{A} \otimes_{\mathbb{Z}_p} V \longrightarrow 0.$$

and thus

$$\begin{aligned} H^0(G_{\mathbf{E}_K}, V) &= V^{G_{\mathbf{E}_K}} = \mathbf{D}(V)^{\varphi=1} \\ H^1(G_{\mathbf{E}_K}, V) &= \frac{\mathbf{D}(V)}{(\varphi-1)(\mathbf{D}(V))}. \end{aligned}$$

Example 4.1.3. We can verify part of Theorem 4.1.2 on a simple explicit example. Let $K = \mathbb{Q}_p$ and $V = \mathbb{F}_p$ with trivial Galois action. Then $\mathbf{D}(V) = (\mathbf{E} \otimes_{\mathbb{F}_p} \mathbb{F}_p)^{G_{\mathbf{E}_{\mathbb{Q}_p}}} = \mathbf{E}_{\mathbb{Q}_p} = \mathbb{F}_p((\pi))$. The φ -action then becomes

$$\varphi(\pi) = (1 + \pi)^p - 1 = \pi^p$$

and trivial on the coefficients. We see that φ fixes only the coefficients, and thus we obtain

$$H^0(G_{\mathbf{E}_{\mathbb{Q}_p}}, V) = \mathbf{D}(V)^{\varphi=1} = \mathbb{F}_p.$$

4.2 The Herr complex

This section closely follows Herr [Her98], who first introduced the description of Galois cohomology using (φ, Γ_K) -modules. Since Γ_K is an open subgroup of $\mathbb{Z}_p^*$ it is in particular procyclic, and we fix a topological generator γ of Γ_K for the entire chapter.

Let V be a $\mathbb{Z}_p$ -representation of G_K . We have

$$\begin{aligned} H^0(G_K, V) &= H^0(\Gamma_K, H^0(H_K, V)) = H^0(\Gamma_K, \mathbf{D}(V)^{\varphi=1}) \\ &= \mathbf{D}(V)^{\varphi=1, \gamma=1} = \mathbf{D}(V)^{\varphi=1, \gamma=1}. \end{aligned} \quad (4.1)$$

Rather surprisingly, the following very simple complex of (φ, Γ_K) -modules $M = \mathbf{D}(V)$ calculates the Galois cohomology of V . We will proceed to call this complex the *Herr complex*.

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M \longrightarrow 0 \\ & & x \longmapsto & & ((\varphi - 1)x, (\gamma - 1)x) & & \\ & & & & (x, y) \longmapsto & & (\gamma - 1)x - (\varphi - 1)y \end{array}$$

We can directly see by (4.1) that the 0-th cohomology group of the Herr complex indeed coincides with the Galois cohomology of V . In order to show that also higher cohomology groups agree, we will make use of the fact that in the category $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G_K)$ Galois cohomology is given as a derived functor.

Our goal for this section is to prove the following theorem:

Theorem 4.2.1 ([Her00], Theorem 4). Let $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ and $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ denote the follow complex

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M \longrightarrow 0 \\ & & x \longmapsto & & ((\varphi - 1)x, (\gamma - 1)x) & & \\ & & & & (x, y) \longmapsto & & (\gamma - 1)x - (\varphi - 1)y \end{array}$$

For $(H^i)_{i \in \mathbb{N}}$ the cohomological functor we obtain a functor $(\mathcal{H}^i := H^i \circ \mathcal{C}_{\varphi, \gamma}^\bullet)_{i \in \mathbb{N}}$ from $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ to the category of abelian groups.

Then the functor $(\mathcal{H}^i \circ \mathbf{D})_{i \in \mathbb{N}}$ from $\text{Rep}_{p\text{-tor}}(G_K)$ to the category of abelian groups coincides with the Galois cohomology functor $(H^i(G_K, \cdot))_{i \in \mathbb{N}}$.

We introduce notation for $\gamma - 1$ and $\varphi - 1$: For $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}}$ or $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$ and $x \in M$ let

$$\begin{aligned} \tau(x) &:= (\gamma - 1)(x) \\ \rho(x) &:= (\varphi - 1)(x). \end{aligned}$$

Remark 4.2.2. The Herr complex depends on the choice of γ . Let γ' be another topological generator of Γ_K . We view M as a $\Lambda[\Gamma_K]$ -module as seen in Proposition 2.4.4. By Remark 2.4.3 then $\frac{\gamma'-1}{\gamma'-1}$ is a well-defined homomorphism from M to itself. We then obtain the following morphism of complexes

$$\begin{array}{ccccccccc} \mathcal{C}_{\varphi, \gamma}^{\bullet}(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M & \longrightarrow & 0 & \longrightarrow & \dots \\ & \downarrow 0 & & \downarrow \frac{\gamma-1}{\gamma-1} & & \downarrow \frac{\gamma-1}{\gamma'-1} \otimes \text{id} & & \downarrow \text{id} & & \downarrow 0 & & \\ \mathcal{C}_{\varphi, \gamma'}^{\bullet}(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M & \longrightarrow & 0 & \longrightarrow & \dots \end{array}$$

It is easy to see that the diagram commutes and also that $\frac{\gamma-1}{\gamma'-1}$ is an isomorphism with inverse $\frac{\gamma'-1}{\gamma-1}$. Thus the diagram describes a quasi-isomorphism of complexes.

We now turn to the proof of Theorem 4.2.1. We will write

$$\mathcal{H}_{\gamma}^i(M) := H^i(\mathcal{C}_{\varphi, \gamma}^{\bullet}(M)).$$

Definition 4.2.3. Let $\mathcal{A}, \mathcal{B}$ be abelian categories. A δ -functor from $\mathcal{A}$ to $\mathcal{B}$ is given by the following data:

1. a collection $T^n : \mathcal{A} \longrightarrow \mathcal{B}$, $n \geq 0$ of additive functors, and
2. for every short exact sequence $0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$ of $\mathcal{A}$ a collection $\delta : T^n(C) \longrightarrow T^{n+1}(A)$, $n \geq 0$ of morphisms of $\mathcal{B}$.

such that

- (i) for every short exact sequence as above we have a long exact sequence

$$0 \longrightarrow T^0(A) \longrightarrow T^0(B) \longrightarrow T^0(C) \xrightarrow{\delta} T^1(A) \longrightarrow \dots$$

- (ii) for every morphism $(0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0) \rightarrow (0 \rightarrow A' \rightarrow B' \rightarrow C' \rightarrow 0)$ of short exact sequences of $\mathcal{A}$ the diagrams

$$\begin{array}{ccc} T^n(C) & \xrightarrow{\delta} & T^{n+1}(A) \\ \downarrow & & \downarrow \\ T^n(C') & \xrightarrow{\delta} & T^{n+1}(A') \end{array}$$

commute.

Definition 4.2.4. A functor $F : \mathcal{C} \longrightarrow \mathcal{D}$ between two abelian categories is called *effaceable* if for every M in $\mathcal{C}$ and every $x \in F(M)$ there exists a monomorphism u between M and some object N in $\mathcal{C}$, such that $F(u)(x) = 0$ in $F(N)$. We call a δ -functor $(T^n)_{n \geq 0}$ effaceable if T^n is effaceable for all $n \geq 0$.

The following theorem from Grothendieck's famous Tôhoku paper will be our main tool for proving Theorem 4.2.1.

Proposition 4.2.5 ([Gro57], Proposition 2.2.1). If $\mathcal{C}, \mathcal{D}$ are abelian categories and $\mathcal{C}$ has enough injectives, then every effaceable δ -functor from $\mathcal{C}$ to $\mathcal{D}$ is universal.

Since we already know that the group cohomology functor $H^i(G_K, \cdot)$ is an effaceable δ -functor, it will now suffice to prove that our newly defined cohomological functor of the Herr complex is also effaceable. However, in order to use Proposition 4.2.5 we first need to check that the cohomological functor $\mathcal{H}_\gamma^i$ indeed defines a δ -functor from $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$. To see that, let

$$0 \longrightarrow M \xrightarrow{f} N \xrightarrow{g} P \longrightarrow 0$$

be a short exact sequence in $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$, then we get maps of complexes

$$\begin{array}{ccccccc} & 0 & & 0 & & 0 & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & M & \xrightarrow{\alpha_M} & M \oplus M & \xrightarrow{\beta_M} & M \longrightarrow 0 \\ & & \downarrow f & & \downarrow (f, f) & & \downarrow f \\ 0 & \longrightarrow & N & \xrightarrow{\alpha_N} & N \oplus N & \xrightarrow{\beta_N} & N \longrightarrow 0 \\ & & \downarrow g & & \downarrow (g, g) & & \downarrow g \\ 0 & \longrightarrow & P & \xrightarrow{\alpha_P} & P \oplus P & \xrightarrow{\beta_P} & P \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 0 & & 0 & & 0 \end{array}$$

where we denote the maps from the complex $\mathcal{C}_{\varphi, \gamma}^\bullet(\cdot)$ by α and β . Using some basic homological algebra we can define the maps $\delta_0 : \mathcal{H}_\gamma^0(P) \longrightarrow \mathcal{H}_\gamma^1(M)$ and $\delta_1 : \mathcal{H}_\gamma^1(P) \longrightarrow \mathcal{H}_\gamma^2(M)$ such that we have a long exact sequence in the cohomology and also condition (ii) from Definition 4.2.3 is satisfied. In particular, the maps are given by

$$\delta_0 = [(f, f)^{-1} \circ \alpha_N \circ g^{-1}]$$

and

$$\delta_1 = [f^{-1} \circ \beta_N \circ (g, g)^{-1}]$$

and we can show that they are well-defined on the cohomology groups of the complexes. Using these maps we obtain our long exact sequence of cohomology

$$0 \longrightarrow \mathcal{H}_\gamma^0(M) \longrightarrow \mathcal{H}_\gamma^0(N) \longrightarrow \mathcal{H}_\gamma^0(P) \xrightarrow{\delta} \mathcal{H}_\gamma^1(M) \longrightarrow \mathcal{H}_\gamma^1(N) \longrightarrow \dots$$

and also for a morphism of short exact sequences $(0 \rightarrow M \rightarrow N \rightarrow P \rightarrow 0) \longrightarrow (0 \rightarrow$

$M' \rightarrow N' \rightarrow P' \rightarrow 0$) we get that

$$\begin{array}{ccc} \mathcal{H}_\gamma^0(P) & \xrightarrow{\delta_0} & \mathcal{H}_\gamma^1(M) \\ \downarrow & & \downarrow \\ \mathcal{H}_\gamma^0(P') & \xrightarrow{\delta_0} & \mathcal{H}_\gamma^1(M') \end{array}$$

and

$$\begin{array}{ccc} \mathcal{H}_\gamma^1(P) & \xrightarrow{\delta_1} & \mathcal{H}_\gamma^2(M) \\ \downarrow & & \downarrow \\ \mathcal{H}_\gamma^1(P') & \xrightarrow{\delta_1} & \mathcal{H}_\gamma^2(M) \end{array}$$

commute.

Proof of 4.2.1. We see from 4.2.5 that if $\mathcal{C}$ has enough injectives, then for $(T^n)_{n \geq 0}$ an effaceable delta functor, T^0 is left exact and T^i is isomorphic to the i -th derived functor $R^i T^0$. We will work in the category $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$ or $\text{Rep}_{p\text{-tor}}^{\text{discrete}}(G_K)$ since it has enough injectives.

We already checked that $\mathcal{H}^i \circ \mathbf{D}$ agrees with $H^i(G_K, \cdot)$ in degree zero. Further, we know that $H^i(G_K, \cdot)$ is the right i -th derived functor of $H^0(G_K, \cdot)$ and $\mathbf{D}$ is an equivalence of categories, thus it suffices to show that $\mathcal{H}^i$ is effaceable for each $i \geq 1$. In fact, since we see that the functor is zero for higher degrees, it suffices to show that $\mathcal{H}^i$ is effaceable for $i = 1, 2$. By definition, this means that for a given $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$, for every $x \in \mathcal{H}_\gamma^1(M)$ ($(x, y) \in \mathcal{H}_\gamma^2(M)$) there exists an $N \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$ and a monomorphism $u : M \hookrightarrow N$ such that $\bar{u}(x) = 0 \in \mathcal{H}_\gamma^1(N)$ ($\bar{u}(x, y) = 0 \in \mathcal{H}_\gamma^2(N)$), where $\bar{u}$ denotes the induced maps on cohomology groups of the Herr complex.

We will need the following Lemma:

Lemma 4.2.6 ([Her98], Lemma 2.2). Let M be an element of $\Phi\Gamma M_{\mathbf{A}_K}^{\acute{\text{e}}\text{t}, p\text{-tor}}$ and $x \in M$, then there exists an element N_x in $\Phi\Gamma M_{\mathbf{A}_K}^{\acute{\text{e}}\text{t}, p\text{-tor}}$ and an embedding $M \hookrightarrow N_x$ such that $x \in (\varphi - 1)N_x$.

Proof. See [Her98], Lemma 2.2. □

For an inductive limit $M = \varinjlim M_i$ of objects in $\Phi\Gamma M_{\mathbf{A}_K}^{\acute{\text{e}}\text{t}, p\text{-tor}}$ and $x \in M$, without loss of generality $x \in M_0$, let N_x^i be given by the lemma with M_i for $i \geq 0$. We then obtain a directed system $(N_x^0 \sqcup \dots \sqcup N_x^i)_i$. Taking the inverse limit gives us some N_x in $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$ with the same properties as in the lemma.

Let $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$. $i = 2$: For $x \in M$, the previous discussion gives us N_x in $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-}\acute{\text{e}}\text{t}, p\text{-tor}}$ and an embedding $M \hookrightarrow N_x$ such that $x \in (\varphi - 1)N_x$. But then we

have that x maps to zero in $\mathcal{H}_\gamma^2(N_x)$ since for some $y \in N_x$ we have $x = \rho(y)$ and thus x is the image of $(0, -y)$ under the boundary map $(x, y) \mapsto \tau(x) - \rho(y)$.

$i = 1$: Now let $(y, z) \in M \oplus M$ such that $\rho(y) = \tau(z)$. Let N_y be given by Lemma 4.2.6 and let $x \in N_y$ such that $\rho(x) = y$. Furthermore, since M is p -torsion, we can choose $s \in \mathbb{N}$ be such that

$$p^s y = p^s z = 0.$$

Let R denote a free $\mathbf{A}_K/p^s \mathbf{A}_K$ module with basis v and define $N_{y,z} := N_y \oplus R$, with semilinear actions given by

$$\begin{aligned}\varphi(v) &= v \\ \gamma(v) &= v + \tau(x) - z\end{aligned}$$

We see that

$$\varphi(\gamma(v)) = \varphi(v + \tau(x) - z) = \varphi(v) + \tau(x) - z = \gamma(\varphi(v))$$

where we used that $\rho(\tau(x) - z) = 0$, i.e. φ fixes $\tau(x) - z$. Further, $(\gamma - 1)(v) \in N_y$ and the Γ_K -action is continuous on N_y , so we also have a continuous Γ_K -action on $N_{y,z}$.

and the action of γ is continuous on $N_{y,z}$ so $N_{y,z}$ defines an element in $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$. By construction, we have an embedding $M \hookrightarrow N_{y,z}$. Furthermore, we have $y = \rho(x - v)$ and $z = \tau(x - v)$. This means that (y, z) maps to zero in $\mathcal{H}_\gamma^1(N_{y,z})$. $\square$

If we now want to derive the statement for a general $\mathbb{Z}_p$ -representation V , we can use Corollary 2.3.11 which tells us that

$$H^i(G_K, V) = \varprojlim H^i(G_K, V/p^n V).$$

We now further want to show that for M an arbitrary (φ, Γ_K) -module over $\mathbf{A}_K$ we have

$$\mathcal{H}_\gamma^i(M) = \varprojlim \mathcal{H}_\gamma^i(M/p^n M)$$

Proposition 4.2.7. Let M be an étale (φ, Γ_K) -module over $\mathbf{A}_K$ and let $M_n := M/p^n M$. Then we have that $(\mathcal{H}_\gamma^i(M_n))_{n \in \mathbb{N}}$ satisfies the Mittag-Leffler conditions for $i = 0, 1, 2$ and

$$\mathcal{H}_\gamma^i(M) = \varprojlim \mathcal{H}_\gamma^i(M/p^n M).$$

Proof. To ease notation let us denote

$$\begin{aligned}\alpha_n : M_n &\longrightarrow M_n \oplus M_n \\ x &\longmapsto ((\varphi - 1)x, (\gamma - 1)x) \\ \beta_n : M_n \oplus M_n &\longrightarrow M_n \\ (x, y) &\longmapsto (\gamma - 1)x - (\varphi - 1)y\end{aligned}$$

and similarly

$$\begin{aligned}\alpha : M &\longrightarrow M \oplus M \\ x &\longmapsto ((\varphi - 1)x, (\gamma - 1)x) \\ \beta : M \oplus M &\longrightarrow M \\ (x, y) &\longmapsto (\gamma - 1)x - (\varphi - 1)y.\end{aligned}$$

We obtain a commutative diagram

$$\begin{array}{ccccccc} & \cdots & & \cdots & & \cdots & \\ & \downarrow & & \downarrow & & \downarrow & \\ 0 & \longrightarrow & M_n & \xrightarrow{\alpha_n} & M_n \oplus M_n & \xrightarrow{\beta_n} & M_n \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & M_{n+1} & \xrightarrow{\alpha_{n+1}} & M_{n+1} \oplus M_{n+1} & \xrightarrow{\beta_{n+1}} & M_{n+1} \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & \cdots & & \cdots & & \cdots \end{array} \quad (4.2)$$

$$0 \longrightarrow M \xrightarrow{\alpha} M \oplus M \xrightarrow{\beta} M \longrightarrow 0$$

We can directly note that $\ker(\alpha) = \varprojlim \ker(\alpha_n)$ and $\ker(\beta) = \varprojlim \ker(\beta_n)$. Note that the transition maps of $(M_n)_{n \in \mathbb{N}}$ are given by quotient maps and are thus surjective, in particular, $(M_n)_{n \in \mathbb{N}}$ and $(M_n \oplus M_n)_{n \in \mathbb{N}}$ satisfy the Mittag-Leffler conditions. Further, $\ker(\alpha_n) = H^0(G_K, V/p^n V)$ for a $\mathbb{Z}_p$ -representations V of G_K and $H^0(G_K, V/p^n V)$ satisfies the Mittag-Leffler conditions by Corollary 2.3.11 since it is an inverse system of finite abelian groups.

Consider the short exact sequence of inverse systems

$$0 \longrightarrow (\ker(\alpha_n)) \longrightarrow (M_n) \longrightarrow (\operatorname{im}(\alpha_n)) \longrightarrow 0. \quad (4.3)$$

As $(\ker(\alpha_n))$ satisfies the Mittag-Leffler conditions, by Proposition 2.1.12 we obtain the following short exact sequence

$$0 \longrightarrow \ker(\alpha) \longrightarrow M \longrightarrow \varprojlim \operatorname{im}(\alpha_n) \longrightarrow 0.$$

so we can deduce that

$$\operatorname{im}(\alpha) = \varprojlim \operatorname{im}(\alpha_n)$$

and the same way that

$$\operatorname{im}(\beta) = \varprojlim \operatorname{im}(\beta_n).$$

Further, we have that since the transition maps of (M_n) are surjective, the short exact sequence (4.3) implies that $(\operatorname{im}(\alpha_n))$ also has surjective transition maps and satisfies the Mittag-Leffler conditions. By applying Proposition 2.1.12 to

$$0 \longrightarrow (\operatorname{im}(\alpha_n)) \longrightarrow (\ker(\beta_n)) \longrightarrow (\ker(\beta_n)/\operatorname{im}(\alpha_n)) \longrightarrow 0$$

we obtain a short exact sequence

$$0 \longrightarrow \operatorname{im}(\alpha) \longrightarrow \ker(\beta) \longrightarrow \varprojlim \ker(\beta_n)/\operatorname{im}(\alpha_n) \longrightarrow 0.$$

We deduce that

$$\ker(\beta)/\operatorname{im}(\alpha) = \varprojlim \ker(\beta_n)/\operatorname{im}(\alpha_n).$$

This suffices to show that the cohomology groups of the last row of the diagram (4.2) are the inverse limits of the cohomology groups $\mathcal{H}_\gamma^i(M_n)$. $\square$

Corollary 4.2.8. Let V be a $\mathbb{Z}_p$ -representation of G_K . Then we have

$$H^i(G_K, V) = H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))).$$

Proof. This follows directly from our previous results via

$$H^i(G_K, V) = \varprojlim H^i(G_K, V/p^n V) = \varprojlim H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V/p^n V))) = H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))).$$

Note that we are using that

$$\mathbf{D}(V/p^n V) = (\mathbf{A}_K \otimes_{\mathbb{Z}_p} V/p^n V)^{H_K} = \mathbf{D}(V)/p^n \mathbf{D}(V). \quad \square$$

Corollary 4.2.9 ([Her98], Theorem A). Let V be a $\mathbb{Z}_p$ -representation of G_K . Then we have

$$H^i(G_K, V) = 0$$

for $i \geq 3$.

Example 4.2.10. We can now use the Herr complex to partially re-derive the results from Example 2.3.8. For example, let $K = \mathbb{Q}_p$ and consider the trivial mod p representation $V = \mathbb{F}_p$. We have that $\mathbf{D}(V) = (\mathbf{A} \otimes_{\mathbb{Z}_p} \mathbb{F}_p)^{H_{\mathbb{Q}_p}} \cong (\mathbf{E}^{\text{sep}})^{H_{\mathbb{Q}_p}} \cong \mathbb{F}_p((\pi))$ and

$$\begin{aligned} \varphi(\pi) &= \pi^p \\ \gamma(\pi) &= (1 + \pi)^{\chi(\gamma)} - 1 \end{aligned}$$

We first note that just like in Example 4.1.3, the only elements that are fixed under $\varphi - 1$ are the constant elements without π -terms. Since these are fixed under γ , we deduce

$$H^0(G_{\mathbb{Q}_p}, \mathbb{F}_p) = \mathbb{F}_p.$$

Further, we have already seen that $H^2(G_{\mathbb{Q}_p}, \mathbb{F}_p)$ should be trivial, i.e., the map

$$\begin{aligned} \mathbf{D}(V) \oplus \mathbf{D}(V) &\longrightarrow \mathbf{D}(V) \\ (x, y) &\longmapsto (\gamma - 1)x - (\varphi - 1)y \end{aligned}$$

is surjective. Explicitly checking that is quite difficult, but we can easily verify that all π^n for $n \in \mathbb{N}$ lie in $(\gamma - 1)\mathbf{D}(V) - (\varphi - 1)\mathbf{D}(V)$. To see that, let $x = \sum_{k \geq 0} \pi^{p^k} \in \mathbf{E}_{\mathbb{Q}_p}$ then we have

$$\varphi(x) = \sum_{k \geq 0} \varphi(\pi)^{p^k} = \sum_{k \geq 0} \pi^{p^{k+1}}$$

and thus

$$(\varphi - 1)(x) = \sum_{k \geq 0} \pi^{p^{k+1}} - \sum_{k \geq 0} \pi^{p^k} = -\pi,$$

i.e., $\pi \in (\varphi - 1)(\mathbf{D}(V))$. Similarly, for $n \in \mathbb{N}$, $x_n = \sum_{k \geq 0} \pi^{np^k}$ we have $(\varphi - 1)(x) = -\pi^n$.

The dual representation given by $V = \mu_p$ corresponds to the (φ, Γ_K) -module $\mathbf{D}(V) = (\mathbf{A} \otimes_{\mathbb{Z}_p} \mu_p)^{H_{\mathbb{Q}_p}} \cong \mathbb{F}_p((\pi)) \cdot v$ with actions

$$\begin{aligned} \varphi(v) &= v \\ \varphi(\pi) &= \pi^p \\ \gamma(v) &= \chi(\gamma) \cdot v \\ \gamma(\pi) &= (1 + \pi)^{\chi(\gamma)} - 1 \end{aligned}$$

The only elements that are fixed under $\varphi - 1$ are the elements $\lambda \cdot v$ for $\lambda \in \mathbb{F}_p$, but

$$\gamma(\lambda \cdot v) = \lambda \chi(\gamma) \cdot v$$

so they are not fixed under γ , so we deduce

$$H^0(G_{\mathbb{Q}_p}, \mathbb{F}_p) = 0.$$

4.3 An explicit 1-cocycle

Let V be a $\mathbb{Z}_p$ -representation of G_K and let $M = \mathbf{D}(V)$. Now that we have already established that

$$H^i(G_K, V) = H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$$

on an abstract level, we want to further explore what this implies on a more explicit level for $i = 1$. We closely follow [CC99].

$H^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$: Set

$$\begin{aligned} Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(M)) &= \ker((x, y) \mapsto (\gamma - 1)x - (\varphi - 1)y) \\ B^1(\mathcal{C}_{\varphi, \gamma}^\bullet(M)) &= \text{im}(x \mapsto ((\varphi - 1)x, (\gamma - 1)x)) \end{aligned}$$

as the 1-cocycles and 1-cocboundaries of the Herr complex. Then elements of $H^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$ are given as

$$H^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))) = \frac{Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))}{B^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))}.$$

$H^1(G_K, M)$: Recall that we also have an explicit description of the Galois cohomology groups using a cochain complex. We have that

$$H^1(G_K, M) = \frac{Z^1(G_K, M)}{B^1(G_K, M)}.$$

where $Z^1(G_K, M)$ are the 1-cocycles and $B^1(G_K, M)$ the 1-coboundaries of the cochain complex of G_K .

Our goal is to establish an explicit isomorphism for V a $\mathbb{Z}_p$ -representation

$$H^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))) = \frac{Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))}{B^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))} \xrightarrow{\sim} \frac{Z^1(G_K, \mathbf{D}(V))}{B^1(G_K, \mathbf{D}(V))}$$

induced by a map

$$\begin{aligned} Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))) &\longrightarrow Z^1(G_K, \mathbf{D}(V)) \\ (x, y) &\longmapsto c_{(x, y)}. \end{aligned}$$

By Remark 2.4.4 we can view $\mathbf{D}(V)$ as a $\Lambda[\Gamma_K]$ -module, where $\Lambda[\Gamma_K]$ is the Iwasawa algebra of Γ_K . By Remark 2.4.3 we see that for any $\sigma \in \Gamma_K$, $\frac{\sigma-1}{\gamma-1} \in \Lambda[\Gamma_K]$, thus, for $y \in \mathbf{D}(V)$ we have $\frac{\sigma-1}{\gamma-1}y \in \mathbf{D}(V)$.

Remark 4.3.1. Recall that $\sigma - 1 : \mathbf{A} \longrightarrow \mathbf{A}$ is surjective. This is part of the lifted Artin-Schreier sequence. Then $\varphi - 1 : \mathbf{A} \otimes_{\mathbb{Z}_p} V \longrightarrow \mathbf{A} \otimes_{\mathbb{Z}_p} V$, $a \otimes v \longmapsto (\sigma - 1)(a) \otimes v$ is also surjective.

Proposition 4.3.2. Let $(x, y) \in Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$ and let $b \in \mathbf{A} \otimes_{\mathbb{Z}_p} V$ satisfy $(\varphi - 1)b = x$. Then the map

$$\begin{aligned} c_{(x, y)} : G_K &\longrightarrow V \\ c_{(x, y)}(g) &= \frac{g-1}{\gamma-1}y - (g-1)b \end{aligned}$$

defines a 1-cocycle and

$$\begin{aligned} \iota_{\varphi, \gamma} : Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V))) &\longrightarrow Z^1(G_K, V) \\ (x, y) &\longmapsto c_{(x, y)} \end{aligned}$$

induces an isomorphism on cohomology.

Proof. $\iota_{\varphi, \gamma}$ is well-defined: By the previous discussion, we see that $\frac{g-1}{\gamma-1}y \in \mathbf{D}(V)$ for any $g \in G_K$. Thus $\frac{g-1}{\gamma-1}y - (g-1)b$ is an element of $\mathbf{A} \otimes_{\mathbb{Z}_p} V \cong \mathbf{A} \otimes_{\mathbf{A}_K} \mathbf{D}(V)$ (see (3.2)). To show it is in V we simply show it is invariant under φ . Remember that since $(x, y) \in Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$, we have $(\gamma-1)x = (\varphi-1)y$.

$$\begin{aligned} (\varphi-1) \left(\frac{g-1}{\gamma-1}y - (g-1)b \right) &= \frac{g-1}{\gamma-1}(\varphi-1)y - (g-1)(\varphi-1)b \\ &= \frac{g-1}{\gamma-1}(\gamma-1)x - (g-1)x \\ &= 0 \end{aligned}$$

$c_{(x, y)}$ defines a cocycle: We have that for $g, h \in G_K$

$$\begin{aligned} c_{(x, y)}(gh) &= \frac{gh-1}{\gamma-1}y - (gh-1)b \\ &= \frac{gh-g}{\gamma-1}y + \frac{g-1}{\gamma-1}y - (gh-g)b - (g-1)b \\ &= gc_{(x, y)}(h) + c_{(x, y)}(g). \end{aligned}$$

Maps from G_K to V that satisfy $c(gh) = gc(h) + c(g)$ are precisely the 1-cocycles (so-called *crossed homomorphisms*).

$\iota_{\varphi, \gamma}$ induces a group homomorphism on cohomology groups: First, it is clear that coboundaries are being sent to coboundaries. We first show that $\iota_{\varphi, \gamma}$ is independent of choice of b . Indeed, if b and b' both satisfy $(\varphi-1)b = (\varphi-1)b' = x$ then $b-b' \in V$. We then have

$$\frac{g-1}{\gamma-1}y - (g-1)b - \frac{g-1}{\gamma-1}y - (g-1)b' = (g-1)(b-b')$$

and $g \mapsto (g-1)(b-b')$ is an element of $B^1(G_K, V)$ (so-called *principal crossed homomorphisms*). Now we can verify $(x, y) \mapsto c_{(x, y)}$ defines a group homomorphism on cohomology groups. Let $(x, y), (x', y') \in Z^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$ and $b, b' \in \mathbf{A} \otimes_{\mathbb{Z}_p} V$ satisfying $(\varphi-1)(b) = x$ and $(\varphi-1)(b') = x'$ then

$$(c_{(x, y)} + c_{(x', y')})(g) = \frac{g-1}{\gamma-1}(y+y') - (g-1)(b+b')$$

and $(\varphi-1)(b+b') = x+x'$. Therefore $c_{(x, y)} + c_{(x', y')}$ and $c_{(x+x', y+y')}$ define the same element in $\frac{Z^1(G_K, V)}{B^1(G_K, V)}$.

$\iota_{\varphi, \gamma}$ is injective: Assume $c_{(x, y)}$ is a coboundary. Then there exists a $z \in V$ such that for all $g \in G_K$ we have $c_{(x, y)}(g) = (g-1)(z)$, i.e., $\frac{g-1}{\gamma-1}y = (g-1)(b+z)$, in particular for all $g \in H_K$

$$0 = \frac{g-1}{\gamma-1}y = (g-1)(b+z)$$

so $b+z \in (\mathbf{A} \otimes_{\mathbb{Z}_p} V)^{H_K} = \mathbf{D}(V)$. Furthermore, we have

$$y = \frac{\gamma-1}{\gamma-1}y = (\gamma-1)(b+z)$$

and $(\varphi - 1)b = x$ so $(x, y) = ((\varphi - 1)(b + z), (\gamma - 1)(b + z)) \in B^1(\mathcal{C}_{\varphi, \gamma}^\bullet(\mathbf{D}(V)))$.

$\iota_{\varphi, \gamma}$ is surjective: Let $c \in H^1(G_K, V)$ and let $g \mapsto c_g$ be a 1-cocycle representing c . We define an extension V' of $\mathbb{Z}_p$ by V as the direct sum of V and a free $\mathbb{Z}_p$ module with generator e , i.e. $V' = V \oplus \mathbb{Z}_p \cdot e$, with G_K -action $g(e) = e + c_g$ and G_K -action on V as defined by the representation. This is a well-defined group action since for $g, h \in G_K$

$$\begin{aligned} (gh)(e) &= e + c_{gh} = e + gc_h + c_g \\ g(h(e)) &= g(e + c_h) = e + c_g + gc_h \end{aligned}$$

where we used that $c_{gh} = gc_h + c_g$ since $g \mapsto c_g$ is a cocycle. We now have a short exact sequence of G_K -modules

$$0 \longrightarrow V \longrightarrow V' \xrightarrow{\kappa} \mathbb{Z}_p \longrightarrow 0.$$

Next, we apply the functor $\mathbf{D}$ to obtain a short exact sequence of (φ, Γ_K) -modules

$$0 \longrightarrow \mathbf{D}(V) \longrightarrow \mathbf{D}(V') \xrightarrow{\kappa} \mathbf{A}_K \longrightarrow 0.$$

Let $\tilde{e} \in \mathbf{D}(V')$ be a lift of $1 \in \mathbb{Z}_p \subset \mathbf{A}_K$. Define $x = (\varphi - 1)\tilde{e} \in \mathbf{D}(V')$ and $y = (\gamma - 1)\tilde{e} \in \mathbf{D}(V')$. Note that $\kappa(x) = (\varphi - 1)\kappa(\tilde{e}) = 0$ and $\kappa(y) = (\gamma - 1)\kappa(\tilde{e}) = 0$ since φ and Γ_K act trivially on $\mathbb{Z}_p$. Therefore, x, y define elements in $\mathbf{D}(V)$ and since φ and γ commute, $(x, y) \in Z^1(G_K, V)$. Finally, $b = \tilde{e} - e \in \mathbf{A} \otimes_{\mathbb{Z}_p} V'$ satisfies $(\varphi - 1)b = (\varphi - 1)(\tilde{e} - e) = x$ since $e \in V'$ is fixed under Frobenius, so we have

$$c_{(x, y)}(g) = \frac{g-1}{\gamma-1}y - (g-1)b = (g-1)\tilde{e} - (g-1)b = (g-1)e = c_g.$$

We deduce that $\iota_{\varphi, \gamma}$ is surjective. $\square$

4.4 The ψ -operator

We closely follow [Her98] and [CC98]. It is possible to define a one-sided inverse to the Frobenius σ , which will give an alternative version of the Herr complex.

Remark 4.4.1. The Frobenius σ on $\mathbf{A}$ or $\mathbf{A}_K$ is the lift of the absolute Frobenius on the non-perfect residue fields. We can see in the following paper [DK14] Remark 1.5 that σ is injective but not bijective, since the residue fields are not perfect. We can show that $\sigma(\mathbf{B}) \subset \mathbf{B}$ and $\sigma(\mathbf{B}_K) \subset \mathbf{B}_K$ are purely inseparable field extensions of degree p . In both cases $1, [\epsilon], \dots, [\epsilon]^{p-1}$ defines a basis for $\mathbf{B}$ or $\mathbf{B}_K$ over $\sigma(\mathbf{B})$ or $\sigma(\mathbf{B}_K)$.

Definition 4.4.2. Define the ψ -operator to be

$$\begin{aligned} \psi : \mathbf{B} &\longrightarrow \mathbf{B} \\ \psi(x) &= \frac{1}{p}\sigma^{-1}(\text{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x)) \end{aligned}$$

Remark 4.4.3. We view $\mathbf{B}$ as a p -dimensional vector space over $\sigma(\mathbf{B})$. Then $\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x)$ is simply the trace of the $\sigma(\mathbf{B})$ -linear map $\mathbf{B} \rightarrow \mathbf{B}$, $y \mapsto xy$. We can clearly see that for $x \in \sigma(\mathbf{B})$ for any basis the matrix of $y \mapsto xy$ is given by

$$\begin{pmatrix} x & & & \\ & \ddots & & \\ & & \ddots & \\ & & & x \end{pmatrix}$$

so $\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x) = px$. Thus, we easily see that ψ indeed defines a one-sided inverse to σ :

$$\psi(\sigma(x)) = \frac{1}{p}\sigma^{-1}(\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(\sigma(x))) = \frac{1}{p}\sigma^{-1}(p(\sigma(x))) = x$$

Remark 4.4.4. We can also give an explicit description of ψ in the basis $1, [\epsilon], \dots, [\epsilon]^{p-1}$ defines a basis for $\mathbf{B}$ over $\sigma(\mathbf{B})$. For an element $x = x_0 + x_1[\epsilon] + \dots + x_{p-1}[\epsilon]^{p-1}$ with $x_i \in \sigma(\mathbf{B})$, we see that the matrix of the linear map $y \mapsto xy$ has the form

$$\begin{pmatrix} x_0 & x_{p-1}[\epsilon]^p & \dots & x_1[\epsilon]^p \\ x_1 & x_0 & \dots & \vdots \\ \vdots & \ddots & \ddots & x_{p-1}[\epsilon]^p \\ x_{p-1} & \dots & x_1 & x_0 \end{pmatrix}$$

using that $[\epsilon]^p = \sigma([\epsilon]) \in \sigma(\mathbf{B})$. Thus $\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x_0 + x_1[\epsilon] + \dots + x_{p-1}[\epsilon]^{p-1}) = px_0$. We deduce the formula

$$\begin{aligned} \psi(x_0 + x_1[\epsilon] + \dots + x_{p-1}[\epsilon]^{p-1}) &= \frac{1}{p}\sigma^{-1}(\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x_0 + x_1[\epsilon] + \dots + x_{p-1}[\epsilon]^{p-1})) \\ &= \sigma^{-1}(x_0). \end{aligned}$$

The ψ -operator commutes with the action of G_K on $\mathbf{A}$. Therefore, $a \otimes v \mapsto \psi(a) \otimes v$ defines an operator on any $\mathbf{D}(V)$ for any $\mathbb{Z}_p$ -representation V . Thus by the equivalence of categories, on any (φ, Γ_K) -module over $\mathbf{A}_K$ there exists an additive, surjective operator

$$\psi = \psi_M : M \longrightarrow M$$

that satisfies

$$\begin{aligned} \psi_M(a\varphi(x)) &= \psi(a)x, \quad a \in \mathbf{A}_K, x \in M \\ \psi_M(\sigma(a)x) &= a\psi_M(x), \quad a \in \mathbf{A}_K, x \in M. \end{aligned}$$

In particular, we have $\psi_M \circ \varphi = id_M$ and ψ is surjective. We will generally just write ψ for ψ_M .

The next lemma states that $\gamma - 1$ is invertible on the kernel of the ψ -operator. This result is in fact very deep, and requires a lot of work to prove.

Lemma 4.4.5. Let V be a p -torsion representation of G_K , then $\gamma - 1 : \mathbf{D}(V)^{\psi=0} \rightarrow \mathbf{D}(V)^{\psi=0}$ is bijective.

Proof. Since the proof is extremely long, see Appendix A. $\square$

The goal of this section is to show the following proposition.

Proposition 4.4.6 ([Her98], Prop.4.1). Let M be a p -torsion (φ, Γ_K) -module over $\mathbf{A}_K$ and let $\mathcal{C}_{\psi, \gamma}^\bullet(M)$ denote the follow complex

$$\begin{array}{ccccccc} 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M \longrightarrow 0 \\ & & x \longmapsto & & ((\psi - 1)x, (\gamma - 1)x) & & \\ & & & & (x, y) \longmapsto & & (\gamma - 1)x - (\psi - 1)y \end{array}$$

Furthermore, let $f : M \oplus M \rightarrow M \oplus M$ denote the map $(x, y) \mapsto (-\psi(x), y)$. Then the morphism of complexes

$$\begin{array}{ccccccc} \mathcal{C}_{\varphi, \gamma}^\bullet(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow M \longrightarrow 0 \longrightarrow \dots \\ & \downarrow 0 & & \downarrow id & & \downarrow f & & \downarrow -\psi & & \downarrow 0 \\ \mathcal{C}_{\psi, \gamma}^\bullet(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow M \longrightarrow 0 \longrightarrow \dots \end{array}$$

is a quasi-isomorphism.

In particular, if $M = \mathbf{D}(V)$ for some representation V , the cohomology of the complex $\mathcal{C}_{\psi, \gamma}^\bullet(M)$ also calculates the Galois cohomology of V .

Proof. The commutativity of the diagram follows directly from the fact that $\psi \circ \varphi = id_M$. Now

$$\mathcal{B}^\bullet : 0 \longrightarrow 0 \longrightarrow \ker(\psi) \oplus 0 \xrightarrow{\gamma-1} \ker(\psi) \longrightarrow 0$$

defines a subcomplex of $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ since

$$\begin{array}{ccccccc} \mathcal{B}^\bullet : & 0 & \longrightarrow & 0 & \longrightarrow & \ker(\psi) \oplus 0 & \longrightarrow \ker(\psi) \longrightarrow 0 \longrightarrow \dots \\ & \downarrow 0 & & \downarrow 0 & & \downarrow incl & & \downarrow incl & & \downarrow 0 \\ \mathcal{C}_{\varphi, \gamma}^\bullet(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow M \longrightarrow 0 \longrightarrow \dots \end{array}$$

commutes. Therefore, we can look at the quotient complex $\mathcal{C}_{\varphi, \gamma}^\bullet(M)/\mathcal{B}^\bullet$

$$0 \longrightarrow M \longrightarrow \frac{M}{\ker(\psi)} \oplus M \longrightarrow \frac{M}{\ker(\psi)} \longrightarrow 0.$$

We now get a commutative diagram

$$\begin{array}{ccccccccccc}
 \mathcal{C}_{\varphi, \gamma}^{\bullet}(M)/\mathcal{B}^{\bullet} : & 0 & \longrightarrow & M & \longrightarrow & \frac{M}{\ker(\psi)} \oplus M & \longrightarrow & \frac{M}{\ker(\psi)} & \longrightarrow & 0 & \longrightarrow \dots \\
 & \downarrow 0 & & \downarrow id & & \downarrow f & & \downarrow -\psi & & \downarrow 0 & \\
 \mathcal{C}_{\psi, \gamma}^{\bullet}(M) : & 0 & \longrightarrow & M & \longrightarrow & M \oplus M & \longrightarrow & M & \longrightarrow & 0 & \longrightarrow \dots
 \end{array}$$

where the vertical maps are all isomorphisms since ψ is surjective. This shows that we have an isomorphism from $\mathcal{C}_{\varphi, \gamma}^{\bullet}(M)/\mathcal{B}^{\bullet}$ to $\mathcal{C}_{\psi, \gamma}^{\bullet}(M)$. It remains to show that $\mathcal{B}^{\bullet}$ is acyclic, which follows since $\gamma - 1 : \ker \psi \rightarrow \ker \psi$ is bijective (see Lemma 4.4.5). $\square$

4.5 Euler-Characteristic Formula

The goal of this section is to use (φ, Γ_K) -modules and the explicit descriptions of the Galois cohomology groups to prove the following theorem, known as the Euler-Characteristic formula.

Theorem 4.5.1 ([Her98], Theorem B). Let V be a p -torsion representation of G_K . Then $H^i(G_K, V)$ are finite and

$$\chi(V) := \frac{\#H^0(G_K, V) \cdot \#H^2(G_K, V)}{\#H^1(G_K, V)} = p^{-[K:\mathbb{Q}_p]\ell(V)}$$

where $\ell(V)$ is the length of V as a $\mathbb{Z}_p$ -module.

By passing to the inverse limit we obtain:

Corollary 4.5.2. Let V be a free $\mathbb{Z}_p$ -representation of G_K . Then $H^i(G_K, V)$ are finitely generated and

$$\chi(V) := \text{rank } H^0(G_K, V) - \text{rank } H^1(G_K, V) + \text{rank } H^2(G_K, V) = -[K:\mathbb{Q}_p]\text{rank } V$$

where rank denotes the rank as a $\mathbb{Z}_p$ -module.

4.5.1 The filtration associated to $\mathcal{C}_{\psi, \gamma}^{\bullet}$

We introduce some properties of the complex $\mathcal{C}_{\psi, \gamma}^{\bullet}$ that will help us prove the Euler-Characteristic formula for (φ, Γ_K) -modules. Let M be a p -torsion (φ, Γ_K) -module.

We regard a filtration of the complex $\mathcal{C}_{\psi, \gamma}^{\bullet}$:

$$\mathcal{C}_{\psi, \gamma}(M) = \mathcal{C}^0(M) \supset \mathcal{C}^1(M) \supset \mathcal{C}^2(M) \supset \mathcal{C}^3(M)$$

where

$$\begin{aligned}
 \mathcal{C}^1(M) : & 0 \longrightarrow M^{\psi=1} \xrightarrow{\gamma-1} M^{\psi=1} \longrightarrow 0 \longrightarrow 0, \\
 \mathcal{C}^2(M) : & 0 \longrightarrow M^{\varphi=1} \xrightarrow{\gamma-1} M^{\varphi=1} \longrightarrow 0 \longrightarrow 0, \\
 \mathcal{C}^3(M) : & 0 \longrightarrow 0 \longrightarrow 0 \longrightarrow \dots
 \end{aligned}$$

Note that $M^{\psi=1} \supset M^{\varphi=1}$ since if $\varphi(x) = x$ then $\psi(x) = \psi(\varphi(x)) = x$. We also define the grading

$$\mathrm{gr}^i \mathcal{C}(M) = \mathcal{C}^i(M) / \mathcal{C}^{i+1}(M)$$

for $i \in \{0, 1, 2\}$. For $M = \mathbf{D}(V)$, $M^{\varphi=1} = (A \otimes_{\mathbb{Z}_p} V)^{H_K, \varphi=1} = V^{H_K}$. Therefore,

$$\mathrm{gr}^2 \mathcal{C}(M) = \mathcal{C}^2(M) : 0 \longrightarrow V^{H_K} \xrightarrow{\gamma-1} V^{H_K} \longrightarrow 0 \longrightarrow 0$$

Furthermore

$$\mathrm{gr}^1 \mathcal{C}(M) : 0 \longrightarrow M^{\psi=1} / M^{\varphi=1} \xrightarrow{\gamma-1} M^{\psi=1} / M^{\varphi=1} \longrightarrow 0 \longrightarrow 0$$

and

$$\mathrm{gr}^0 \mathcal{C}(M) : 0 \longrightarrow M / M^{\psi=1} \xrightarrow{\gamma-1} M \oplus M / M^{\psi=1} \longrightarrow M \longrightarrow 0$$

Lemma 4.5.3 ([Her98], Lemma 4.2). The complex $\mathrm{gr}^0 \mathcal{C}(M)$ is quasi-isomorphic to

$$0 \longrightarrow 0 \longrightarrow \frac{M}{(\psi-1)M} \xrightarrow{\gamma-1} \frac{M}{(\psi-1)M} \longrightarrow 0 \longrightarrow 0.$$

Proof. This is the quotient complex of $\mathrm{gr}^0 \mathcal{C}(M)$ and the subcomplex

$$0 \longrightarrow \frac{M}{M^{\psi=1}} \longrightarrow (\psi-1)M \oplus \frac{M}{M^{\psi=1}} \longrightarrow (\psi-1)M \longrightarrow 0 \longrightarrow 0.$$

It remains to show this complex is acyclic, which is fairly straightforward.

Degree zero: We show that

$$\begin{aligned} \frac{M}{M^{\psi=1}} &\longrightarrow (\psi-1)M \oplus \frac{M}{M^{\psi=1}} \\ x &\longmapsto ((\psi-1)x, (\gamma-1)x) \end{aligned}$$

is injective. But this is clear since if x is fixed by ψ then by definition $x \in M^{\psi=1}$.

Degree one: Let $(x, y) \in (\psi-1)M \oplus \frac{M}{M^{\psi=1}}$ with $(\gamma-1)x = (\psi-1)y$. Since $x \in (\psi-1)M$, there exists $z \in M$ such that $(\psi-1)z = x$. Then we also have that $(\gamma-1)(\psi-1)z = (\psi-1)y$ i.e., $(\gamma-1)z - y \in M^{\psi=1}$. Thus in the quotient $\frac{M}{M^{\psi=1}}$, $(\gamma-1)z$ and y represent the same element. It follows that $((\psi-1)z, (\gamma-1)z) = (x, y)$ in $(\psi-1)M \oplus \frac{M}{M^{\psi=1}}$.

Degree two: Finally, the map

$$\begin{aligned} (\psi-1)M \oplus \frac{M}{M^{\psi=1}} &\longrightarrow (\psi-1)M \\ (x, y) &\longmapsto (\gamma-1)x - (\psi-1)y \end{aligned}$$

is clearly surjective since for $y \in (\psi-1)M$ with $(\psi-1)z = y$ we have $(0, -z)$ maps to y . $\square$

To simplify notation, let $H^i(\Gamma_K, N)$ denote the cohomology groups of the complex

$$0 \longrightarrow N \xrightarrow{\gamma-1} N \longrightarrow 0 \longrightarrow 0.$$

In particular, $H^i(\mathcal{C}^1(M)) = H^i(\Gamma_K, M^{\psi=1})$ and $H^i(\mathcal{C}^2(M)) = H^i(\Gamma_K, M^{\varphi=1})$.

Theorem 4.5.4 ([Her98], Theorem 4.3). For all $i \in \mathbb{N}$ we have

- (i) $H^i(\mathrm{gr}^0 \mathcal{C}(M)) = H^{i-1}(\Gamma_K, \frac{M}{(\psi-1)M})$
- (ii) $H^i(\mathrm{gr}^1 \mathcal{C}(M)) = H^i(\Gamma_K, (\varphi-1)(M^{\psi=1}))$ and in particular $H^i(\mathrm{gr}^1 \mathcal{C}(M)) = \{0\}$ for $i \neq 1$
- (iii) $H^i(\mathrm{gr}^2 \mathcal{C}(M)) = H^i(\Gamma_K, M^{\varphi=1}) = H^i(\Gamma_K, V^{H_K})$

The filtration gives us

$$\mathcal{H}^i(M) \supset H^i(\Gamma_K, M^{\psi=1}) \supset H^i(\Gamma_K, M^{\varphi=1})$$

with

$$\frac{\mathcal{H}^i(M)}{H^i(\Gamma_K, M^{\psi=1})} = H^i(\mathrm{gr}^0 \mathcal{C}(M)) \quad \text{and} \quad \frac{H^i(\Gamma_K, M^{\psi=1})}{H^i(\Gamma_K, M^{\varphi=1})} = H^i(\mathrm{gr}^1 \mathcal{C}(M)).$$

Proof. Everything is obvious except (ii). For (ii) note that $\varphi-1$ induces a Γ_K -invariant isomorphism

$$\varphi-1 : \frac{M^{\psi=1}}{M^{\varphi=1}} \longrightarrow (\varphi-1)(M^{\psi=1}).$$

So we have a quasi-isomorphism

$$\begin{array}{ccccccc} 0 & \longrightarrow & \frac{M^{\psi=1}}{M^{\varphi=1}} & \xrightarrow{\gamma-1} & \frac{M^{\psi=1}}{M^{\varphi=1}} & \longrightarrow & 0 \longrightarrow 0 \\ & & \downarrow \varphi-1 & & \downarrow \varphi-1 & & \\ 0 & \longrightarrow & (\varphi-1)(M^{\psi=1}) & \xrightarrow{\gamma-1} & (\varphi-1)(M^{\psi=1}) & \longrightarrow & 0 \longrightarrow 0. \end{array}$$

Finally note that $(\varphi-1)(M^{\psi=1}) \subset \ker(\psi)$ and we already know that $\gamma-1$ is injective on $\ker(\psi)$ (see Lemma 4.4.5), which implies $H^i(\mathrm{gr}^1 \mathcal{C}(M)) = \{0\}$ for $i \neq 1$. $\square$

4.5.2 Proof of the Euler-Characteristic Formula

For the proof we need two more lemmas, that are proved in [Her98].

Lemma 4.5.5 ([Her98], Proposition 3.6). Let $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$. Then $\frac{M}{(\psi-1)M}$ is finite.

Lemma 4.5.6 ([Her98], Cor. 3.11). Let $M = \mathbf{D}(V) \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$. Let $c(M) := (\varphi-1)M^{\psi=1} = (\varphi-1)M \cap \ker(\psi)$ then $\frac{c(M)}{(\gamma-1)c(M)}$ is a finite group of order $p^{[K:\mathbb{Q}_p]\ell(V)}$, where $\ell(V)$ is the length of V as a $\mathbb{Z}_p$ -module.

Proof of Theorem 4.5.1. We make heavy use of Theorem 4.5.4. Also note that if N is a finite Γ_K -module and $H^i(\Gamma_K, N)$ as before denote the cohomology groups of

$$0 \longrightarrow N \xrightarrow{\gamma-1} N \longrightarrow 0 \longrightarrow 0$$

we have

- (i) $H^i(\Gamma_K, N) = 0$ for $i \notin \{0, 1\}$
- (ii) $\#H^0(\Gamma_K, N) = \#H^1(\Gamma_K, N)$ and both are finite.

Both statements follow directly from basic group theory.

Let $M = \mathbf{D}(V)$. We know that $H^i(\mathrm{gr}^0\mathcal{C}(M))$ and $H^i(\Gamma_K, M^{\varphi=1}) = H^i(\Gamma_K, V^{H_K})$ are finite since $\frac{M}{(\psi-1)M}$ (by Lemma 4.5.5) and V^{H_K} are finite and $H^i(\mathrm{gr}^1\mathcal{C}(M))$ are finite by Lemma 4.5.6. Using Theorem 4.5.4 we can directly deduce that $\mathcal{H}^i(M)$ are finite.

We now have

$$\begin{aligned}
 \chi(V) &= \frac{\#H^0(G_K, V) \cdot \#H^2(G_K, V)}{\#H^1(G_K, V)} \\
 &= \frac{\#\mathcal{H}^0(M) \cdot \#\mathcal{H}^2(M)}{\#\mathcal{H}^1(M)} \\
 &= \frac{\#H^0(\mathrm{gr}^0\mathcal{C}(M))\#H^0(\Gamma_K, M^{\psi=1}) \cdot \#H^2(\mathrm{gr}^0\mathcal{C}(M))\#H^2(\Gamma_K, M^{\psi=1})}{\#H^1(\mathrm{gr}^0\mathcal{C}(M))\#H^1(\Gamma_K, M^{\psi=1})} \\
 &\stackrel{(1)}{=} \frac{\#H^0(\Gamma_K, M^{\psi=1}) \cdot \#H^2(\mathrm{gr}^0\mathcal{C}(M))}{\#H^1(\mathrm{gr}^0\mathcal{C}(M))\#H^1(\Gamma_K, M^{\psi=1})} \\
 &\stackrel{(2)}{=} \frac{\#H^0(\Gamma_K, M^{\psi=1})}{\#H^1(\Gamma_K, M^{\psi=1})} \\
 &\stackrel{(3)}{=} \frac{\#H^0(\mathrm{gr}^1\mathcal{C}(M))\#H^0(\Gamma_K, M^{\varphi=1})}{\#H^1(\mathrm{gr}^1\mathcal{C}(M))\#H^1(\Gamma_K, M^{\varphi=1})} \\
 &\stackrel{(4)}{=} \frac{1}{\#H^1(\mathrm{gr}^1\mathcal{C}(M))} \\
 &\stackrel{(5)}{=} p^{-[K:\mathbb{Q}_p]\ell(V)}
 \end{aligned}$$

where we used

- (1) $\#H^0(\mathrm{gr}^0\mathcal{C}(M))$ and $H^2(\Gamma_K, M^{\psi=1})$ are trivial
- (2) $\#H^2(\mathrm{gr}^0\mathcal{C}(M)) = \#H^1(\Gamma_K, \frac{M}{(\psi-1)M}) = \#H^0(\Gamma_K, \frac{M}{(\psi-1)M}) = \#H^1(\mathrm{gr}^0\mathcal{C}(M))$ using Theorem 4.5.4 (i) and that $\frac{M}{(\psi-1)M}$ is finite
- (3) $\frac{H^i(\Gamma_K, M^{\psi=1})}{H^i(\Gamma_K, M^{\varphi=1})} = H^i(\mathrm{gr}^1\mathcal{C}(M))$ for $i = 0, 1$ by Theorem 4.5.4
- (4) $H^i(\Gamma_K, M^{\varphi=1}) = H^i(\Gamma_K, V^{H_K})$ and V^{H_K} is finite thus $\#H^0(\Gamma_K, M^{\varphi=1}) = \#H^1(\Gamma_K, M^{\varphi=1})$, further, $\#H^0(\mathrm{gr}^1\mathcal{C}(M))$ is trivial.

(5) $\#H^1(\mathrm{gr}^1\mathcal{C}(M))) = \frac{(\varphi-1)M^{\psi=1}}{(\gamma-1)(\varphi-1)M^{\psi=1}} = \frac{c(M)}{(\gamma-1)c(M)}$ by Theorem 4.5.4 (ii). The statement then follows directly from Lemma 4.5.6. $\square$

4.6 Local Tate Duality

The goal of the section is to prove the following theorem using (φ, Γ_K) -modules. For the entire section, we fix a topological generator γ , and by writing $\mathcal{H}^i$ we mean the cohomological functor of the Herr complex with the chosen γ . In order to ease notation we will only present the proof of local Tate duality for the case $K = K_0$, however, the proof works almost analogously for K a possibly ramified extension (see [Her01]). For the rest of the section, always assume $K = K_0$. Note that if $K = K_0$, then K_∞ will have the same residue field as K .

Theorem 4.6.1 ([Her98], Theorem C). Let V be a p -torsion representation of G_K and let $V^*(1) = \mathrm{Hom}(V, \mu_{p^\infty})$. Then there is a canonical isomorphism

$$H^2(G_K, \mu_{p^\infty}) \xrightarrow{\sim} \mathbb{Q}_p/\mathbb{Z}_p.$$

Further, for $0 \leq i \leq 2$, the cup-product

$$H^i(G_K, V) \times H^{2-i}(G_K, V^*(1)) \longrightarrow H^2(G_K, \mu_{p^\infty}) \cong \mathbb{Q}_p/\mathbb{Z}_p$$

is a perfect pairing.

First, we want to find an explicit description for the cup-product in terms of (φ, Γ_K) -modules. To do so, we first need a description of the tensor-product in the category of $\Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$.

Definition 4.6.2. Let $M, N \in \Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$. We define a tensor product in the category $\Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$, given as $M \otimes_{\mathbf{A}_K} N$ with diagonal Γ_K and φ -action.

We also define $\mathrm{Hom}_{\mathbf{A}_K}(M, N) \in \Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$ with

$$\begin{aligned} \varphi_{\mathrm{Hom}_{\mathbf{A}_K}(M, N)}(f)(\varphi_M(m)) &= \varphi_N(f(m)) \\ \gamma(f)(\gamma(m)) &= \gamma(f(m)). \end{aligned} \tag{4.4}$$

for $f \in \mathrm{Hom}_{\mathbf{A}_K}(M, N)$. Note that since M is étale, $\varphi_M(M)$ generates M over $\mathbf{A}_K$ and thus the formula determines $\varphi_{\mathrm{Hom}_{\mathbf{A}_K}(M, N)}(f) \in \mathrm{Hom}_{\mathbf{A}_K}(M, N)$ uniquely.

Lemma 4.6.3 ([Her01], Proposition 3.5). Let V, W be two p -torsion representations of G_K . The natural maps

$$\begin{aligned} \mathbf{D}(V) \otimes_{\mathbf{A}_K} \mathbf{D}(W) &\longrightarrow \mathbf{D}(V \otimes_{\mathbb{Z}_p} W) \\ \mathrm{Hom}_{\mathbf{A}_K}(\mathbf{D}(V), \mathbf{D}(W)) &\longrightarrow \mathbf{D}(\mathrm{Hom}_{\mathbb{Z}_p}(V, W)) \end{aligned} \tag{4.5}$$

are isomorphisms of (φ, Γ_K) -modules over $\mathbf{A}_K$.

Proposition 4.6.4 ([Her01]). Let (M, φ_M) and (N, φ_N) be objects in $\Phi\Gamma M_{\mathbf{A}_K}^{\text{ind-ét}, p\text{-tor}}$. Let $\bar{x}$ denote the cohomology class of an element x . Then we find the following description of the cup-product

$$\begin{aligned}
 \text{(i)} \quad & \mathcal{H}^0(M) \times \mathcal{H}^0(N) \longrightarrow \mathcal{H}^0(M \otimes_{\mathbf{A}_K} N) \\
 & (x, y) \longmapsto x \otimes y \\
 \text{(ii)} \quad & \mathcal{H}^0(M) \times \mathcal{H}^1(N) \longrightarrow \mathcal{H}^1(M \otimes_{\mathbf{A}_K} N) \\
 & (x, \overline{(y, z)}) \longmapsto \overline{(x \otimes y, x \otimes z)} \\
 \text{(iii)} \quad & \mathcal{H}^1(M) \times \mathcal{H}^0(N) \longrightarrow \mathcal{H}^1(M \otimes_{\mathbf{A}_K} N) \\
 & (\overline{(y, z)}, x) \longmapsto \overline{(y \otimes x, z \otimes x)} \\
 \text{(iv)} \quad & \mathcal{H}^0(M) \times \mathcal{H}^2(N) \longrightarrow \mathcal{H}^2(M \otimes_{\mathbf{A}_K} N) \\
 & (x, \overline{y}) \longmapsto \overline{x \otimes y} \\
 \text{(v)} \quad & \mathcal{H}^2(M) \times \mathcal{H}^0(N) \longrightarrow \mathcal{H}^2(M \otimes_{\mathbf{A}_K} N) \\
 & (\overline{x}, y) \longmapsto \overline{x \otimes y} \\
 \text{(vi)} \quad & \mathcal{H}^1(M) \times \mathcal{H}^1(N) \longrightarrow \mathcal{H}^2(M \otimes_{\mathbf{A}_K} N) \\
 & (\overline{(x, y)}, \overline{(a, b)}) \longmapsto \overline{y \otimes \gamma(a) - x \otimes \varphi_N(b)}
 \end{aligned}$$

Proof. We only show that the given maps are indeed well-defined. For the verification that these maps fulfil the characteristic properties of the cup-product, as given in Proposition 2.1.14, we refer to [Her01] Proposition 4.4.

(i) For $x \in \mathcal{H}_\gamma^0(M)$, $y \in \mathcal{H}_\gamma^0(N)$ we have

$$\tau(x \otimes y) = \tau(x) \otimes \gamma(y) + x \otimes \tau(y) = 0$$

and

$$\rho_{M \otimes N}(x \otimes y) = \rho_M(x) \otimes \varphi_N(y) + x \otimes \rho_N(y) = 0$$

thus $x \otimes y$ indeed defines an element in $\mathcal{H}_\gamma^0(M \otimes_{\mathbf{A}_K} N)$. Bi-additivity is clear from the formula.

(ii) Let $x \in \mathcal{H}_\gamma^0(M)$, $\overline{(y, z)} \in \mathcal{H}_\gamma^1(N)$.

We show $(x \otimes y, x \otimes z)$ defines an element in $\mathcal{H}_\gamma^1(M \otimes_{\mathbf{A}_K} N)$: We have

$$\begin{aligned}
 & \tau(x \otimes y) - \rho_{M \otimes N}(x \otimes z) \\
 &= \tau(x) \otimes \gamma(y) + x \otimes \tau(y) - (\rho_M(x) \otimes \varphi_N(z) + x \otimes \rho_N(z)) \\
 &= x \otimes \tau(y) - (x \otimes \rho_N(z)) \\
 &= 0.
 \end{aligned}$$

It is clear from the formula that the pairing is bi-additive.

Now let $x \in \mathcal{H}_\gamma^0(M)$ and $y \in N$. We show $(x, (\rho(y), \tau(y)))$ gets mapped to zero in $\mathcal{H}_\gamma^1(M \otimes_{\mathbf{A}_K} N)$: Since $\gamma(x) = x$ and $\varphi_M(x) = x$, we have

$$\begin{aligned} \rho(x \otimes y) &= \varphi_M(x) \otimes \varphi_N(y) - x \otimes y = x \otimes \rho(y) = \\ \rho(x \otimes y) &= \gamma(x) \otimes \gamma(y) - x \otimes y = \tau(x \otimes y). \end{aligned}$$

(iii) Entirely analogous to (ii).

(iv) For $x \in \mathcal{H}_\gamma^0(M)$, $y \in N$ a 2-coboundary, i.e. satisfying $y = \tau(z) - \rho(z')$ for $z, z' \in N$, we have

$$x \otimes y = \tau(x \otimes z) - \rho(x \otimes z')$$

so $x \otimes y$ is again a 2-coboundary.

(v) Entirely analogous to (iv). The bi-additivity of the pairing follows directly from the bi-additivity of the tensor-product and the additivity of γ and φ_N .

Let $\overline{(x, y)} \in \mathcal{H}_\gamma^1(M)$ and $a \in N$. We show $((x, y), (\rho(a), \tau(a)))$ maps to zero:

$$y \otimes \gamma(\rho(a)) - x \otimes \varphi_N(\tau(a)) = (y - x) \otimes \varphi_N(\gamma(a)) - y \otimes \gamma(a) + x \otimes \varphi_N(a)$$

if we now use that $y - x = \varphi_M(y) - \gamma(x)$ we get

$$\begin{aligned} (\varphi_M(y) - \gamma(x)) \otimes \varphi_N(\gamma(a)) - y \otimes \gamma(a) + x \otimes \varphi_N(a) \\ = \tau(x \otimes \varphi_N(a)) - \rho_{M \otimes N}(y \otimes \gamma(a)). \end{aligned}$$

Let $\overline{(a, b)} \in \mathcal{H}_\gamma^1(N)$ and $x \in N$. Finally, we show $((\rho(x), \tau(x)), (a, b))$ maps to zero:

$$\tau(x) \otimes \gamma(a) - \rho(x) \otimes \varphi_N(b) = \gamma(x \otimes a) - \varphi_{M \otimes N}(x \otimes b) + x \otimes (\varphi_N(b) - \gamma(a))$$

using $b - a = \varphi_N(b) - \gamma(a)$ we get

$$\begin{aligned} \gamma(x \otimes a) - \varphi_{M \otimes N}(x \otimes b) + x \otimes (b - a) \\ = \tau(x \otimes a) - \rho_{M \otimes N}(x \otimes b). \end{aligned}$$

□

Since the proof of local Tate duality is very long, we will give a vague proof sketch first.

Sketch of proof of Theorem 4.6.1. Let V be a p -torsion representation of G_K and $M = \mathbf{D}(V)$. For convenience, we also fix a topological generator γ of Γ_K . However, under the isomorphism described in Remark 4.2.2 the constructed maps will coincide.

1. We introduce the module of differential 1-forms Ω_K^1 over $\mathbf{A}_K$ which is a free rank one $\mathbf{A}_K$ -module generated by $d\pi$. Then we show that we can equip Ω_K^1 with the structure of a (φ, Γ_K) -module and show it is isomorphic to $\mathbf{D}(\mathbb{Z}_p(1))$. We further define the residue map satisfying $\text{Res}((\sum_{n \in \mathbb{Z}} a_n \pi^n) d\pi) = a_{-1}$.
2. We establish the Tate isomorphism

$$\mathcal{T}_K : \mathcal{H}_\gamma^2(\Omega_K^1) \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$$

which is induced by the map $\text{TR}_K = \text{Tr}_{K/\mathbb{Q}_p} \circ \text{Res}$. We further define $\tilde{M} = \text{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ such that $\mathbf{D}(\tilde{V}) = \tilde{M}$ where $\tilde{V} = \text{Hom}_{\mathbb{Z}_p}(V, \mu_{p^\infty})$. It remains now to show that

$$\mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(\tilde{M}) \xrightarrow{\cup} \mathcal{H}_\gamma^2(\Omega_K^1) \xrightarrow{\mathcal{T}_K} \mathbb{Q}_p/\mathbb{Z}_p$$

defines a perfect pairing.

3. We establish an isomorphism between $\tilde{M}$ and the Pontryagin dual $M^\vee = \text{Hom}_{\mathbb{Z}_p, c}(M, \mathbb{Q}_p/\mathbb{Z}_p)$ of M . We then construct an explicit isomorphism between $\mathcal{H}_\gamma^i(M^\vee) \xrightarrow{\sim} [\mathcal{H}_\gamma^{2-i}(M)]^\vee$ using Alexander-Pontryagin-Duality. Since this gives an explicit pairing, we can directly check that these formulas coincide under isomorphism with the cup-product formulas from Proposition 4.6.4 composed with $\mathcal{T}_K$. $\square$

4.6.1 Differential forms

Definition 4.6.5. Let $\text{Der}_{\mathbf{A}_K}$ denote the category consisting of pairs (M, d_M) , where M is a finitely generated $\mathbf{A}_K$ -module and d_M is a continuous $\mathcal{O}_K$ -linear derivation $\mathbf{A}_K \rightarrow M$.

We define $(\mathbf{A}_K, \frac{d}{d\pi}) \in \text{Der}_{\mathbf{A}_K}$ by

$$\frac{d}{d\pi} \sum_{n \in \mathbb{Z}} a_n \pi^n = \sum_{n \in \mathbb{Z}} (n+1) a_{n+1} \pi^n$$

where $a_n \rightarrow 0$ for $n \rightarrow -\infty$.

Given some $(M, d_M) \in \text{Der}_{\mathbf{A}_K}$ and $\lambda = \sum_{n \in \mathbb{Z}} a_n \pi^n \in \mathbf{A}_K$ we have

$$d_M(\lambda) = d_M(\sum_{n \in \mathbb{Z}} a_n \pi^n \in \mathbf{A}_K) = \sum_{n \in \mathbb{Z}} n a_n \pi^{n-1} d_M(\pi) = \frac{d}{d\pi}(\lambda) d_M(\pi)$$

by product rule, i.e., d_M is determined by $d_M(\pi)$. Any two free rank one $\mathbf{A}_K$ -modules M, N with derivations d_M, d_N , such that the images of d_M and d_N generate M and N , are thus isomorphic via a $\mathbf{A}_K$ -linear map that sends $d_M(\pi)$ to $d_N(\pi)$. Up to isomorphism we therefore have a unique free rank one $\mathbf{A}_K$ -module with continuous

derivation d , such that the image of d generates the module. We denote this module by (Ω_K^1, d) . Since by assumption $d\pi$ generates Ω_K^1 over $\mathbf{A}_K$, we can write Ω_K^1 as $\mathbf{A}_K d\pi$ with elements of the form $\sum_{n \in \mathbb{Z}} a_n \pi^n \cdot d\pi$.

Definition 4.6.6. We call $(\Omega_K^1, d) \in \text{Der}_{\mathbf{A}_K}$ the $\mathbf{A}_K$ -module of continuous differential forms of $\mathbf{A}_K$ over $\mathcal{O}_K$.

We define a residue map $\text{Res} : \Omega_K^1 \longrightarrow \mathcal{O}_K$ by

$$\text{Res}\left(\sum_{n \in \mathbb{Z}} a_n \pi^n \cdot d\pi\right) = a_{-1}.$$

Fontaine shows in [Fon07] 2.2.3 that the residue map is surjective and independent of choice of uniformizer π . We can endow Ω_K^1 with the structure of a (φ, Γ_K) -module by setting

$$\begin{aligned} \varphi(\lambda d\mu) &= \sigma(\lambda) \frac{1}{p} d(\sigma(\mu)) \\ \gamma(\lambda d\mu) &= \gamma(\lambda) d(\gamma(\mu)) \end{aligned}$$

for all $\lambda, \mu \in \mathbf{A}_K$, where σ is the usual Frobenius on $\mathbf{A}_K$.

Let $\mathbb{Z}_p(1)$ denote the $\mathbb{Z}_p$ -representation of G_K given by $\varprojlim \mu_{p^n}$. A basis for $\mathbb{Z}_p(1)$ is given by $\epsilon = (1, \zeta_p, \zeta_{p^2}, \dots)$, where we have the structure of a $\mathbb{Z}_p$ -module via $\alpha \cdot \epsilon = (1, \zeta_p^\alpha, \zeta_{p^2}^\alpha, \dots) = \epsilon^\alpha$ for $\alpha \in \mathbb{Z}_p$.

Theorem 4.6.7 ([Her01], Theorem 3.7). Ω_K^1 endowed with the structure of a (φ, Γ_K) -module as described above is naturally isomorphic to $\mathbf{D}(\mathbb{Z}_p(1))$.

We first prove the following lemma:

Lemma 4.6.8 ([Her01], Lemma 3.6). The $\mathbf{A}$ -linear map

$$\begin{aligned} \kappa_K : \mathbf{A} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p(1) &\longrightarrow \mathbf{A} \otimes_{\mathbf{A}_K} \Omega_K^1 \\ \lambda \otimes \epsilon &\longmapsto \lambda \otimes \frac{d[\epsilon]}{[\epsilon]} \end{aligned}$$

is an isomorphism of $\mathbf{A}$ -modules that does not depend on the choice of generator ϵ of $\mathbb{Z}_p(1)$.

Proof. The only thing we really need to show is that $d[\epsilon]$ is a basis of Ω_K^1 . But we know that $d\pi = d([\epsilon] - 1) = d[\epsilon]$. Therefore, ρ_K is an isomorphism and clearly $\mathbf{A}$ -linear. Also note that $\frac{1}{[\epsilon]} = \sum_{i \geq 0} (-\pi)^i \in \mathbf{A}_K$ thus the map is well-defined.

Consider a different generator of $\mathbb{Z}_p(1)$ given by $\epsilon' = \epsilon^\alpha$ for $\alpha \in \mathbb{Z}_p^*$, which essentially means we choose different compatible p^i -th roots of unity ζ_{p^i} . Defining

$$\begin{aligned} \kappa'_K : \mathbf{A} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p(1) &\longrightarrow \mathbf{A} \otimes_{\mathbf{A}_K} \Omega_K^1 \\ \lambda \otimes \epsilon' &\longmapsto \lambda \otimes \frac{d[\epsilon']}{[\epsilon']} \end{aligned}$$

we see that

$$\begin{aligned}
 \kappa'_K(\lambda \otimes \epsilon) &= \kappa'_K\left(\frac{\lambda}{\alpha} \otimes \epsilon^\alpha\right) \\
 &= \frac{\lambda}{\alpha} \otimes \frac{d[\epsilon^\alpha]}{[\epsilon^\alpha]} \\
 &= \frac{\lambda}{\alpha} \otimes \frac{\alpha[\epsilon]^{\alpha-1} d[\epsilon]}{[\epsilon]^\alpha} \\
 &= \lambda \otimes \frac{d[\epsilon]}{[\epsilon]}.
 \end{aligned}$$

Thus κ_K and κ'_K define the same map. $\square$

If we equip Ω_K^1 with the G_K -action given by

$$g(\lambda d\mu) = g(\lambda) d(g(\mu))$$

we can see that κ_K is G_K -equivariant if we consider $\mathbf{A} \otimes_{\mathbf{A}_K} \Omega_K^1$ with diagonal G_K -action.

Proof of Theorem 4.6.7. Using the G_K -equivariant map κ_K we obtain

$$\mathbf{D}(\mathbb{Z}_p(1)) = (\mathbf{A} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p(1))^{H_K} \xrightarrow{\sim} (\mathbf{A} \otimes_{\mathbf{A}_K} \Omega_K^1)^{H_K} \xleftarrow{\sim} \Omega_K^1$$

where the second arrow is induced by the inclusion, using that Ω_K^1 is indeed invariant under action of H_K since $\mathbf{A}_K$ is invariant under H_K . Both arrows are therefore Γ_K -equivariant.

Finally, we check that $\mathbf{D}(\mathbb{Z}_p(1))$ and Ω_K^1 have equivalent Frobenius actions. If $\lambda \otimes \epsilon \in (\mathbf{A} \otimes_{\mathbb{Z}_p} \mathbb{Z}_p(1))^{H_K}$ then $\lambda \in \mathbf{A}_K$ since ϵ is fixed under H_K . Then $\lambda \otimes \epsilon$ corresponds to $\lambda \frac{d[\epsilon]}{[\epsilon]} \in \Omega_K^1$ under the mentioned isomorphisms. Furthermore, we have

$$\varphi(\lambda \otimes \epsilon) = \sigma(\lambda) \otimes \epsilon$$

in $\mathbf{D}(\mathbb{Z}_p(1))$, corresponding to $\sigma(\lambda) \frac{d[\epsilon]}{[\epsilon]} \in \Omega_K^1$ under the above isomorphisms. We conclude by noting that

$$\varphi\left(\lambda \frac{d[\epsilon]}{[\epsilon]}\right) = \sigma\left(\frac{\lambda}{[\epsilon]}\right) \frac{1}{p} d(\sigma([\epsilon])) = \frac{\sigma(\lambda)}{[\epsilon]^p} \frac{1}{p} d([\epsilon]^p) = \frac{\sigma(\lambda)}{[\epsilon]^p} \frac{1}{p} p[\epsilon]^{p-1} d[\epsilon] = \sigma(\lambda) \frac{d[\epsilon]}{[\epsilon]}$$

so the Frobenius actions correspond to each other under the isomorphisms.

We deduce that $\mathbf{D}(\mathbb{Z}_p(1))$ and Ω_K^1 are isomorphic as (φ, Γ_K) -modules over $\mathbf{A}_K$. $\square$

4.6.2 The Tate-isomorphism

Definition 4.6.9. We define the $\mathbb{Z}_p$ -linear map

$$\mathrm{TR}_K : \Omega_K^1 \longrightarrow \mathbb{Z}_p$$

given as the composition $\mathrm{TR}_K = \mathrm{Tr}_{K/\mathbb{Q}_p} \circ \mathrm{Res}$.

Lemma 4.6.10 ([Her01], Lemma 5.1). For all $\omega \in \Omega_K^1$ and $\gamma \in \Gamma_K$ we have

$$\begin{aligned}\mathrm{TR}_K(\gamma(\omega) - \omega) &= 0 \\ \mathrm{TR}_K(\varphi(\omega) - \omega) &= 0\end{aligned}$$

Proof. For the first one, notice that γ commutes with the residue map since

$$\gamma\left(\sum_{n \in \mathbb{Z}} a_n \pi^n \cdot d\pi\right) = \sum_{n \in \mathbb{Z}} \gamma(a_n) \gamma(\pi)^n \cdot d\gamma(\pi)$$

and since $\gamma(\pi)$ is another uniformizer and the residue map does not depend on the choice of uniformizer we see that the residue is precisely $\gamma(a_{-1})$. Since γ fixes $\mathcal{O}_K$, $\mathrm{Tr}_{K/\mathbb{Q}_p}(\gamma(a_{-1}) - a_{-1}) = 0$ for all $a_{-1} \in \mathcal{O}_K$. The second statement follows similarly, for details, check [Her01], Lemma 5.1. $\square$

We now introduce the Tate-isomorphism. First, let $\Omega_{K,n}^1 := \frac{\Omega_K^1}{p^n \Omega_K^1}$, which then defines a p -torsion (φ, Γ_K) -module over $\mathbf{A}_K$ corresponding to the p -torsion $\mathbb{Z}_p$ -representation μ_{p^n} . Reducing modulo p^n we obtain a map

$$\mathrm{TR}_{K,n} : \Omega_{K,n}^1 \longrightarrow \mathbb{Z}/p^n \mathbb{Z}.$$

Proposition 4.6.11. $\mathrm{TR}_{K,n}$ factors through

$$\Omega_{K,n}^1 / ((\gamma - 1)(\Omega_{K,n}^1) - (\varphi - 1)(\Omega_{K,n}^1)) = \mathcal{H}^2(\Omega_{K,n}^1).$$

Proof. Let $x, y \in \Omega_{K,n}^1$. Then

$$\mathrm{TR}_{K,n}((\gamma - 1)(x) - (\varphi - 1)(y)) = \mathrm{TR}_{K,n}(\gamma(x) - x) - \mathrm{TR}_{K,n}(\varphi(y) - y) = 0$$

by the previous Lemma. $\square$

Therefore, we obtain a map

$$\mathrm{TR}_{K,n} : H^2(G_K, \mu_{p^n}) = \mathcal{H}^2(\Omega_{K,n}^1) \longrightarrow \mathbb{Z}/p^n \mathbb{Z}.$$

Since direct limits preserve exact sequences, we have that the direct limit commutes with our cohomological functor of the Herr complex, similar to Proposition 4.2.7. By passing to the direct limit on both sides, we obtain a map

$$\mathcal{H}^2(\varinjlim \Omega_{K,n}^1) \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p.$$

Proposition 4.6.12. As elements of $\Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ind-ét}, p\text{-tor}}$, $\varinjlim \Omega_{K,n}^1 = \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1$.

Proof. We have a Γ_K -equivariant $\mathbf{A}_K$ -linear isomorphism

$$\begin{aligned}\varinjlim \Omega_{K,n}^1 &\longrightarrow \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1 \\ xd[\epsilon] \in \Omega_{K,n}^1 &\longmapsto \frac{x}{p^n} \otimes d[\epsilon].\end{aligned}$$

which is also compatible with the Frobenius action since

$$\varphi_{\Omega_{K,n}^1}(xd[\epsilon]) = \sigma(x)\frac{1}{p}d(\sigma([\epsilon])) = \sigma(x)\frac{1}{p}d[\epsilon]^p = \sigma(x)[\epsilon]^{p-1}d[\epsilon]$$

gets mapped to

$$\frac{\sigma(x)[\epsilon]^{p-1}}{p^n} \otimes d[\epsilon] = \frac{\sigma(x)}{p^n} \otimes \sigma(x)[\epsilon]^{p-1}d[\epsilon] = \varphi\left(\frac{x}{p^n} \otimes d[\epsilon]\right) \quad \square$$

Theorem 4.6.13 ([Her01], Theorem 5.2). For each $n \geq 1$ we have isomorphisms

$$\mathrm{TR}_{K,n} : \mathcal{H}_\gamma^2(\Omega_{K,n}^1) \longrightarrow \mathbb{Z}/p^n \mathbb{Z}.$$

Passing to the inductive limit gives us a map $\mathcal{T}_K$, which is a natural isomorphism, called *Tate isomorphism*,

$$\mathcal{T}_K : \mathcal{H}^2(\varinjlim \Omega_{K,n}^1) = \mathcal{H}^2\left(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1\right) \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p.$$

Proof. It suffices to show that

$$\mathrm{TR}_{K,n} : \mathcal{H}_\gamma^2(\Omega_{K,n}^1) \longrightarrow \mathbb{Z}/p^n \mathbb{Z}.$$

define group isomorphisms for all $n \in \mathbb{N}$. Since the residue map and the trace map $\mathrm{Tr}_{K/\mathbb{Q}_p} : \mathcal{O}_K \longrightarrow \mathbb{Z}_p$ are surjective, we see that TR_K is surjective and thus $\mathrm{TR}_{K,n}$ is surjective. It suffices now to show that the cardinality of $\mathcal{H}_\gamma^2(\Omega_{K,n}^1) = H^2(G_K, \mu_{p^n})$ is precisely p^n .

To see this, we use the Euler-Characteristic formula (Theorem 4.5.1). We first calculate the cardinalities of $H^0(G_K, \mu_{p^n})$ and $H^1(G_K, \mu_{p^n})$. Clearly $H^0(G_K, \mu_{p^n})$ is trivial. We now use the Kummer sequence

$$0 \longrightarrow \mu_{p^n} \longrightarrow \bar{K}^* \xrightarrow{\cdot p^n} \bar{K}^* \longrightarrow 0.$$

Together with $H^1(G_K, \bar{K}^*) = 0$, from Hilbert 90, and the long exact sequence of cohomology we get $H^1(G_K, \mu_{p^n}) \cong K^*/(K^*)^{p^n}$. From [Lan86] Chapter 2 Proposition 6 we know that $|K^*/(K^*)^{p^n}| = p^{n[K:\mathbb{Q}_p]+n}$ for K an unramified extension of $\mathbb{Q}_p$. The Euler-Characteristic then gives

$$\#H^2(G_K, V) = \frac{p^{-[K:\mathbb{Q}_p]\ell(V)} \#H^1(G_K, V)}{\#H^0(G_K, V)} = p^n,$$

and we deduce that the $\mathrm{TR}_{K,n}$ are isomorphisms. Passing to the inductive limit yields an isomorphism between $\mathcal{H}^2(\varinjlim \Omega_{K,n}^1)$ and $\mathbb{Q}_p/\mathbb{Z}_p$.

Further note that this isomorphism does not depend on the choice of γ in the sense that for γ' another topological generator of Γ_K under the quasi-isomorphism of Herr complexes $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ and $\mathcal{C}_{\varphi, \gamma'}^\bullet(M)$ given in Remark 4.2.2, $\mathcal{T}_K$ is mapped to itself. Indeed, the quasi-isomorphism in Remark 4.2.2 gives a map from $\mathcal{H}_\gamma^2(M)$ to $\mathcal{H}_{\gamma'}^2(M)$, that is simply induced by the identity map on M . $\square$

For a (φ, Γ_K) -module M , let $\tilde{M} = \text{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ and for a $\mathbb{Z}_p$ -representation V of G_K define $\tilde{V} = \text{Hom}_{\mathbb{Z}_p}(V, \mu_{p^\infty})$. Now we see that

$$\mathbf{D}(\mu_{p^\infty}) = \varinjlim_{n \geq 1} \mathbf{D}(\mu_{p^n}) \cong \varinjlim_{n \geq 1} \Omega_{K,n}^1 \cong \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1.$$

Therefore we have

$$\begin{aligned} \mathbf{D}(\tilde{V}) &= \mathbf{D}(\text{Hom}_{\mathbb{Z}_p}(V, \mu_{p^\infty})) \cong \text{Hom}_{\mathbf{A}_K}(\mathbf{D}(V), \mathbf{D}(\mu_{p^\infty})) \\ &= \text{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) = \tilde{M}. \end{aligned}$$

where we used (4.5) for the isomorphism. We can now translate the statement of Tate duality into the context of (φ, Γ_K) -modules.

$$\begin{array}{ccccccc} \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(\tilde{M}) & \xrightarrow{\cup} & \mathcal{H}_\gamma^i(M \otimes_{\mathbf{A}_K} \tilde{M}) & \longrightarrow & \mathcal{H}_\gamma^i(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) & \xrightarrow{\mathcal{T}_K, \gamma} & \mathbb{Q}_p/\mathbb{Z}_p \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H^i(G_K, V) \times H^{2-i}(G_K, \tilde{V}) & \xrightarrow{\cup} & H^i(G_K, V \otimes_{\mathbb{Z}_p} \tilde{V}) & \longrightarrow & H^i(G_K, \mu_{p^\infty}) & \xrightarrow{\mathbf{v}(\mathcal{T}_K)} & \mathbb{Q}_p/\mathbb{Z}_p \end{array}$$

Remark 4.6.14. Recall that

$$\text{TR}_K = \text{Tr}_{K/\mathbb{Q}_p} \circ \text{Res} : \Omega_K^1 \longrightarrow \mathbb{Z}_p$$

and

$$\text{TR}_{K,n} : \Omega_{K,n}^1 \longrightarrow \mathbb{Z}/p^n\mathbb{Z}$$

is the map obtained by reduction modulo p . We will from now on abuse notation and also denote the induced map

$$\varinjlim \text{TR}_{K,n} : \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1 \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$$

by TR_K . Note that this map factors through $\mathcal{H}^2(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ and becomes the Tate isomorphism. We specifically denote the Tate isomorphism, i.e., the induced map on the second cohomology group, by $\mathcal{T}_K : \mathcal{H}^2(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$.

4.6.3 Pontryagin Duality

Let $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ and denote by $M^\vee := \text{Hom}_{\mathbb{Z}_p, c}(M, \mathbb{Q}_p/\mathbb{Z}_p)$ the continuous homomorphisms from M to $\mathbb{Q}_p/\mathbb{Z}_p$. Since M is p -torsion and finitely generated over $\mathbf{A}_K$, since $\mathbf{A}_K^+$ is a compact neighbourhood of 0 in $\mathbf{A}_K$, we deduce M is locally compact and $M^\vee$ is the Pontryagin dual of M .

Lemma 4.6.15 ([Her01], Lemma 1.3). The map

$$\mathrm{TR}_K : \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1 \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$$

induces a group isomorphism between $\tilde{M} = \mathrm{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ and $M^\vee = \mathrm{Hom}_{\mathbb{Z}_p, c}(M, \mathbb{Q}_p/\mathbb{Z}_p)$.

Proof. We show that the map

$$\begin{aligned} \tilde{M} = \mathrm{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) &\longrightarrow M^\vee = \mathrm{Hom}_{\mathbb{Z}_p, c}(M, \mathbb{Q}_p/\mathbb{Z}_p) \\ f &\longmapsto \mathrm{TR}_K \circ f \end{aligned}$$

defines an isomorphism of groups. Note that it is well-defined since any element of $\mathrm{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ is continuous and TR_K is also continuous and $\mathbb{Z}_p$ -linear.

reduction to case of M a simple module: First note that $\mathbf{A}_K$ is a PID and $\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1$ is divisible as an $\mathbf{A}_K$ -module since for all $\alpha \in \mathbf{A}_K$ non-zero and $x \in \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1$ we have $\frac{1}{\alpha}x \in \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1$. We know that every divisible module over a PID is injective, and in particular, $\mathrm{Hom}(-, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ is an injective functor (from modules over $\mathbf{A}_K$ to modules over $\mathbf{A}_K$). Further, we know that taking the Pontryagin dual is a left-exact functor. For a short exact sequence of elements in $\Phi\Gamma M_{\mathbf{A}_K}^{\mathrm{ét}, p\text{-tor}}$

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

we now get the commutative diagram with exact rows

$$\begin{array}{ccccccc} \tilde{M}'' & \longrightarrow & \tilde{M} & \longrightarrow & \tilde{M}' & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & (M'')^\vee & \longrightarrow & M^\vee & \longrightarrow & (M')^\vee \end{array}$$

The snake lemma now lets us deduce that the middle vertical arrow defines an isomorphism if the left and right vertical arrows do. By choosing $M' = pM$ and $M'' = M/pM$ we can inductively reduce to the case where M has length one over $\mathbf{A}_K$, i.e. a one-dimensional $\mathbf{A}_K/(p) = \mathbf{E}_K$ -vector space. By choosing a basis we can assume $M = \mathbf{E}_K$. Note that using the fact that $\mathbf{E}_K$ has characteristic p , we have

$$\begin{aligned} \tilde{\mathbf{E}}_K &= \mathrm{Hom}_{\mathbf{A}_K}(\mathbf{E}_K, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) \cong \mathrm{Hom}_{\mathbf{E}_K}(\mathbf{E}_K, \Omega_K^1/p\Omega_K^1) \\ \mathbf{E}_K^\vee &= \mathrm{Hom}_{\mathbb{Z}_p, c}(\mathbf{E}_K, \mathbb{Q}_p/\mathbb{Z}_p) \cong \mathrm{Hom}_{\mathbb{F}_p, c}(\mathbf{E}_K, \mathbb{F}_p) \end{aligned}$$

where $\mathbb{F}_p$ carries the discrete topology.

the residue map induces an isomorphism: We show that

$$\begin{aligned} \mathrm{Hom}_{\mathbf{E}_K}(\mathbf{E}_K, \Omega_K^1/p\Omega_K^1) &\longrightarrow \mathrm{Hom}_{k, c}(\mathbf{E}_K, k) \\ f &\longmapsto \mathrm{Res} \circ f \end{aligned}$$

defines an isomorphism of groups. Since $\text{im}(f)$ is a one-dimensional $\mathbf{E}_K$ -vector space we have that it necessarily contains some elements that is not in the kernel of the residue map and we deduce injectivity of the map above. For surjectivity, let $u \in \text{Hom}_{k,c}(\mathbf{E}_K, k)$ and let $\bar{\pi} = \epsilon - 1$ be the class of π in $\mathbf{E}_K$. Note that $\bar{\pi}^n \rightarrow 0$ for $n \rightarrow +\infty$. We define $\alpha_n = u(\bar{\pi}^n) \in \mathbb{F}_p$ for all $n \in \mathbb{Z}$. By continuity $\alpha_n \rightarrow 0$ for $n \rightarrow +\infty$, but since $\mathbb{F}_p$ carries the discrete topology we have that there is a $N \in \mathbb{N}$ such that $\alpha_n = 0$ for all $n \geq N$. Let $\omega = (\sum_{n \geq -N} \alpha_{-n-1} \pi^n) d\pi \in \Omega_K^1$ and let $\bar{\omega}$ denote its class in $\Omega_K^1/p\Omega_K^1$. Now define $f \in \text{Hom}_{\mathbf{E}_K}(\mathbf{E}_K, \Omega_K^1/p\Omega_K^1)$ be the $\mathbf{E}_K$ -linear map satisfying $f(1) = \bar{\omega}$. Then $\text{Res} \circ f = u$ since $\text{Res}(\pi^n \omega) = \alpha_n$.

the trace map induces an isomorphism: We show that

$$\begin{aligned} \text{Hom}_{k,c}(\mathbf{E}_K, k) &\longrightarrow \text{Hom}_{\mathbb{F}_p,c}(\mathbf{E}_K, \mathbb{F}_p) \\ f &\longmapsto \text{Tr}_{k/\mathbb{F}_p} \circ f \end{aligned}$$

defines an isomorphism. Note that since $K/\mathbb{Q}_p$ is unramified, the trace maps $\text{Tr}_{K/\mathbb{Q}_p}$ and $\text{Tr}_{k/\mathbb{F}_p}$ correspond to each other modulo p . Injectivity follows since any non-zero $f \in \text{Hom}_{\mathbb{F}_p,c}(\mathbf{E}_K, \mathbb{F}_p)$ is necessarily surjective and $\text{Tr}_{K/\mathbb{Q}_p}$ is not the zero map. For surjectivity, let $u \in \text{Hom}_{\mathbb{F}_p,c}(\mathbf{E}_K, \mathbb{F}_p)$. Note that the trace map defines a symmetric non-degenerate bilinearform, and thus a perfect pairing

$$(x, y) \longmapsto \text{Tr}_{k/\mathbb{F}_p}(xy).$$

Therefore there exists an $\alpha_n \in k$ such that the linearform $\alpha \mapsto u(\alpha \bar{\pi}^n)$ is equal to $\alpha \mapsto \text{Tr}_{k/\mathbb{F}_p}(\alpha_n \alpha)$. Define $f \in \text{Hom}_{k,c}(\mathbf{E}_K, k)$ be the k -linear map satisfying $f(\bar{\pi}^n) = \alpha_n$. Then $\text{Tr}_{k/\mathbb{F}_p} \circ f = u$ since $\text{Tr}_{k/\mathbb{F}_p}(f(\alpha \bar{\pi}^n)) = \text{Tr}_{k/\mathbb{F}_p}(\alpha_n \alpha) = u(\alpha \bar{\pi}^n)$ for all $\alpha \in \mathbb{F}_p$ and $n \in \mathbb{N}$. $\square$

Using the isomorphism, we can equip $M^\vee$ with the corresponding structure of a (φ, Γ_K) -module over $\mathbf{A}_K$ of $\tilde{M}$. Using the formulas for the Γ_K and φ -action on $\tilde{M} = \text{Hom}_{\mathbf{A}_K}(M, \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ as seen in Definition 4.6.2 and the formulas in Lemma 4.6.10, we derive the following formulas for $f \in M^\vee$:

$$\begin{aligned} \lambda f &= f(\lambda \cdot) && \text{for all } \lambda \in \mathbf{A}_K \\ \gamma(f) \circ \gamma &= f && \text{for all } \gamma \in \Gamma_K \\ \varphi_{M^\vee}(f) \circ \varphi_M &= f. \end{aligned} \tag{4.6}$$

We know state Alexander-Pontryagin-Duality. Which will be our tool to relate homology groups of the Pontryagin dual to cohomology groups of the dual complex of the Herr complex.

Definition 4.6.16. A continuous homomorphism $f : G \longrightarrow G'$ between two topological groups is called strict if the induced map $\bar{f} : G/\ker(f) \longrightarrow G'$ is a homeomorphism.

Lemma 4.6.17 ([Her01], Lemma 5.6). Let $\mathcal{C} = (M^i, d^i : M^i \rightarrow M^{i+1})_{i \in \mathbb{Z}}$ be a cochain complex of locally compact Hausdorff abelian groups. Assume d^i are strict, with closed image.

Then $\mathcal{C}^\vee = (N_i := M^{i\vee}, d_i := {}^t d^{i-1} : N_i \rightarrow N_{i-1})_{i \in \mathbb{Z}}$ is a chain complex of locally compact separable abelian groups. The d_i are strict with closed image and for all i we have natural isomorphisms

$$H_i(\mathcal{C}^\vee) \cong H^i(\mathcal{C})^\vee$$

where both sides are locally compact separable abelian groups.

Remark 4.6.18. The isomorphisms in the previous lemma are natural in a way that the class of $f \in \ker(d_i) \subset N_i$ is being sent to precisely the class of the homomorphism $H^i(\mathcal{C}) \rightarrow \mathbb{Q}_p/\mathbb{Z}_p$ that sends $x \in \ker(d^i) \subset M_i$ to $f(x)$.

Proposition 4.6.19. The complex $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ satisfies the hypotheses of Lemma 4.6.17.

Proof. We know that continuous surjective homomorphisms between second countable, Hausdorff, locally compact groups are open maps. In particular, since M and $M \oplus M$ are second countable, Hausdorff, locally compact, the boundary maps of $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ are strict. It remains to show that the images of the boundary maps are closed. We will make use of the following lemma.

Lemma 4.6.20 ([Her01], Lemma 5.7). Let M be a p -torsion (φ, Γ_K) -module over $\mathbf{A}_K$. Then M contains a compact neighbourhood L stable under φ such that ρ is bijective on it. By compactness and Hausdorffness of L , ρ is a homeomorphism from L to L .

Proof. Choose $n \geq 1$ such that $p^n \cdot M = 0$. Note that $\sigma(\pi) \equiv \pi^p \pmod{p}$ so since $p^n \cdot M = 0$ we have $\sigma(\pi^{p^{n-1}}) \equiv \pi^{p^n} \pmod{p^n}$. Let $e_1, \dots, e_d$ be a system of generators of M over $\mathbf{A}_K$. Let a_{ij} be such that

$$\varphi(e_i) = \sum_{j, j} a_{ij} e_j.$$

Recall that the open neighbourhood of $\mathbf{A}_K$ are given by $\pi^{p^{n-1}} \mathbf{A}_K^+ + p^n \mathbf{A}_K$. Choose $s \geq 0$ large enough such that for all i, j

$$\pi^s a_{ij} \in \pi^{p^{n-1}} \mathbf{A}_K^+ + p^n \mathbf{A}_K$$

Further, choose $r \in \mathbb{N}$ such that $p^{n-1}(p-1)r \geq s$. Then we have

$$\varphi(\pi^{p^{n-1}r} e_i) = \sigma(\pi^{p^{n-1}r}) \varphi(e_i) = \pi^{p^n r} \sum_j a_{ij} e_j = \sum_j (\pi^{p^{n-1}(p-1)r} a_{ij}) \pi^{p^{n-1}r} e_j. \quad (4.7)$$

Let

$$f_i = \pi^{p^{n-1}r} e_i.$$

and L be the sub- $\mathbf{A}_K^+$ -module generated by the f_i 's. We see that L is a compact neighbourhood of 0, since $\mathbf{A}_K^+$ is a compact neighbourhood of 0, and further, by (4.7), L is stable under φ . We now show that for all $x \in L$

$$\lim_{r \rightarrow \infty} \varphi^r(x) = 0.$$

For $x = \sum_i x_i f_i \in L$, we have

$$\varphi(x) = \sum_i \sigma(x_i) \left(\sum_j (\pi^{p^{n-1}(p-1)r} a_{ij}) f_j \right) = \sum_j \left(\sum_i \sigma(x_i) (\pi^{p^{n-1}(p-1)r} a_{ij}) \right) f_j.$$

We deduce that $\varphi(x) \in \pi^{p^{n-1}} L$ and by semi-linearity $\varphi(\pi^{tp^{n-1}} L) \subset \pi^{(1+tp)p^{n-1}} L$ for all $t \in \mathbb{N}$. Thus $\lim_{r \rightarrow \infty} \varphi^r(x) = 0$ and $\sum_{n \geq 0} (-\varphi)^n(x)$ converges in L . We deduce that ρ is invertible on L with inverse given by

$$\rho^{-1} = \sum_{n \geq 0} (-\varphi)^n. \quad \square$$

We deduce that the image of ρ contains an open neighbourhood of 0 and thus $\text{Im}(\rho) - \text{Im}(\tau)$ is an open and therefore closed subgroup of M . Finally the subgroup,

$$C = \{(\rho(x), \tau(x)) \mid x \in M\}$$

is closed since if $(\rho(x_n), \tau(x_n))$ with $x_n \in M$ converges to 0 then we see that $\rho(x_n)$ converges to 0 in M . Since ρ is a homeomorphism onto some neighbourhood of 0 this implies x_n converges to 0 in M . This shows that any convergence sequence $(\rho(x_n), \tau(x_n))$ converges to $(\rho(x), \tau(x))$ for some $x \in M$ and thus C is closed. $\square$

We deduce that by Lemma 4.6.17

$$H_i([\mathcal{C}_{\varphi, \gamma}^\bullet(M)]^\vee) \cong H^i(\mathcal{C}_{\varphi, \gamma}^\bullet(M))^\vee.$$

Recall that we defined a one-sided inverse to the Frobenius, denoted by ψ .

Lemma 4.6.21. Let $M \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ then the operator $\psi_{M^\vee}$ satisfies

$$\psi_{M^\vee}(f) \circ \psi_M = f.$$

In other words, $\psi_{M^\vee}(f) = f \circ \varphi_M$.

Proof. Since $\psi_{M^\vee}$ is uniquely characterized as the additive operator satisfying

$$\psi_{M^\vee}(a\varphi_{M^\vee}(f)) = \psi(a)f$$

for all $f \in M^\vee$ and $a \in \mathbf{A}_K$ it suffices to show that $\Psi : f \mapsto f \circ \varphi_M$ satisfies the equation. To that end, recall that we obtained the (φ, Γ_K) -structure of $M^\vee$ from $\tilde{M}$ under the isomorphism induced by TR_K . Since M is p -torsion, we have $M^\vee \cong \mathrm{Hom}_{\mathbb{Z}_p}(M, \mathbb{Z}/p^n\mathbb{Z})$ for some n and we have that the isomorphism induced by TR_K is given by

$$\begin{aligned} \tilde{M} = \mathrm{Hom}_{\mathbf{A}_K}(M, \Omega_{K,n}^1) &\longrightarrow \mathrm{Hom}_{\mathbb{Z}_p}(M, \mathbb{Z}/p^n\mathbb{Z}) \\ g &\longrightarrow \mathrm{TR}_{K,n} \circ g. \end{aligned}$$

Fix $f \in M^\vee$ and let $g \in \mathrm{Hom}_{\mathbf{A}_K}(M, \Omega_{K,n}^1)$ such that $\mathrm{TR}_{K,n} \circ g = f$. We have

$$\begin{aligned} \Psi(a\varphi_{M^\vee}(f))(m) &= (\psi(a)f)(m) \\ \iff (a\varphi_{M^\vee}(f) \circ \varphi_M)(m) &= f(\psi(a)m) \\ \iff \varphi_{M^\vee}(f)(a\varphi_M(m)) &= f(\psi(a)m) \\ \iff \varphi_{M^\vee}(\mathrm{TR}_{K,n} \circ g)(a\varphi_M(m)) &= (\mathrm{TR}_{K,n} \circ g)(\psi(a)m) \\ \iff (\mathrm{TR}_{K,n} \circ \varphi_{\tilde{M}}(g))(a\varphi_M(m)) &= (\mathrm{TR}_{K,n} \circ g)(\psi(a)m) \\ \iff \mathrm{TR}_{K,n}(a\varphi_{\Omega_{K,n}^1}(g(m))) &= \mathrm{TR}_{K,n}(\psi(a)g(m)) \end{aligned}$$

where we repeatedly used the formulas (4.6). Now since

$$\psi_{\Omega_{K,n}^1}(a\varphi_{\Omega_{K,n}^1}(g(m))) = \psi(a)g(m)$$

it suffices to show that

$$\mathrm{TR}_{K,n}(\psi_{\Omega_{K,n}^1}(\omega) - \omega) = 0$$

for all $\omega \in \Omega_{K,n}^1$. Write

$$\omega = \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega)) + (\omega - \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega)))$$

where $\omega - \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega)) \in \ker(\psi_{\Omega_{K,n}^1})$. Since $\gamma - 1$ is bijective on $\ker(\psi_{\Omega_{K,n}^1})$ we can write $\omega - \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega)) = (\gamma - 1)x$ where $x \in \ker(\psi_{\Omega_{K,n}^1})$. We conclude by noting that

$$\begin{aligned} \mathrm{TR}_{K,n}(\psi_{\Omega_{K,n}^1}(\omega) - \omega) &= \mathrm{TR}_{K,n}(\psi_{\Omega_{K,n}^1}(\omega) - \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega)) + (\gamma - 1)x) \\ &= \mathrm{TR}_{K,n}(\psi_{\Omega_{K,n}^1}(\omega) - \varphi_{\Omega_{K,n}^1}(\psi_{\Omega_{K,n}^1}(\omega))) + \mathrm{TR}_{K,n}(\gamma(x) - x) \\ &= 0 \end{aligned}$$

by the formulas in Lemma 4.6.10. $\square$

We define a slightly modified complex to the one in section 5.4. Let $\mathcal{C}_{\psi, \gamma-1}^\bullet(M^\vee)$ denote the follow complex

$$\begin{aligned} 0 &\longrightarrow M^\vee \longrightarrow M^\vee \oplus M^\vee \longrightarrow M^\vee \longrightarrow 0 \\ f &\longmapsto ((\gamma^{-1} - 1)f, (1 - \psi_{M^\vee})f) \\ (g, h) &\longmapsto (\psi_{M^\vee} - 1)g + (\gamma^{-1} - 1)h \end{aligned}$$

Lemma 4.6.22 ([Her01], 5.8). Let

$$\begin{aligned} F_1 : M^\vee \oplus M^\vee &\longrightarrow M^\vee \oplus M^\vee \\ (g, h) &\longmapsto (-\gamma^{-1}(h), \psi_{M^\vee}(g)) \end{aligned}$$

and

$$\begin{aligned} F_2 : M^\vee &\longrightarrow M^\vee \\ f &\longmapsto -\gamma^{-1} \circ \psi_{M^\vee}(f) \end{aligned}$$

then the morphism of complexes

$$\begin{array}{ccccccccc} \mathcal{C}_{\varphi, \gamma}^\bullet(M^\vee) : & 0 & \longrightarrow & M^\vee & \longrightarrow & M^\vee \oplus M^\vee & \longrightarrow & M^\vee & \longrightarrow & 0 \\ & \downarrow 0 & & \downarrow id=F_0 & & \downarrow F_1 & & \downarrow F_2 & & \downarrow 0 \\ \mathcal{C}_{\psi, \gamma^{-1}}^\bullet(M^\vee) : & 0 & \longrightarrow & M^\vee & \longrightarrow & M^\vee \oplus M^\vee & \longrightarrow & M^\vee & \longrightarrow & 0 \end{array}$$

is a quasi-isomorphism. We will denote the induced isomorphisms on cohomology groups by $\bar{F}_i$.

Proof. Since the F_i are surjective, we can see that $\mathcal{C}_{\psi, \gamma^{-1}}^\bullet$ is isomorphic to the quotient complex of $\mathcal{C}_{\varphi, \gamma}^\bullet$ and the kernel of the morphism of complexes, given by

$$0 \longrightarrow 0 \longrightarrow \ker(\psi_{M^\vee}) \oplus 0 \xrightarrow{\gamma^{-1}} \ker(\psi_{M^\vee}) \longrightarrow 0.$$

Using Lemma 4.4.5, i.e. the fact that γ is bijective on $\ker(\psi_{M^\vee})$, we see that this complex is acyclic, and we are done. $\square$

Definition 4.6.23. We define the natural isomorphisms

$$\begin{aligned} v_0 &= id : M^\vee \longrightarrow M^\vee \\ v_1 : M^\vee \oplus M^\vee &\longrightarrow (M \oplus M)^\vee \\ (g, h) &\longmapsto g \oplus h \\ v_2 &= id : M^\vee \longrightarrow M^\vee \end{aligned}$$

Remark 4.6.24. We can see that v_1 is indeed an isomorphism since for any $\phi \in (M \oplus M)^\vee$, we have $v_1(\phi(x, 0), \phi(0, y)) = \phi(x, y)$. In particular, every element in $(M \oplus M)^\vee$ can be written as $g \oplus h$ for $g, h \in M^\vee$.

We can show that v_i induces isomorphisms between the cohomology groups $H^i(\mathcal{C}_{\psi, \gamma^{-1}}^\bullet(M^\vee))$ and the homology groups $H_{2-i}([\mathcal{C}_{\varphi, \gamma}^\bullet(M)]^\vee)$ of the dual complex of $\mathcal{C}_{\varphi, \gamma}^\bullet(M)$ (as described in Lemma 4.6.17) for $i \in \{0, 1, 2\}$. To see that, note that the following diagram commutes

$$\begin{array}{ccccccccc} \mathcal{C}_{\psi, \gamma^{-1}}^\bullet(M^\vee) : & 0 & \xrightarrow{f \mapsto ((\gamma^{-1}-1)f, (1-\psi_{M^\vee})f)} & M^\vee & \xrightarrow{(g, h) \mapsto (\psi_{M^\vee}-1)g + (\gamma^{-1}-1)h} & M^\vee \oplus M^\vee & \xrightarrow{} & M^\vee & \xrightarrow{} & 0 \\ & \downarrow 0 & & \downarrow v_0=id & & \downarrow v_1 & & \downarrow v_2=id & & \downarrow 0 \\ [\mathcal{C}_{\varphi, \gamma}^\bullet(M)]^\vee : & 0 & \xrightarrow{f \mapsto f \circ (\gamma-1) \oplus f \circ (1-\varphi)} & M^\vee & \xrightarrow{g \oplus h \mapsto g \circ (\varphi-1) + h \circ (\gamma-1)} & (M \oplus M)^\vee & \xrightarrow{} & M^\vee & \xrightarrow{} & 0 \end{array}$$

using that $\gamma^{-1}(f) = f \circ \gamma$ and $\psi_{M^\vee}(f) = f \circ \varphi$. We will denote the isomorphisms induced on cohomology groups by $\bar{v}_i$.

For $i \in \{0, 1, 2\}$ we thus obtain canonical isomorphisms

$$u_i(M) : \mathcal{H}_\gamma^i(M^\vee) \xrightarrow{\bar{F}_i} H_\gamma^i(\mathcal{C}'_{\psi, \gamma^{-1}}(M^\vee)) \xrightarrow{\bar{v}_i} H_{2-i}([\mathcal{C}'_{\varphi, \gamma}(M)]^\vee) \xrightarrow{\sim} [\mathcal{H}_\gamma^{2-i}(M)]^\vee$$

where the last arrow is the isomorphism from Alexander-Pontryagin Duality, i.e. Theorem 4.6.17. This defines the following perfect pairing

$$\begin{aligned} \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(M^\vee) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, y) &\longmapsto [u_{2-i}(M)(y)](x) \end{aligned} \quad (4.8)$$

Using the definitions of F_i and v_i for $i = 0, 1, 2$ we can deduce the following proposition.

Proposition 4.6.25. The pairing (4.8) is given by the following explicit formulas:

$$\begin{aligned} \mathcal{H}_\gamma^0(M) \times \mathcal{H}_\gamma^2(M^\vee) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, \bar{f}) &\longmapsto f(x) \\ \mathcal{H}_\gamma^1(M) \times \mathcal{H}_\gamma^1(M^\vee) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ ((x, y), \overline{(g, h)}) &\longmapsto g(\varphi_M(y)) - h(\gamma(x)) \\ \mathcal{H}_\gamma^2(M) \times \mathcal{H}_\gamma^0(M^\vee) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (\bar{x}, f) &\longmapsto f(x) \end{aligned} \quad (4.9)$$

Proof. (i) For $x \in \mathcal{H}_\gamma^0(M)$ and $\bar{f} \in \mathcal{H}_\gamma^2(M^\vee)$ we have

$$[u_2(M)(f)](x) = -\gamma^{-1} \circ \psi_{M^\vee}(f)(x) = f(\gamma \circ \varphi_M(x)) = f(x)$$

since φ_M and γ both fix x .

(ii) For $\overline{(x, y)} \in \mathcal{H}^1(M)$, $\overline{(g, h)} \in \mathcal{H}(M^\vee)$ we have

$$\begin{aligned} (\gamma - 1)x &= (\varphi_M - 1)(y) \\ (\gamma - 1)g &= (\varphi_{M^\vee} - 1)(h) \end{aligned}$$

and applying $\gamma^{-1} \circ \psi_{M^\vee}$ to both sides we get

$$\psi_{M^\vee}((1 - \gamma^{-1})g) = \gamma^{-1}(1 - \psi_{M^\vee})(h)$$

so we see that

$$\begin{aligned} [u_1(M)(g, h)](x, y) &= -\gamma^{-1}(h)(x) + \psi_M^\vee(g)(y) \\ &= g(\varphi_M(y)) - h(\gamma(x)). \end{aligned}$$

(iii) For $\bar{x} \in \mathcal{H}_\gamma^2(M)$ and $f \in \mathcal{H}_\gamma^0(M^\vee)$ we have

$$[u_0(M)(f)](x) = f(x). \quad \square$$

It remains to verify that these formulas indeed coincide with the cup-product under compositions with the isomorphism $\mathrm{TR}_K : \tilde{M} \rightarrow M^\vee$ and the Tate isomorphism $\mathcal{T}_K$.

Following Lemma 4.6.4 we can find the following formulas for the pairing

$$\begin{aligned} \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(\tilde{M}) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, y) &\longmapsto \mathcal{T}_K[\overline{x \cup y}] \end{aligned} \quad (4.10)$$

where we view $x \cup y \in \mathcal{H}^2(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1)$ under isomorphism $M \otimes \tilde{M} \cong \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1$:

$$\begin{aligned} \mathcal{H}^0(M) \times \mathcal{H}^2(\tilde{M}) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, \bar{f}) &\longmapsto \mathcal{T}_K[\overline{f(x)}] \\ \mathcal{H}^1(M) \times \mathcal{H}^1(\tilde{M}) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ ((\overline{x, y}), \overline{(g, h)}) &\longmapsto \mathcal{T}_K[\overline{g(\varphi_M(y)) - h(\gamma(x))}] \\ \mathcal{H}^2(M) \times \mathcal{H}^0(\tilde{M}) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (\overline{x}, f) &\longmapsto \mathcal{T}_K[\overline{f(x)}] \end{aligned} \quad (4.11)$$

Note that for the second formula we used Proposition 2.1.15 to see that

$$\overline{(x, y)} \cup \overline{(g, h)} = -\overline{(g, h)} \cup \overline{(x, y)} = -h \otimes \gamma(x) + g \otimes \varphi_M(y)$$

since we have the natural isomorphisms $M \otimes \tilde{M} \cong \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1 \cong \tilde{M} \otimes M$.

Proposition 4.6.26. Using the explicit formulas, the pairings (4.8) and (4.10) coincide under compositions with the isomorphism $\mathrm{TR}_K : \tilde{M} \rightarrow M^\vee$. In particular, the pairing

$$\begin{aligned} \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(\tilde{M}) &\longrightarrow \mathbb{Q}_p/\mathbb{Z}_p \\ (x, y) &\longmapsto \mathcal{T}_K[\overline{x \cup y}] \end{aligned}$$

is a perfect pairing.

Proof. We note that this statement is equivalent to saying the following diagram commutes

$$\begin{array}{ccc} \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(\tilde{M}) & \xrightarrow{\cup} & \mathcal{H}_\gamma^2(\frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1) \\ \downarrow (id, (f \mapsto \mathrm{TR}_K \circ f)^{\oplus \binom{2}{i}}) & & \downarrow \mathcal{T}_K \\ \mathcal{H}_\gamma^i(M) \times \mathcal{H}_\gamma^{2-i}(M^\vee) & \xrightarrow{(x, y) \mapsto [u_{2-i}(M)(y)](x)} & \mathbb{Q}_p/\mathbb{Z}_p \end{array}$$

Recall that $\mathrm{TR}_K : \frac{\mathbf{B}_K}{\mathbf{A}_K} \otimes_{\mathbf{A}_K} \Omega_K^1 \longrightarrow \mathbb{Q}_p/\mathbb{Z}_p$ precisely induces the Tate-isomorphism $\mathcal{T}_K$ on cohomology. Therefore, using the explicit formulas (4.9) and (4.11), this is trivial to see.

□

Appendix A

A.1 Proof of Lemma 4.4.5

We want to show the following theorem.

Lemma A.1.1. For V a p -torsion representation of G_K $\gamma - 1 : \mathbf{D}(V)^{\psi=0} \rightarrow \mathbf{D}(V)^{\psi=0}$ is bijective.

In order to prove this, we will first prove the statement for overconvergent representations and then deduce it for p -torsion representations. The main property of overconvergent representations that we will use is that the $\tilde{\mathbf{A}}_r^\dagger$ are complete with respect to the π -adic and the $\sigma(\pi)$ -adic topology. The proof reduces the general case to the case of the trivial $\mathbb{Z}_p$ -representation. The main difficulty of the proof will be showing the surjectivity of $\gamma - 1$ on $\mathbf{D}^\dagger(V)$, which we will show by constructing a contraction with respect to the π -adic topology which will have a fixed point.

Theorem A.1.2 ([CC98], Theorem II.3.2). If V is a overconvergent $\mathbb{Z}_p$ -representation and γ is a topological generator of Γ_K , then $\gamma - 1$ admits a continuous inverse on $\mathbf{D}^\dagger(V)^{\psi=0}$.

Remark A.1.3. The ψ -operator is well-defined on $\mathbf{D}^\dagger(V)$, i.e., it sends $\mathbf{B}^\dagger$ to $\mathbf{B}^\dagger$ since $\sigma(\mathbf{B}^\dagger) \subset \mathbf{B}^\dagger$ and for $x \in \mathbf{B}^\dagger$ we have $\mathrm{Tr}_{\mathbf{B}/\sigma(\mathbf{B})}(x) = \mathrm{Tr}_{\mathbf{B}^\dagger/\sigma(\mathbf{B}^\dagger)}(x)$ since in particular $1, [\epsilon], \dots, [\epsilon]^{p-1}$ still also forms a basis of $\mathbf{B}^\dagger/\sigma(\mathbf{B}^\dagger)$.

We begin by showing the injectivity of $\gamma - 1$ on the kernel of ψ . Then, we prove the theorem for the case of the trivial overconvergent $\mathbb{Z}_p$ -representation for $K = K_0$, and afterwards show how to reduce the general case to the trivial case. Finally, we show how to deduce that statement for mod p representation.

The injectivity of $\gamma - 1$

Lemma A.1.4 ([CC98], Lemma I.3.1). Let K be a finite extension of $\mathbb{Q}_p$ and γ be an element of infinite order of Γ_K . Then $\mathbf{B}_K^{\gamma=1} \subset K_\infty \cap K_0^{\mathrm{nr}}$ and in particular, $\mathbf{B}_K^{\gamma=1}$

is a finite extension of K_0 .

Proposition A.1.5 ([CC98], Lemma I.5.1). Let γ be an element of infinite order of Γ_K and V a free $\mathbb{Z}_p$ -representation of G_K . Then $\gamma - 1$ is injective on $\mathbf{D}(V)^{\psi=0}$.

Proof. Let $V' = V \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ then we know that $\mathbf{D}(V')$ is a finite-dimensional $\mathbf{B}_K$ -vector space and $\mathbf{D}(V')^{\gamma=1}$ a finite-dimensional $\mathbf{B}_K^{\gamma=1}$ vector space. By Lemma A.1.4 $\mathbf{D}(V')^{\gamma=1}$ is a finite-dimensional K_0 -vector space and since φ is an injective K_0 -linear map from $\mathbf{D}(V')^{\gamma=1}$ to $\mathbf{D}(V')^{\gamma=1}$ it is in particular surjective. Let $x \in \mathbf{D}(V')^{\gamma=1} \cap \mathbf{D}(V')^{\psi=0}$ then $x = \varphi(y)$ for some $y \in \mathbf{D}(V')^{\gamma=1}$ so $0 = \psi(x) = \psi(\varphi(y)) = y$ thus also $x = 0$. In particular, $\gamma - 1$ is also injective on $\mathbf{D}(V)^{\psi=0}$. $\square$

Remark A.1.6. Note that this in particular also shows injectivity of $\gamma - 1$ on $\mathbf{D}^\dagger(V)$ for V an overconvergent representation.

Corollary A.1.7. Let γ be an element of infinite order of Γ_K and V a p -torsion representation of G_K . Then $\gamma - 1$ is injective on $\mathbf{D}(V)^{\psi=0}$.

Proof. For V a length one module over $\mathbb{Z}_p$, i.e. a one-dimensional $\mathbb{F}_p$ -vector space, the same argumentation holds when replacing $\mathbf{B}_K$ with $\mathbf{E}_K$. In particular, $\mathbf{D}(V)^{\gamma=1}$ is a finite-dimensional vector space over k and $\varphi : \mathbf{D}(V)^{\gamma=1} \rightarrow \mathbf{D}(V)^{\gamma=1}$ is surjective.

It remains to show that for a p -torsion representation V we can reduce to the case of V having length one over $\mathbb{Z}_p$, i.e., $\mathbf{D}(V)$ having length one over $\mathbf{A}_K$. For $M, M', M'' \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ and a short sequence of (φ, Γ_K) -modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

we can deduce, since ψ is surjective and commutes with maps of (φ, Γ_K) -modules, that

$$0 \longrightarrow \ker(\psi_{M'}) \longrightarrow \ker(\psi_M) \longrightarrow \ker(\psi_{M''}) \longrightarrow 0$$

is also exact. Since the maps of the sequence are Γ_K -equivariant we can directly deduce the injectivity of τ on $\ker(\psi_M)$ from the injectivity of τ on $\ker(\psi_{M'})$ and $\ker(\psi_{M''})$. We can therefore assume $\mathbf{D}(V)$ has length one over $\mathbf{A}_K$, by for a given M for example choosing $M' = pM$ and $M'' = M/pM$ and proceeding inductively. $\square$

The case of V being the trivial representation

Proposition A.1.8 ([CC98], II.5.1). If $m \geq 1$ and γ_m is a generator of $\Gamma_{K_{0m}}$ and $r \geq (p-1)p^{m-1}$, then $\tau_m = \gamma_m - 1$ induces a bijection of $\varphi(\pi)^a(\mathbf{A}_{r, K_0}^\dagger)^{\psi=0}$ onto $\varphi(\pi)^{a+p^{m-1}}(\mathbf{A}_{r, K_0}^\dagger)^{\psi=0}$ for all $a \in \mathbb{Z}$.

Proof. We have already showed injectivity but we can also directly deduce it from the fact that $\mathbf{A}_{K_0}^{\gamma_m=1} = \mathcal{O}_{K_0}$ by Lemma A.1.4. Now σ acts bijective from $\mathcal{O}_{K_0}$ onto

$\mathcal{O}_{K_0}$, thus we have that ψ is non-zero on $\mathcal{O}_{K_0}$ and thus $\mathbf{A}_{K_0}^{\gamma_m=1} \cap \mathbf{A}_{K_0}^{\psi=0} = \{0\}$ or in other words, τ_m is injective on $\mathbf{A}_{K_0}^{\psi=0}$.

Since γ_m is a generator of $\Gamma_{K_{0m}}$ we have that γ_m acts trivially on ζ_{p^i} for $1 \leq i \leq p-1$ and non-trivially on ζ_{p^m} thus we can see that $\chi(\gamma_m) = 1 + p^m u$ for some $u \in \mathbb{Z}_p^*$ and $\chi(\gamma_m) - 1 = p^m u$. Recall that $1, [\epsilon], \dots, [\epsilon]^{p-1}$ forms a basis of $\mathbf{B}_{K_0}$ over $\sigma(\mathbf{B}_{K_0})$ and $\psi(x_0 + \dots + [\epsilon]^{p-1} x_{p-1}) = \sigma^{-1}(x_0)$ for $x_i \in \sigma(\mathbf{A}_{K_0})$. This implies that we can write every element $z \in \mathbf{A}_{K_0}^{\psi=0}$ as $z = \sum_{i=1}^{p-1} [\epsilon]^i z_i$ with $z_i \in \sigma(\mathbf{A}_{K_0})$. Define $f_m : \mathbf{A}_{K_0}^{\psi=0} \longrightarrow \mathbf{A}_{K_0}^{\psi=0}$ via

$$f_m\left(\sum_{i=1}^{p-1} [\epsilon]^i z_i\right) = -\sum_{i=1}^{p-1} [\epsilon]^i \frac{z_i}{1 - [\epsilon]^{p^m i u}}$$

and using $\gamma_m([\epsilon]^i) = [\epsilon]^{i\chi(\gamma_m)} = [\epsilon]^i [\epsilon]^{p^m i u}$ we get

$$\tau_m(z) = \sum_{i=1}^{p-1} \gamma([\epsilon]^i z_i) - z = \sum_{i=1}^{p-1} [\epsilon]^i [\epsilon]^{p^m i u} \gamma(z_i) - z$$

and thus

$$\begin{aligned} f_m(\tau_m(z)) &= f_m\left(\sum_{i=1}^{p-1} [\epsilon]^i [\epsilon]^{p^m i u} \gamma(z_i)\right) - f_m(z) \\ &= -\sum_{i=1}^{p-1} [\epsilon]^i \frac{[\epsilon]^{p^m i u} \gamma(z_i)}{1 - [\epsilon]^{p^m i u}} + \sum_{i=1}^{p-1} [\epsilon]^i \frac{z_i}{1 - [\epsilon]^{p^m i u}} \\ &= -\sum_{i=1}^{p-1} [\epsilon]^i \frac{[\epsilon]^{p^m i u} \gamma(z_i)}{1 - [\epsilon]^{p^m i u}} + \sum_{i=1}^{p-1} [\epsilon]^i \frac{z_i - [\epsilon]^{p^m i u} z_i + [\epsilon]^{p^m i u} z_i}{1 - [\epsilon]^{p^m i u}} \\ &= z - \sum_{i=1}^{p-1} [\epsilon]^i \frac{\tau_m(z_i)}{[\epsilon]^{-p^m i u} - 1}. \end{aligned}$$

The rest of the proof will follow from the next lemmas. The goal is to show that $f_m(\tau_m(z)) - z$ is small with respect to the $\sigma(\pi)$ -adic topology and note that $\mathbf{A}_{K_0}^{\psi=0}$ is complete for that topology. $\square$

Lemma A.1.9. (i) If $a \in \mathbb{Z}_p^*$ and $r \geq (p-1)p^{m-1}$ then $\frac{[\epsilon]^{p^m a} - 1}{\pi^{p^m}}$ and $\frac{[\epsilon]^{p^m a} - 1}{\sigma(\pi)^{p^m - 1}}$ are units in $\mathbf{A}_{r, K_0}^\dagger$.

(ii) If $r \geq (p-1)p^{m-1}$ then $\frac{\tau_m(\pi^k)}{\pi^k} \in \pi^{p^m - 1} \mathbf{A}_{r, K_0}^\dagger$.

Proof. (i) We have that

$$\frac{\sigma(\pi)}{\pi^p} = 1 + \frac{p}{\pi} + \dots + \frac{p}{\pi^{p-1}}$$

is in $\mathbf{A}_{p-1, K_0}^\dagger$ using Lemma 3.6.10 since $(p-1)v_p(p) + i \geq 0$ for all $0 \leq i \leq p-1$. Further, it is a unit in $\mathbf{A}_{p-1, K_0}^\dagger$ by Lemma 3.6.2. We now see that

$$\begin{aligned} \frac{[\epsilon]^{p^m} - 1}{\pi^{p^m}} &= \frac{\sigma^m(\pi)}{\pi^{p^m}} = \frac{\sigma^m(\pi)}{\sigma^{m-1}(\pi^p)} \frac{\sigma^{m-1}(\pi)^p}{\sigma^{m-2}(\pi^p)^p} \cdots \frac{\sigma(\pi)^{p^{m-1}}}{\pi^{p^m}} \\ &= \sigma^{m-1}\left(\frac{\sigma(\pi)}{\pi^p}\right) \cdot \sigma^{m-2}\left(\frac{\sigma(\pi)}{\pi^p}\right)^p \cdots \left(\frac{\sigma(\pi)}{\pi}\right)^{p^{m-1}} \end{aligned}$$

is also a unit in $\mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger$. Further, for $a \in \mathbb{Z}_p^*$ we have that

$$\frac{[\epsilon]^{p^m a} - 1}{[\epsilon]^{p^m} - 1} = a + \sum_{i=1}^{+\infty} \binom{a}{i+1} ([\epsilon]^{p^m} - 1)^i$$

is invertible in $\mathbf{A}_{K_0}^\dagger$ and thus also a unit in $\mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger$. To conclude we have that

$$\begin{aligned} \frac{[\epsilon]^{p^m a} - 1}{\pi^{p^m}} &= \frac{[\epsilon]^{p^m a} - 1}{[\epsilon]^{p^m} - 1} \cdot \frac{\pi^{p^m}}{[\epsilon]^{p^m} - 1} \\ \frac{[\epsilon]^{p^m a} - 1}{\sigma(\pi)^{p^{m-1}}} &= \frac{[\epsilon]^{p^m a} - 1}{\pi^{p^m}} \cdot \left(\frac{\pi^p}{\sigma(\pi)}\right)^{p^{m-1}} \end{aligned}$$

are units in $\mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger$.

(ii) For $k = 1$ we have

$$\begin{aligned} \tau_m(\pi) &= \gamma_m([\epsilon] - 1) - ([\epsilon] - 1) = [\epsilon]([\epsilon]^{p^m u} - 1) \\ &= \pi^{p^m} [\epsilon] \frac{[\epsilon]^{p^m u} - 1}{\pi^{p^m}} \in \pi^{p^m} \mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger \end{aligned}$$

where we used part (i) in the last equation. We deduce that $\frac{\tau_m(\pi)}{\pi} \in \pi^{p^m-1} \mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger$. For $k \geq 1$ we have

$$\frac{\tau_m(\pi^k)}{\pi^k} = \left(\frac{\gamma_m(\pi)}{\pi}\right)^k - 1 = \sum_{j=1}^k \binom{k}{j} \left(\frac{\gamma_m(\pi)}{\pi} - 1\right)^j = \sum_{j=1}^k \binom{k}{j} \left(\frac{\tau_m(\pi)}{\pi}\right)^j$$

thus we see $\frac{\tau_m(\pi^k)}{\pi^k} \in \pi^{p^m-1} \mathbf{A}_{(p-1)p^{m-1}, K_0}^\dagger$.

□

Corollary A.1.10. If $r \geq (p-1)p^{m-1}$ and $a \in \mathbb{Z}$ we have

$$\tau_m(\pi^a \mathbf{A}_{r, K_0}^\dagger) \subset \pi^{a+p^m-1} \mathbf{A}_{r, K_0}^\dagger$$

Proof. We know τ_m is continuous, $\pi^{a+p^m-1} \mathbf{A}_{r, K_0}^\dagger$ is complete with respect to the π -adic topology and $X := \pi^a \mathbf{A}_{r, K_0}^\dagger \cap \mathcal{O}_{K_0}[\pi, \frac{1}{\pi}]$ is dense in $\pi^a \mathbf{A}_{r, K_0}^\dagger$. It now suffices

to check that $\tau_m(X) \subset \pi^{a+p^{m-1}}\mathbf{A}_{r,K_0}^\dagger$. Let $n = \lfloor r \rfloor$, then $X = \pi^a \mathcal{O}_{K_0}[\pi, \frac{p}{\pi^n}]$. X is generated over $\mathcal{O}_{K_0}$ by $e_{i,j} = \pi^{a+i}(\frac{p}{\pi^n})^j$ for $i, j \in \mathbb{N}$. Then

$$\tau_m(e_{i,j}) = p^j \tau_m(\pi^{a+i-jn}) = p^j \pi^{a+i-jn} \frac{\tau_m(\pi^{a+i-jn})}{\pi^{a+i-jn}} = e_{i,j} \pi^{p^{m-1}} x_{i,j}$$

where $x_{i,j} \in \mathbf{A}_{r,K_0}^\dagger$ using Lemma A.1.9 (ii). Since $e_{i,j} \in \pi^a \mathbf{A}_{r,K_0}^\dagger$, we have $\tau_m(e_{i,j}) \in \pi^{a+p^{m-1}}\mathbf{A}_{r,K_0}^\dagger$ which shows the corollary. $\square$

Corollary A.1.11. If $r \geq (p-1)p^{m-1}$ and $z \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ then

$$z - f_m(\tau_m(z)) \in \sigma(\pi)^{a+(p-1)p^{m-1}-1}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$$

Proof. We can write z of the form $z = \sum_{i=1}^{p-1} [\epsilon]^i \sigma(z_i)$ with $z_i \in \pi^a \mathbf{A}_{p^{-1}r,K_0}^\dagger$ and thus $\tau_m(z_i) \in \pi^{a+p^{m-1}}\mathbf{A}_{p^{-1}r,K_0}^\dagger$ by Lemma A.1.9, part (ii). Thus we can write

$$z - f_m(\tau_m(z)) = \sum_{i=1}^{p-1} [\epsilon]^i \frac{\tau_m(\sigma(z_i))}{[\epsilon]^{-p^{m-1}iu} - 1} = \sum_{i=1}^{p-1} [\epsilon]^i \sigma\left(\frac{\tau_m(z_i)}{[\epsilon]^{-p^{m-1}iu} - 1}\right).$$

Recall that $v := \frac{[\epsilon]^{-p^{m-1}iu} - 1}{\pi^{p^{m-1}}}$ is a unit in $\mathbf{A}_{p^{-1}r,K_0}^\dagger$ by Lemma A.1.9, and write

$$z - f_m(\tau_m(z)) = \sum_{i=1}^{p-1} [\epsilon]^i \sigma\left(\frac{\tau_m(z_i)}{\pi^{p^{m-1}}} v\right).$$

We conclude using that $\sigma(\tau_m(z_i)) \in \sigma(\pi)^{a+p^{m-1}}\mathbf{A}_{r,K_0}^\dagger$ and thus $\sigma\left(\frac{\tau_m(z_i)}{\pi^{p^{m-1}}}\right) \in \sigma(\pi)^{a+p^{m-1}-p^{m-1}}\mathbf{A}_{r,K_0}^\dagger = \sigma(\pi)^{a+(p-1)p^{m-1}-1}\mathbf{A}_{r,K_0}^\dagger$. $\square$

Proof of Proposition A.1.8. We now finish the proof of surjectivity of

$$\tau_m : \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0} \longrightarrow \sigma(\pi)^{a+p^{m-1}}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$$

Let $x \in \sigma(\pi)^{a+p^{m-1}}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$. We want to find $y \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ such that $\tau_m(y) = x$. Using that f_m is injective, this is equivalent to the function

$$g_x(y) = y - f_m(\tau_m(y) - x)$$

having a fixed point in $\sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$. We first show that g_x defines a map

$$g_x : \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0} \longrightarrow \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}.$$

Since $x \in \sigma(\pi)^{a+p^{m-1}}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ we can write $x = \sum_{i=0}^{p-1} [\epsilon]^i x_i$ with $x_i \in \sigma(\pi)^{a+p^{m-1}}\mathbf{A}_{r,K_0}^\dagger$, and using that $v := \frac{[\epsilon]^{p^{m-1}a} - 1}{\sigma(\pi)^{p^{m-1}}}$ is a unit in $\mathbf{A}_{r,K_0}^\dagger$ we have

$$f_m\left(\sum_{i=0}^{p-1} [\epsilon]^i x_i\right) = - \sum_{i=1}^{p-1} [\epsilon]^i \frac{x_i}{1 - [\epsilon]^{p^{m-1}iu}} = \sum_{i=1}^{p-1} [\epsilon]^i \frac{x_i}{\sigma(\pi)^{p^{m-1}}} v \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}.$$

Now we have that

$$g_x(y) = y - f_m(\tau_m(y) - x) = y - f_m(\tau_m(y)) - f_m(x)$$

where $y - f_m(\tau_m(y)) \in \sigma(\pi)^{a+(p-1)p^{m-1}-1}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ and $f_m(x) \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$. Thus $g_x(y) \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$. Finally, we show that g_x defines a contraction with respect to the $\sigma(\pi)$ -adic topology. We have that for $y, y' \in \sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ with $y - y' \in \sigma(\pi)^b(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$

$$g_x(y) - g_x(y') = y - y' + (f_m(\tau_m(y') - y)) \in \sigma(\pi)^{b+(p-1)p^{m-1}-1}(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0},$$

by Lemma A.1.11, so with respect to the $\sigma(\pi)$ -adic topology, g_x is a contraction. We finish the proof by noting that $\sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$ is complete with respect to the $\sigma(\pi)$ -adic topology, since $\mathbf{A}_{r,K_0}^\dagger$ is and ψ is continuous, therefore, g_x must have a fixed point in $\sigma(\pi)^a(\mathbf{A}_{r,K_0}^\dagger)^{\psi=0}$. $\square$

The case of V being a general representation

We begin by showing that we can reduce to the case $K = K_0$.

Lemma A.1.12. Let V be a free overconvergent $\mathbb{Z}_p$ -representation of G_K and let γ be a generator of G_{K_0} such that γ^m is a generator of G_K . Then if $\tau_m = \gamma^m - 1$ is invertible on $\mathbf{D}^\dagger(\text{Ind}_L^K(V))^{\psi=0}$, then τ_m is invertible on $\mathbf{D}^\dagger(V)^{\psi=0}$.

Proof. First, note that by Corollary 3.8.4 we have

$$\mathbf{D}^\dagger(\text{Ind}_K^{K_0}(V)) = \text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V)).$$

There is a surjective continuous map of (φ, Γ_{K_0}) -modules that commutes with ψ given by

$$\begin{aligned} \kappa : \text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V)) &\longrightarrow \mathbf{D}^\dagger(V) \\ f &\longmapsto f(1). \end{aligned}$$

Indeed, we have that $\gamma(f)(1) = f(\gamma) = \gamma(f(1))$ and $\varphi(f)(1) = \varphi(f(1))$ by definition of $\text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V))$. The map is surjective since we can define a map of (φ, Γ_{K_0}) -modules

$$\begin{aligned} \iota : \mathbf{D}^\dagger(V) &\longrightarrow \text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V)) \\ x &\longmapsto g \mapsto \begin{cases} g(x), & \text{for } g \in \Gamma_K \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

such that $\kappa \circ \iota = \text{id}$. We now have a commutative diagram

$$\begin{array}{ccc} \text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V))^{\psi=0} & \xrightarrow{\kappa} & \mathbf{D}^\dagger(V)^{\psi=0} \\ \downarrow \tau_m & & \downarrow \tau_m \\ \text{Ind}_{\Gamma_K}^{\Gamma_{K_0}}(\mathbf{D}^\dagger(V))^{\psi=0} & \xrightarrow{\kappa} & \mathbf{D}^\dagger(V)^{\psi=0} \end{array}$$

which shows the surjectivity of τ_m on $\mathbf{D}^\dagger(V)$ and the injectivity follows from Proposition A.1.5. $\square$

Proposition A.1.13. Let γ be a generator of Γ_K and V a free overconvergent $\mathbb{Z}_p$ -representation of G_K , then $\tau = \gamma - 1$ admits a continuous inverse on $\mathbf{D}^\dagger(V)^{\psi=0}$, denoted by $\tau^{-1} : \mathbf{D}^\dagger(V)^{\psi=0} \rightarrow \mathbf{D}^\dagger(V)^{\psi=0}$. More precisely, there exists $c(K, V) \in \mathbb{N}$ and $r(K, V) \in \mathbb{R}_+$ such that for all $r \geq r(K, V)$ and $a \in \mathbb{Z}$ we have

$$\tau^{-1}((\pi^a \mathbf{D}_r^\dagger(V))^{\psi=0}) \subset (\pi^{a-c(K,V)} \mathbf{D}_r^\dagger(V))^{\psi=0}.$$

Proof. Using the previous lemma, we can assume that $K = K_0$ by if necessary replacing V with $\text{Ind}_K^{K_0}(V)$ and we will show that $\gamma^f - 1$ admits a continuous inverse for f large enough. Since for any $k \mid f$ this also implies invertibility of $\gamma^k - 1$ we are able to apply the previous lemma. In particular, we have that the invertibility of $\gamma - 1$ follows from the invertibility of $\gamma^f - 1$ via

$$\tau^{-1} = (\gamma - 1)^{-1} = (1 + \gamma + \dots + \gamma^{f-1})\tau_f^{-1}.$$

Let $d = \text{rank}(V)$. By definition of overconvergent representation, we can find generators $e_1, \dots, e_d$ of $\mathbf{D}(V)$ over $\mathbf{A}_{K_0}$ and $r_0 \in \mathbb{R}_+^*$ such that $e_i \in \mathbf{D}_{r_0}^\dagger(V)$ for $0 \leq i \leq d$. We choose r_0 such that $r_0 \geq \frac{p-1}{p}$.

For $g \in \Gamma_K$ let U_g be the matrix the g -action with respect to the basis $e_1, \dots, e_d$, then

$$\begin{aligned} \Gamma_K &\longrightarrow \text{GL}_d(\mathbf{A}_{r_0, K_0}^\dagger) \\ g &\longmapsto U_g \end{aligned}$$

is a continuous group homomorphism. By continuity there exists a $m \in \mathbb{N}$ such that $g \in \Gamma_{K_{0m}}$ implies $U_g \in 1 + \pi^2 M_d(\mathbf{A}_{r_0, K_0}^\dagger)$. If $f = [K_{0m} : K_0]$ we can take $\gamma_m := \gamma^f$ as a generator of $\Gamma_{K_{0m}}$. Note also that any $g \in \Gamma_{K_{0m}}$ satisfies $\chi(\gamma_m) \in 1 + p^m \mathbb{Z}_p$. Since we may view V as a representation of $G_{K_{0m}}$ we can now replace K by K_{0m} and γ by γ_m . From now on we therefore assume we are in the following situation: γ is a generator of Γ_K , $U = U_\gamma \in 1 + \pi^2 M_d(\mathbf{A}_{r_0, K_0}^\dagger)$, $\chi(\gamma) \in 1 + p^m \mathbb{Z}_p$ and $m = v_p(\chi(\gamma) - 1)$. Further set $c = p^{m-1}$. Our goal is to show that $\tau = \gamma - 1$ admits a continuous inverse on $\mathbf{D}(V)^{\psi=0}$.

Since

$$\frac{\sigma^k(\pi)}{\sigma(\pi)^{p^{k-1}}} = \frac{\sigma^k(\pi)}{\pi^{p^k}} \left(\frac{\pi^p}{\sigma(\pi)} \right)^{p^{k-1}}$$

is a unit $\mathbf{A}_{p^k r, K_0}^\dagger$ for $r \geq 1$ we see that

$$\sigma^k(\pi \mathbf{A}_{r, K_0}^\dagger) \subset \sigma(\pi)^{p^{k-1}} \mathbf{A}_{p^k r, K_0}^\dagger$$

for $r \geq 1$. If we now replace $e_1, \dots, e_d$ by $\varphi^m(e_1), \dots, \varphi^m(e_d)$ we can assume $e_1, \dots, e_d \in \varphi(\mathbf{D}(V))$. Then we see that U changes to $\sigma^m(U)$, where the σ -action

is considered componentwise and with replacing r_0 by $p^m r_0$ with see that our new $U \in 1 + \sigma(\pi)^{2p^{m-1}} M_d(\mathbf{A}_{r_0, K_0}^\dagger) \in 1 + \sigma(\pi)^{c+1} M_d(\mathbf{A}_{r_0, K_0}^\dagger)$. Also note that now $r_0 \geq p^{m-1}(p-1)$ because we originally chose r_0 satisfying $r_0 \geq \frac{p-1}{p}$.

From now on, let $r \geq r_0$ and let $\mathbf{D}_r$ denote the sub- $\mathbf{A}_{r, K_0}^\dagger$ -module of $\mathbf{D}(V)$ generated by $e_1, \dots, e_d$. In summary, we have reduced to the case where we have a basis $e_1, \dots, e_d$ of $\mathbf{D}(V)$ over $\mathbf{A}_{K_0}$ with $e_i \in \mathbf{D}_{r_0}^\dagger(V) \cap \varphi(\mathbf{D}(V))$. Further, since $U \in 1 + \sigma(\pi)^{c+1} M_d(\mathbf{A}_{r_0, K_0}^\dagger)$, we have $\tau(e_i) \in \sigma(\pi)^{c+1} \mathbf{D}_{r_0}$ for $0 \leq i \leq d$.

Lemma A.1.14. An element $x = \sum_{i=1}^d x_i e_i \in \mathbf{D}_r$ is in $\mathbf{D}_r^{\psi=0}$ if and only if $x_i \in (\mathbf{A}_{r, K_0}^\dagger)^{\psi=0}$ for $0 \leq i \leq d$.

Proof. Since φ is injective we have that $\psi(x) = 0$ if and only if $\varphi(\psi(x)) = 0$. Since $e_i \in \varphi(\mathbf{D}(V))$ we have $\varphi(\psi(e_i)) = e_i$ and

$$\varphi(\psi(x)) = \varphi(\psi(\sum_{i=1}^d x_i e_i)) = \sum_{i=1}^d \sigma(\psi(x_i)) e_i$$

and thus $\psi(x) = 0$ if and only if $\psi(x_i) = 0$ for $0 \leq i \leq d$. $\square$

Proposition A.1.15. τ induces a bijection

$$\tau : \sigma(\pi)^a \mathbf{D}_r^{\psi=0} \longrightarrow \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0}$$

for all $r \geq r_0$ and $a \in \mathbb{Z}$.

Proof. Let $r \geq r_0$ and $a \in \mathbb{Z}$. Write $z = \sum_{i=1}^d z_i e_i \in \sigma(\pi)^a \mathbf{D}_r^{\psi=0}$ then $z_i \in \sigma(\pi)^a \mathbf{A}_{r_0, K_0}^\dagger$ and

$$\tau(\sum_{i=1}^d z_i e_i) = \sum_{i=1}^d \tau(z_i) e_i + \sum_{i=1}^d \gamma(z_i) \tau(e_i).$$

We see that by Lemma A.1.10 that $\tau(z_i) \in \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0}$ and we established $\tau(e_i) \in \sigma(\pi)^{c+1} \mathbf{D}_r^{\psi=0}$. Further, $\gamma(z_i) \in \gamma(\sigma(\pi))^a \mathbf{A}_{r_0, K_0}^\dagger \subset \sigma(\pi)^a \mathbf{A}_{r_0, K_0}^\dagger$ since $\gamma(\pi) = (1 + \pi)^{\chi(\gamma)} - 1 \in \pi \mathbf{A}_{K_0}^+$. Thus we see that

$$\tau(\sigma(\pi)^a \mathbf{D}_r^{\psi=0}) \subset \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0}.$$

The injectivity of γ follows from the injectivity of γ on $\mathbf{D}(V)^{\psi=0}$ (see Lemma A.1.5).

We show the surjectivity of τ using the same technique as in Lemma A.1.10. For $z = \sum_{i=1}^d z_i e_i \in \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0}$ we have $z_i \in \sigma(\pi)^{a+c} (\mathbf{A}_{r_0, K_0}^\dagger)^{\psi=0}$ by the previous lemma. Recall that $\tau : \sigma(\pi)^a (\mathbf{A}_{r_0, K_0}^\dagger)^{\psi=0} \longrightarrow \sigma(\pi)^{a+c} (\mathbf{A}_{r_0, K_0}^\dagger)^{\psi=0}$ is invertible by Proposition A.1.8.

Define

$$f : \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0} \longrightarrow \sigma(\pi)^a \mathbf{D}_r^{\psi=0}$$

$$f(z) = \sum_{i=1}^d \tau^{-1}(z_i) e_i$$

then f is injective and we have

$$z - f(\tau(z)) = z - f\left(\sum_{i=1}^d \tau(z_i) e_i + \sum_{i=1}^d \gamma(z_i) \tau(e_i)\right) = -f\left(\sum_{i=1}^d \gamma(z_i) \tau(e_i)\right).$$

For $z \in \sigma(\pi)^b \mathbf{D}_r$, using that $\tau(e_i) \in \sigma(\pi)^{c+1} \mathbf{D}_r$ and $\gamma(z_i) \in \sigma(\pi)^b (\mathbf{A}_{r_0, K_0}^\dagger)^{\psi=0}$ that $\gamma(z_i) \tau(e_i) \in \sigma(\pi)^{b+c+1} \mathbf{D}_r$. We then see that $z - f(\tau(z)) \in \sigma(\pi)^{b+1} \mathbf{D}_r^{\psi=0}$. Completely analogous to before, we can once again see that the function $g_z(y) = y - f(\tau(y) - z)$ defines a contraction on $\sigma(\pi)^a \mathbf{D}_r^{\psi=0}$ with respect to the $\sigma(\pi)$ -adic topology. First, for $z \in \sigma(\pi)^{a+c} \mathbf{D}_r^{\psi=0}$, g_z indeed defines a function from $\sigma(\pi)^a \mathbf{D}_r^{\psi=0} \longrightarrow \sigma(\pi)^a \mathbf{D}_r^{\psi=0}$. Further, from above we see that for $y, y' \in \sigma(\pi)^a \mathbf{D}_r^{\psi=0}$ with $y - y' \in \sigma(\pi)^b \mathbf{D}_r^{\psi=0}$ we have

$$g_z(y) - g_z(y') = y - y' - f(\tau(y - y')) \in \sigma(\pi)^{b+1} \mathbf{D}_r^{\psi=0}$$

so g_z defines a contraction. But $\sigma(\pi)^a \mathbf{D}_r^{\psi=0}$ is complete with regard to the $\sigma(\pi)$ -adic topology, since $\mathbf{D}_r$ is and ψ is continuous, so g_z must have a fixed point. $\square$

Now choose generators $v_1, \dots, v_d$ for V over $\mathbb{Z}_p$. Note that since $\mathbf{D}(V) = (\mathbf{A} \otimes_{\mathbb{Z}_p} V)^{H_K}$, any element $x \in \mathbf{D}_{r_0}^\dagger(V)$ can also be written as $x = \sum_{i=1}^d x_i (1 \otimes v_i)$ with $x_i \in \mathbf{A}_{r_0}^\dagger$. We will write v_i for $1 \otimes v_i$ from now on. By (3.2) we have

$$\mathbf{A} \otimes_{\mathbf{A}_K} \mathbf{D}(V) \xrightarrow{\sim} \mathbf{A} \otimes_{\mathbb{Z}_p} V$$

so $e_1, \dots, e_d$ generate $\mathbf{A} \otimes_{\mathbb{Z}_p} V$ over $\mathbf{A}$. In particular, if we let M be the case change matrix from $v_1, \dots, v_d$ to $e_1, \dots, e_d$ we have $M \in \mathrm{GL}_d(\mathbf{A})$. Further $M \in M_d(\mathbf{A}_{r_0}^\dagger)$ since $e_i \in \mathbf{D}_{r_0}^\dagger(V)$ and thus $M \in M_d(\mathbf{A}_{r_0}^\dagger) \cap \mathrm{GL}_d(\mathbf{A})$. Using the formula $M^{-1} = \det(M)^{-1} \mathrm{adj}(M)$ we can find that M^{-1} must have coefficients in the fraction field of $\mathbf{A}_{r_0}^\dagger$ so in particular in $\mathbf{B}^\dagger$ and thus $M^{-1} \in \mathrm{GL}_d(\mathbf{B}^\dagger) \cap \mathrm{GL}_d(\mathbf{A}) = \mathrm{GL}_d(\mathbf{A}^\dagger)$. By definition of $\mathbf{A}^\dagger$ there exists some $r_1 \geq r_0$ and some $b \in \mathbb{N}$ such that $M^{-1} \in \pi^{-b} M_d(\mathbf{A}_{r_1}^\dagger)$.

As before, we can write any element $x \in \mathbf{D}_{r_1}^\dagger(V)$ as $x = \sum_{i=1}^d x'_i v_i$ with $x'_i \in \mathbf{A}_{r_1}^\dagger$. Changing the base to e_i using M^{-1} we then get $x = \sum_{i=1}^d x_i e_i$ with $x_i \in \pi^{-b} \mathbf{A}_{r_1}^\dagger$ since $M^{-1} \in \pi^{-b} M_d(\mathbf{A}_{r_1}^\dagger)$. Since e_i are fixed under H_K , so are the x_i and thus $x_i \in \pi^{-b} \mathbf{A}_{r_1, K_0}^\dagger \subset \sigma(\pi)^{-b} \mathbf{A}_{r_1, K_0}^\dagger$. Recall that $\mathbf{D}_r$ is the sub- $\mathbf{A}_{r, K_0}^\dagger$ -module generated by $e_1, \dots, e_d$. We deduce that for $r \geq r_1$

$$\sigma(\pi)^b x = \sum_{i=1}^d \sigma(\pi)^b x_i e_i \in \mathbf{D}_r$$

and thus

$$\sigma(\pi)^b \mathbf{D}_r^\dagger(V) \subset \mathbf{D}_r \subset \mathbf{D}_r^\dagger(V). \quad (\text{A.1})$$

We can now see that $\tau : \mathbf{D}^\dagger(V)^{\psi=0} \rightarrow \mathbf{D}^\dagger(V)^{\psi=0}$ is surjective. Every element in $\mathbf{D}^\dagger(V)^{\psi=0}$ lies in some $(\pi^{-pa} \mathbf{D}_r^\dagger(V))^{\psi=0} \subset \sigma(\pi)^{-a} \mathbf{D}_r^\dagger(V)^{\psi=0}$ for $a \in \mathbb{N}$ and $r \geq r_1$. Now Lemma A.1.15 states that

$$\tau : \sigma(\pi)^{-a-b-c} \mathbf{D}_r^{\psi=0} \rightarrow \sigma(\pi)^{-a-b} \mathbf{D}_r^{\psi=0}$$

is bijective, and since $\sigma(\pi)^{-a} \mathbf{D}_r^\dagger(V)^{\psi=0} \subset \sigma(\pi)^{-a-b} \mathbf{D}_r^{\psi=0}$ we can deduce surjectivity of τ . Recall that we already showed injectivity of τ on $\mathbf{D}(V)^{\psi=0}$ in Lemma A.1.5 so we can now denote the inverse of τ by $\tau^{-1} : \mathbf{D}^\dagger(V)^{\psi=0} \rightarrow \mathbf{D}^\dagger(V)^{\psi=0}$.

Finally, we can also see using Lemma A.1.15 that for $a \in \mathbb{Z}$ we have

$$\tau^{-1}(\sigma(\pi)^{a-b} \mathbf{D}_r^{\psi=0}) \subset \sigma(\pi)^{a-b-c} \mathbf{D}_r^{\psi=0}.$$

thus we deduce using the inclusions (A.1) that

$$\tau^{-1}(\sigma(\pi)^a \mathbf{D}_r^\dagger(V)^{\psi=0}) \subset \sigma(\pi)^{a-b-c} \mathbf{D}_r^\dagger(V)^{\psi=0}.$$

If we now use the fact that $\frac{\sigma(\pi)}{\pi^p}$ is a unit in $\mathbf{A}_{r,K_0}^\dagger$ we can deduce the desired statement with $r(K, V) = r_1$ and $c(K, V) = p(b+c+1)$. In particular, this implies the continuity of τ^{-1} . $\square$

We can now also deduce a statement about the invertibility of $\gamma - 1$ for p -torsion representations. We do that by reducing to the case of a length one $\mathbb{Z}_p$ -module, i.e., a one-dimensional mod p representation.

Corollary A.1.16. Let γ be a generator of Γ_K and V a p -torsion representation. Then $\gamma - 1$ is invertible on $\mathbf{D}(V)^{\psi=0}$.

Proof. Precisely as before, we can reduce to the case $K = K_0$ by if necessary considering the representation $\text{Ind}_K^{K_0}(V)$.

We can deduce the statement for a one-dimensional mod p representations by using the same proof as for free $\mathbb{Z}_p$ representations. Alternatively, we can use that for a free $\mathbb{Z}_p$ representation V the following diagram commutes since ψ fixes p :

$$\begin{array}{ccc} \mathbf{D}^\dagger(V)^{\psi=0} & \xrightarrow{\tau} & \mathbf{D}^\dagger(V)^{\psi=0} \\ \downarrow & & \downarrow \\ \mathbf{D}^\dagger(V/pV)^{\psi=0} & \xrightarrow{\tau} & \mathbf{D}^\dagger(V/pV)^{\psi=0} \end{array}$$

where the vertical maps are induced by the projection $V \twoheadrightarrow V/pV$ and are in particular surjective. Since for a one-dimensional mod p representation W we can define a free

rank one $\mathbb{Z}_p$ -representation V such that $W = V/pV$, we deduce from the diagram that τ is surjective on $\mathbf{D}^\dagger(W)^{\psi=0}$. The injectivity of τ is Corollary A.1.7. Finally, note that for W a mod p representation, we have $\mathbf{D}^\dagger(W) = \mathbf{D}(W)$ since $\mathbf{A}_K/p\mathbf{A}_K \subset \mathbf{A}_K^\dagger$ and therefore $\mathbf{A}_K/p\mathbf{A}_K = \mathbf{A}_K^\dagger/p\mathbf{A}_K^\dagger$.

Like before, we can show that for a p -torsion representation V we can reduce to the case of V having length one over $\mathbb{Z}_p$, i.e., $\mathbf{D}(V)$ having length one over $\mathbf{A}_K$. For $M, M', M'' \in \Phi\Gamma M_{\mathbf{A}_K}^{\text{ét}, p\text{-tor}}$ and a short sequence of (φ, Γ_K) -modules

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

we can deduce, since ψ is surjective and commutes with maps of (φ, Γ_K) -modules, that

$$0 \longrightarrow \ker(\psi_{M'}) \longrightarrow \ker(\psi_M) \longrightarrow \ker(\psi_{M''}) \longrightarrow 0.$$

Since the maps of the sequence are Γ_K -equivariant we can directly deduce the bijectivity of τ on $\ker(\psi_M)$ from the bijectivity of τ on $\ker(\psi_{M'})$ and $\ker(\psi_{M''})$. We can therefore assume $\mathbf{D}(V)$ has length one over $\mathbf{A}_K$. $\square$

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